A Capability-Based, Event-Driven Linux Desktop: Knowledge Graph Architecture and Design Rationale

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Abstract Desktop operating systems accumulate rich contextual data about user activity, but this data is trapped in application silos with no shared query layer and no consistent permission model for cross-application access. This work presents the design and architecture of a Linux-based desktop operating system built around a structured, local-first Knowledge Graph as a first-class infrastructure component.

The Knowledge Graph records system activity as typed nodes and relationships queryable across application boundaries and time. A two-layer write architecture separates high-frequency event ingestion (SQLite) from semantic graph traversal (an embedded property graph database), connected by an asynchronous promotion pipeline. All components access the graph through a single privileged daemon using cryptographically scoped capability tokens; no component holds elevated trust by default.

The architecture covers the full system stack: a forked Wayland compositor, a Tauri-based application framework with a shared design token system, an event pipeline driven by eBPF, a provider-agnostic AI layer with explicit MCP integration, profile isolation via systemd-homed, and a layered security model with an explicit threat model. Each design decision is documented with its rationale and trade-offs, including honest treatment of unresolved risks.

The system is currently in the design phase. The contribution of this work is the architecture itself: a concrete, implementable blueprint that treats the Knowledge Graph as operating system infrastructure rather than an application feature.

Keywords operating system, knowledge graph, capability tokens, eBPF, event-driven architecture, Linux desktop, AI integration, local-first

Contents

1 Introduction **4**
1.1 Motivation . . . 4
1.2 Target Audience . . . 5
1.3 Design Principles . . . 5
1.4 Positioning . . . 6
1.5 What This Project Is Not . . . 6

2 System Overview **7**
2.1 Architecture at a Glance . . . 7
2.2 Layer Responsibilities . . . 8
2.3 Why Rust . . . 9
2.4 Why Wayland . . . 10
2.5 Why Tauri . . . 10
2.6 Why cosmic-comp (Forked) . . . 11
2.7 Component Boundaries . . . 13
2.8 Communication Model . . . 14
2.8.1 Unix Sockets . . . 14
2.8.2 Protocol Buffers . . . 14
2.8.3 Knowledge Graph Rationale . . . 15
2.9 D-Bus Coexistence . . . 16

3 OS Foundation **17**
3.1 What is a Linux Distribution . . . 17
3.2 Base Distribution . . . 17
3.3 Update Strategy . . . 18
3.4 System Infrastructure . . . 20
3.5 Hardware Requirements . . . 22
3.6 Configuration Format . . . 23

4 Event System and Knowledge Graph **24**
4.1 Overview . . . 24
4.2 Wayland as an Event Source . . . 25
4.3 Kernel Monitoring with eBPF . . . 25
4.4 Event Bus . . . 27
4.5 Write Architecture . . . 28
4.6 Knowledge Graph . . . 30
4.7 Project System . . . 31
4.8 Retention Policy . . . 36
4.9 Temporal Filesystem View . . . 37

5 AI Layer **38**
5.1 Overview . . . 38
5.2 Cold-Start Behavior . . . 38
5.3 Provider Abstraction . . . 39

5.4 MCP Integration . . . 40
5.5 Interaction Model . . . 41
5.6 Project Context . . . 42
5.7 MCP Permission Model . . . 43
5.8 System Explanation Mode . . . 44

6 Desktop Environment and Design Language 44
6.1 Architecture and IPC . . . 44
6.2 Compositor and Window Manager . . . 45
6.3 Shell and Application Framework . . . 46
6.4 Shell API . . . 50
6.5 App Menu Context Popup . . . 60
6.6 Default Applications . . . 61
6.7 Waypointer . . . 62
6.8 Focus Mode . . . 64
6.9 Settings . . . 65
6.10 Notification System . . . 69
6.11 Screen Lock . . . 69
6.12 Multi-Monitor and HiDPI . . . 70
6.13 Design Language and Theming . . . 70

7 Apps, Modules, and the Store 78
7.1 App Ecosystem . . . 78
7.2 Notification System . . . 79
7.3 Permission System . . . 80
7.4 Integration Packages . . . 82
7.5 Install Daemon . . . 85
7.6 Store . . . 90
7.7 MCP Safety and External Connections . . . 92

8 Platform, Profiles, and Security 93
8.1 Account System . . . 93
8.2 Profile System . . . 96
8.3 Gaming and Windows Compatibility . . . 99
8.4 Security and Privacy . . . 100
8.4.1 Threat Model . . . 100
8.4.2 Full Disk Encryption . . . 101
8.4.3 Secure Boot, Unified Kernel Images, and Measured Boot . . . 102
8.4.4 App Sandboxing . . . 102
8.4.5 Knowledge Graph Security . . . 103
8.4.6 AI Security . . . 107
8.4.7 Audit Log . . . 108
8.4.8 Anomaly Detection . . . 110
8.4.9 Network Security . . . 110
8.4.10 Physical Security . . . 111
8.4.11 Memory Safety . . . 112

8.4.12 Multi-User Isolation 113
8.5 Managed Environments 113

9 Companion App and Health **115**
9.1 Companion App 115
9.2 Health and Wellbeing 117

10 Developer Experience, Sustainability, and Roadmap **119**
10.1 Developer Experience and Infrastructure 119
10.2 Sustainability and Business Model 126
10.3 Roadmap 128

1 Introduction

1.1 Motivation

Most Linux distributions are either conservative remixes of existing systems or highly opinionated tools for a specific niche. Neither describes this project.

The dominant desktop operating systems have a structural problem. Windows works against its own users: forced updates, advertisements in the start menu, and background telemetry that nobody enabled and cannot be fully disabled. The German Federal Office for Information Security (BSI) has studied this directly: as part of the SiSyPHuS research programme, the BSI found that required diagnostic data collection in Windows 10 and 11 runs continuously and cannot be turned off by the user [26]. macOS presents a different trade-off: its privacy posture is meaningfully better than Windows, but the ecosystem is deliberately constructed to keep users paying and dependent. A 2024 study published at CHI found that even opting out of Apple's analytics during device setup does not prevent data collection by individual system applications, and that the setup process itself involves at least eight distinct data-sharing decisions presented in ways that are not straightforward to review [67]. Every Linux distribution, however technically sound, has failed to reach people who simply want to use a computer without becoming a system administrator. The gap between technically capable and actually usable has not closed in thirty years.

This project exists to close that gap. It is a Linux-based operating system designed around a different set of priorities: one that integrates a system-wide knowledge graph, a structured project system, AI interfaces, and modern tooling not as features bolted on top, but as foundational infrastructure. It takes a clear stance on user sovereignty, independence from dominant platform vendors, and not being annoying to use. If an operating system is being built in 2026 anyway, there is no reason to repeat patterns from 1995.

1.2 Target Audience

The primary audience is people switching from macOS or Windows who want a genuinely good alternative: modern in appearance, functional out of the box, and respectful of their autonomy. Consumer-first, not enterprise-first. The macOS user tired of lock-in but unwilling to compromise on quality. The Windows user whose computer feels like it belongs to Microsoft more than to them. They do not want to configure. They want to arrive and work, and feel like the operating system is actually theirs.

Business deployment is a supported use case. The architecture accommodates managed environments and administrator policies. The user remains the primary stakeholder regardless of deployment context: an administrator can configure defaults and restrict software; an administrator cannot surveil users or override their personal data ownership. The system is not enterprise-first; it is user-first by design, in every context.

1.3 Design Principles

The following principles govern every design decision in this project. They are not aspirational language; they are constraints that break ties and override convenience.

The user's interests come first. No advertisements, no tracking, no dark patterns, no telemetry without explicit consent. This applies in managed environments too: an organization can configure and restrict, but cannot surveil. All system intelligence runs locally. The knowledge graph, AI context, and activity history do not leave the machine without an explicit user action. The system works fully offline, without registration, and without depending on services controlled by companies in other jurisdictions. Cloud features are opt-in extensions, not prerequisites. This is a technical commitment. It means the architecture enforces it, not just the privacy policy.

The system has to be genuinely good to use. Everything works on first boot. No hunting for drivers, no reading wiki pages to get audio working. The interface must look better than what users are coming from, because a poor interface is a barrier to entry regardless of what is underneath it. Small things matter more than large ones: a system that remembers what you were working on, groups your files and activity into projects without requiring manual organization, or switches themes everywhere in one action, earns trust faster than an impressive feature nobody uses. Coherence matters too: shared configuration, shared theming, shared interfaces across all components. No part of the system should look like it was designed in a different decade by a different team.

New technology gets used when it makes sense, not because it is trending. AI and the knowledge graph are here because they genuinely change what a personal computer can do for the user. The goal is a useful, trustworthy system, not a clean architecture. Where established standards exist, this project builds compatibility bridges rather than replacements. Native applications are distributed as RPM packages; Flatpak serves as a compatibility layer for third-party applications not distributed through the Store. Standard hardware interfaces work; nothing is deliberately broken for the sake

of purity. The system can explain what it is doing, in plain language, in a place the user can see and act on it.

1.4 Positioning

Figure 1 maps the design space along two axes: how easy the system is to use for a non-technical user, and how much sovereignty the user has over their own data and software environment. These two properties have historically traded off against each other. Mainstream commercial systems prioritize usability but compromise on sovereignty, typically through telemetry, ecosystem lock-in, or both. The more privacy-conscious Linux distributions give users genuine control but impose a real usability cost. The upper-right quadrant, high usability *and* high user sovereignty, is where this project aims to land. No system targeting mainstream usability currently occupies it.

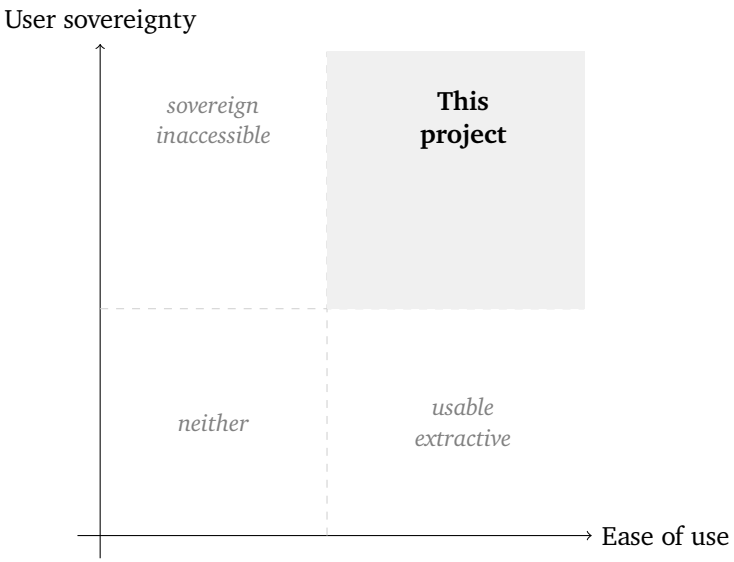

Figure 1 Design space: ease of use versus user sovereignty. The shaded quadrant is the project target. Mainstream commercial systems land in the lower-right (usable but extractive); privacy-oriented Linux distributions in the upper-left (sovereign but inaccessible). No mainstream system currently occupies the upper-right.

1.5 What This Project Is Not

Clarifying what this project is not is as important as describing what it is.

It is not a minimal system stripped to its essentials. There is real complexity here, and that is intentional: a system that works for non-technical users requires more infrastructure, not less.

It is not a fully declarative system where every piece of software is managed through configuration files. Reproducibility is a goal; mandatory declarative configuration is not.

It is not a distro aimed at advanced Linux users only. New users are explicitly part of the target. Advanced users are welcome and well-served, but they are not the primary design constraint.

It is not an enterprise product that happens to support consumers. The design constraint is the non-technical user; business features exist where they do not compromise that.

It is not a Windows or macOS clone. The goal is a better alternative, not a familiar imitation.

2 System Overview

2.1 Architecture at a Glance

The system is organized as a layered stack. Lower layers provide services to higher layers; nothing reaches past its immediate neighbor. The bottom four layers are always present: the OS base, the desktop environment, the messaging and integration infrastructure, and the Knowledge Graph. Above those sit the system's consumers: the applications, daemons, and optional AI daemon that read from and write to the graph. Figure 2 shows the full structure.

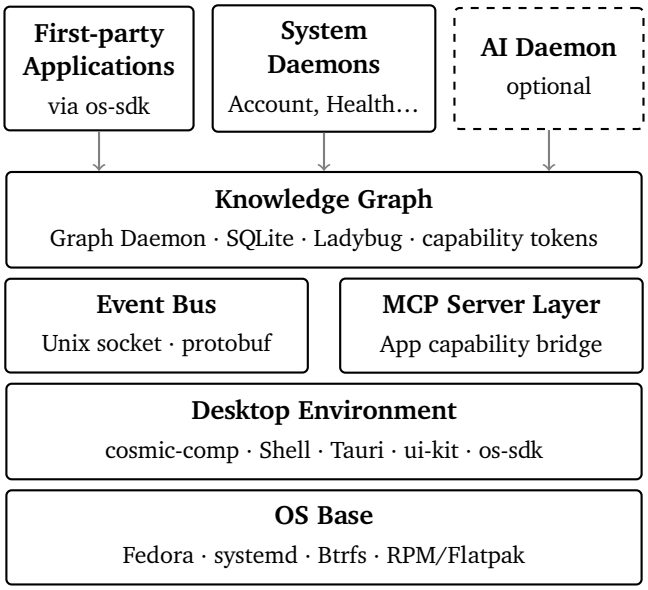

Figure 2 System layer structure. The bottom four layers are always present and initialize in order from bottom to top. The top row shows the three classes of graph consumer: first-party applications (connected via os-sdk), system daemons (Account, Health, Audit, Install, and others), and the AI daemon (dashed, optional). All three connect to the Graph Daemon over a Unix socket with per-application capability tokens; none of them bypasses the daemon or accesses the graph database directly. The AI daemon is not architecturally privileged over the others.

2.2 Layer Responsibilities

OS Base. The foundation is a standard Linux distribution providing the kernel, init system, package manager, and base userspace tools. Nothing at this layer is custom-built. The system builds on top of it; it does not replace it. The choice of base distribution, the update strategy, and the concrete packages involved are covered in detail in Section 3.

Desktop Environment. The Wayland compositor and the shell sit at this layer. This is what the user directly interacts with: windows, the taskbar, the Waypointer launcher, notifications, and the lock screen. All user-facing applications in this project are built with *Tauri* [106], a framework described in Section 2.5. The compositor is based on a fork of *cosmic-comp* [103], described in Section 2.6.

Event Bus. The internal message bus handles all communication between custom system components. It receives, validates, and routes structured events from kernel monitoring, the display system, and first-party applications. It runs as a background service over *Unix sockets* (see Section 2.8.1) using *Protocol Buffers* for schema-enforced serialization (see Section 2.8.2). Its role is purely to move and validate data; it does not interpret or store anything.

MCP Server Layer. The *Model Context Protocol* [2] (MCP) is an open standard published by Anthropic in 2024 for exposing application capabilities as callable tools that AI models can invoke. In this system first-party applications expose MCP servers as part of their standard implementation via os-sdk, giving any MCP-compatible client a structured, permission-gated way to act on the system. The MCP layer is not specific to the AI daemon; any client that implements the protocol can use it.

Knowledge Graph. The graph database is the system's long-term memory: a local, structured index of activity and context, queryable across time, application boundaries, and data types. The *Graph Daemon* is the sole reader and writer of the underlying database; no other process opens the database files directly. Every component that needs graph data connects to the daemon over a Unix socket and presents a cryptographically signed capability token that encodes its permitted query scope. Why a graph database rather than a relational one, and why *Ladybug* [69] specifically, is explained in Section 2.8.3.

First-party applications. Every first-party application written for this system connects to the Graph Daemon via the typed client bundled in *os-sdk*. Opening the File Manager, searching in Waypointer, or checking System Monitor all translate to parameterized Cypher queries over a Unix socket. The application receives structured results and renders them; it never touches the graph database files directly, and the capability token it presents limits which node types and relations it can read.

System daemons. Background daemons such as the Account Daemon, Health Daemon, and Audit Daemon are also graph clients. They write structured data into the graph

(account events, health observations, audit records) and read from it when their function requires cross-domain context. Like applications, each daemon holds a capability token with the minimum scope it needs; graph access is not elevated for daemons by default.

AI daemon (optional). The AI daemon is a system daemon like the others. It connects to the Graph Daemon via the same Unix socket interface, presents the same style of capability token, and receives the same enforcement from the daemon. What distinguishes it is not a privileged protocol position but a broader read permission at higher user-configured access levels (see Section 8.4.6). It additionally orchestrates tool calls via the MCP layer and mediates between user requests and model providers, local or cloud, behind a common Rust trait interface (see Section 2.3). Disabling the AI daemon removes AI features and nothing else; the graph, the applications, and the other daemons are unaffected.

2.3 Why Rust

All system daemons and Tauri application backends in this project are written in *Rust* [111], a systems programming language developed by Mozilla Research with its 1.0 release in 2015. Understanding why requires a brief look at what systems programming has historically looked like.

C and C++ have dominated operating system components, daemons, and performance-critical software for decades. Their advantage is direct control over memory and hardware with minimal overhead. Their liability is that this control is entirely manual: the programmer allocates memory and is responsible for freeing it correctly, without using it afterward, without reading past its bounds, and without sharing it unsafely across threads. Memory safety bugs (use-after-free, buffer overflows, data races) are the single largest source of exploitable vulnerabilities in system software. Google's Project Zero and Microsoft's Security Response Center have independently reported that roughly 70% of serious security vulnerabilities in their codebases are memory safety issues [70, 78].

Go is a more recent alternative with garbage collection and a simpler concurrency model, but its garbage collector introduces unpredictable pause latency unsuitable for low-level system components, and its runtime overhead is significant compared to C or Rust.

Rust enforces memory safety through a compile-time ownership and borrowing system. An entire class of bugs that would be exploitable vulnerabilities in C simply cannot compile in Rust; the compiler rejects them. This is not a runtime check with performance overhead; it is a zero-cost compile-time guarantee. Rust has no garbage collector and achieves performance comparable to C and C++.

For this project the consequence is concrete: every daemon runs continuously, handles untrusted input from applications, and manages sensitive data including credentials and graph queries. Writing them in Rust eliminates an entire vulnerability class without sacrificing performance.

A *Rust trait* is the language's mechanism for defining a shared interface that multiple concrete types can implement, analogous to an interface in Java or a pure abstract class in C++. When this document refers to the AI provider abstraction as a Rust trait, it means that every AI backend implements the same set of methods, and the rest of the codebase calls only those methods without knowing which concrete backend is active at runtime.

2.4 Why Wayland

Wayland [122] is a display server protocol: the specification that defines how applications communicate with the component responsible for drawing windows on the screen.

X11 (the X Window System) was designed in 1984 and has been the foundation of graphical Linux desktops ever since. Its architecture reflects its era: the display server is a separate process, and by default any application can inspect the input events and window contents of any other application on the same display. One X11 application can read keystrokes typed into another application's window [123]. This is a keylogger-class capability that cannot be removed without a protocol redesign, and it makes X11 structurally incompatible with the isolation guarantees this project requires.

Wayland redesigns this from scratch: each application communicates only with the compositor, not with other applications. Input events are delivered only to the window that currently has focus. An application cannot read another application's keystrokes or window contents. This isolation is the default behaviour, not an optional hardening measure.

Wayland also has a cleaner model for multi-monitor and HiDPI configurations: each output carries its own scale factor, and applications receive this information directly and render at the correct resolution automatically.

The compatibility cost is real. Wayland addresses it through *XWayland*, an X11 compatibility server that runs as a Wayland client. Legacy X11 applications work inside XWayland without modification. X11's security limitations apply within the XWayland boundary but are contained there and do not affect native Wayland applications.

2.5 Why Tauri

Tauri is a framework for building desktop applications with a Rust backend and a web-based user interface. A Tauri application consists of a Rust process that handles system interactions, file access, and IPC, and a *WebView* that renders the UI using standard web technologies: HTML, CSS, and TypeScript. The two halves communicate through a typed command bridge: the frontend calls Rust functions via `invoke()`, and Rust pushes events to the frontend via Tauri's event emitter. No ad-hoc serialization is needed; the bridge handles it.

The dominant alternative is *Electron*, which takes the same split but bundles an entire Chromium browser and a Node.js runtime with every application. A minimal Electron application ships roughly 150 MB of runtime dependencies and consumes

around 300 MB of RAM at idle [108]. On a system with several Electron applications running, the memory cost is substantial.

Tauri uses the operating system's existing WebView implementation instead of bundling one. On Linux this is *WebKitGTK*, the GTK port of the WebKit rendering engine. WebKitGTK is already present on any modern Linux desktop; a Tauri application does not add it. A Tauri application binary is typically 5–10 MB and uses a fraction of the RAM that an equivalent Electron application would require.

Building the UI in a native toolkit such as *GTK4* or *Qt* was considered and rejected for two reasons. First, maintaining visual consistency across components developed independently is significantly harder with a native toolkit than with a shared CSS design system where a token change propagates everywhere. Second, the developer pool for TypeScript and CSS is larger than for GTK or Qt development, which matters for an open-source project expecting community contributions. The trade-off is that WebView rendering is slightly less efficient than native rendering for highly dynamic content, but this is not a relevant concern for the settings panels, file browsers, and launchers that make up this system's UI.

Tauri version 2.0, released in October 2024, is used throughout this project. Version 2.0 introduced a revised permission system for Tauri commands (replacing the allowlist model of 1.x with a finer-grained capability model), official mobile support, and improved IPC ergonomics. The capability model in 2.0 aligns well with this project's general approach to least-privilege permissions.

2.6 Why cosmic-comp (Forked)

The compositor is based on a fork of *cosmic-comp* [103], the Wayland compositor built by System76 for their COSMIC desktop environment. cosmic-comp is written in Rust on top of *Smithay* [100], a Rust library that provides the low-level compositor plumbing: Wayland protocol handling, DRM/KMS interfaces for direct display hardware access, input processing via *libinput*, and buffer management including GPU-side zero-copy paths via `dmabuf`.

Why fork rather than build from scratch. Building a production Wayland compositor from Smithay primitives is a multi-year effort. cosmic-comp has already solved the hard problems: correct multi-monitor handling with per-output scale factors, tiling and floating window management, XWayland integration, screen capture via `wlr-screencopy`, input method framework support for non-Latin input, and the full Wayland protocol extension set a modern desktop requires. Forking inherits all of this. Starting from scratch would reproduce it, worse, over a longer timeline.

Why not adopt COSMIC wholesale. The COSMIC desktop ships as an integrated product: cosmic-comp is coupled to cosmic-panel, cosmic-launcher, cosmic-settings, and the rest of the COSMIC shell via custom Wayland protocol extensions and D-Bus interfaces designed specifically for that stack. Adopting COSMIC wholesale would mean inheriting its shell design, its theming system based on *iced*, and its release

cadence, none of which align with this project. The goal is the compositor kernel, not the product.

What the fork retains. The fork keeps everything below the shell boundary: the Smithay-based window management engine, the input pipeline, the multi-monitor and HiDPI infrastructure, the XWayland bridge, and the Wayland protocol extension implementations. Window placement logic, tiling and floating behaviour, focus rules, and workspace management all come from cosmic-comp and are modified only where this project's behaviour explicitly differs.

What the fork replaces. The entire shell layer is replaced. cosmic-comp's custom IPC protocols for communicating with cosmic-panel and cosmic-launcher are removed. In their place the compositor exposes the standard wlr- Wayland protocol extensions (wlr-layer-shell, wlr-foreign-toplevel-management, wlr-workspace-management), which the Tauri-based shell consumes exactly as any other compliant shell client would. A minimal custom Unix socket channel handles events that have no standard Wayland equivalent: theme updates, Focus Mode signals, and graph notifications from the Event Bus. The shell is not a privileged internal component of the compositor; it is a Wayland client with the same protocol access any shell would have.

Maintenance posture and divergence cost. Forking an actively developed project carries a real and ongoing cost that is acknowledged directly. System76 develops cosmic-comp actively for their own hardware and their own COSMIC desktop. Their upstream receives regular commits: bug fixes, Smithay version bumps, new Wayland protocol extensions, and COSMIC-specific features. Every upstream change must be evaluated and either merged, adapted, or consciously rejected. As the fork diverges over time, this evaluation becomes more expensive because the diff between fork and upstream grows.

The concrete maintenance burden is: tracking upstream releases on a defined schedule, running the integration test suite after each upstream merge, resolving conflicts where upstream changes touch code that the fork has also modified, and communicating breaking changes to the rest of the project when an upstream merge requires downstream adaptation. This is not free work; it requires at least one contributor who understands both the Smithay internals and the fork's shell boundary well enough to evaluate upstream changes correctly.

An alternative was considered: contributing required features upstream to cosmic-comp rather than maintaining a fork, and using extensions or protocol additions to achieve the shell separation. This is rejected for one concrete reason. System76 is a commercial company with its own roadmap, hardware, and product direction. Contributing features upstream makes them subject to System76's review, acceptance criteria, and merge timeline. This is a different kind of dependency, not a lighter one. It would mean the project's compositor development velocity is gated by a company with no obligation to prioritise or accept the contributions. The fork trades upstream merge cost for development independence, and for a project where the compositor shell boundary is a fundamental architectural decision, that trade is correct.

The practical mitigation for merge cost is minimising the diff. The fork does not change window management logic, input handling, multi-monitor behaviour, or any subsystem that cosmic-comp maintains well. Changes are concentrated at the shell boundary: removing COSMIC-specific IPC and replacing it with standard Wayland protocols. A smaller diff means lower merge cost and a more tractable review process for each upstream bump.

2.7 Component Boundaries

The system's components divide into two categories: background services and user-facing applications.

Background services are called *daemons* in Unix terminology: processes with no visible window that are started at boot by *systemd* and automatically restarted if they crash. systemd is the standard init system and service manager on modern Linux distributions. It handles starting services in dependency order at boot, logging their output to the journal, and enforcing resource and sandboxing constraints via its service unit configuration. All daemons in this project are written in Rust and run under tight systemd sandboxing: each service unit declares exactly the filesystem paths, syscalls, and network access it requires, and everything else is denied at the kernel level.

Table 1 Background services (daemons)

Service	Responsibility
Event Bus	Receives, validates, and routes internal events between components
Graph Daemon	Sole reader/writer of the Knowledge Graph database
Account Daemon	Credential management and login token handling
AI Daemon	Model providers, MCP orchestration, graph queries (optional)
Module Runtime	Manages the lifecycle of shell extensions
Transfer Daemon	Controls data movement between user profiles
Audit Daemon	Sole writer to the append-only audit log
Health Daemon	Aggregates and stores health and wellbeing data
Install Daemon	Tracks app installations across all package managers; matches and installs Integration Packages automatically
Compositor	Draws windows and manages the display (Wayland compositor)

User-facing applications are built with Tauri and share two common libraries, described below; Table 2 lists them.

Every first-party application is built on two shared libraries. *ui-kit* is a TypeScript component library based on *shadcn/ui* (a set of accessible, unstyled components built on Radix UI primitives) styled with the project's shared design token system using *Tailwind CSS*. Tailwind is a utility-first CSS framework where styles are applied via class names rather than separate stylesheets; combined with CSS custom properties for design tokens, a theme change propagates to all running applications without a reload. *os-sdk* is a Rust crate that provides ready-made clients for the Graph Daemon

Table 2 User-facing applications

Application	Notes
Shell	Taskbar, Waypointer launcher, notification center, lock screen
Settings	System-wide configuration: theming, permissions, AI level, accounts
File Manager	First-party file browser with graph-aware features
Store	App and module discovery, installation, and updates
Terminal	First-party terminal emulator with MCP integration
Wine Manager	Windows compatibility management UI

and Event Bus, MCP server boilerplate, and the shared TOML configuration system. A new first-party application starts with these two libraries and has graph access, event emission, an MCP server, and a consistent UI without writing any infrastructure code.

2.8 Communication Model

2.8.1 Unix Sockets

All inter-component communication uses *Unix domain sockets*, a communication mechanism built into the Linux kernel. A Unix socket behaves like a network socket in its API but exists only on the local machine: it is represented as a file path in the filesystem (for example `/run/graph-daemon/graph.sock`), and only processes on the same machine can connect to it. There is no network stack involved; data passes directly through the kernel with latency in the range of a few microseconds.

D-Bus, the standard Linux IPC mechanism discussed in Section 2.9, was not chosen for internal component communication: it introduces a central broker process as an intermediary, adds marshalling overhead, and its type system is not well suited to the structured binary messages used here. TCP loopback sockets were also considered and rejected: they go through the full network stack, carry more overhead, and expose a port reachable by any process on the machine.

2.8.2 Protocol Buffers

Data transmitted over the internal sockets is serialized with *Protocol Buffers* (protobuf) [52], a binary serialization format developed and open-sourced by Google. Serialization is the process of converting a data structure in memory into a byte sequence for transmission, and deserialization is the reverse.

The alternative most developers reach for is JSON, a human-readable text format. JSON has two disadvantages for a high-frequency internal bus. First, it carries no schema: any JSON object is syntactically valid, so a receiver cannot distinguish a malformed message from a valid one without application-level checks. Second, text encoding is inefficient: a 64-bit integer written as its decimal string representation takes 1–20 bytes; protobuf encodes the same value in 1–10 bytes using variable-length encoding.

Protobuf defines message types in `.proto` schema files, which are compiled to Rust code. A message that does not match the schema is rejected by the generated parser before it reaches application logic, catching schema violations at the boundary rather than deep inside a handler.

MessagePack was considered as an alternative: it is a binary JSON variant with compact encoding but no schema enforcement. *Cap'n Proto* and *FlatBuffers* offer zero-copy deserialization and faster parsing for large messages, but their Rust ecosystems are less mature than protobuf's. Protobuf has the best combination of schema enforcement, Rust library quality, and tooling.

Table 3 Inter-component communication paths

Path	Transport
Daemon ⟷ Daemon	Unix socket (protobuf)
App backend ⟷ Daemon	Unix socket (protobuf)
App backend ⟷ App frontend	Tauri command bridge (typed)
Compositor ⟷ Shell	Wayland protocols + Unix socket

A concrete example: a user action in the UI that queries the Knowledge Graph follows this path:

`UI event → App backend → Unix socket → Graph Daemon → Database`

Four steps, all local to the machine. The full round-trip is imperceptible to the user.

2.8.3 Knowledge Graph Rationale

A relational database such as *PostgreSQL* or *SQLite* organizes data into tables with rows and columns. Relationships between entities are expressed as foreign key references and queried using `JOIN` operations, which become progressively more expensive and harder to write as the number of hops between entities increases.

A *graph database* stores relationships as first-class data: entities are *nodes* and the connections between them are typed *edges*, both with their own properties. Traversing a relationship is a direct pointer follow, not a table scan or index lookup. Querying "which files did the user work on last Tuesday, via which applications, and are any of those applications still running?" is a single graph traversal. In SQL it requires at minimum three joins, a time-range filter, and a correlated subquery for the running-process check.

For an operating system where everything is connected (processes open files, files belong to projects, projects span sessions, sessions involve multiple applications), a graph is the natural data model.

Ladybug is the query-facing graph database in this system. It is an embedded graph database written in C++ with first-class Rust bindings, released under the MIT license. *Embedded* means it runs as a library inside the Graph Daemon process with no separate server process, no network port, and no additional service to manage. The database is a directory of files on disk that only the Graph Daemon process opens.

Ladybug is an OLAP system: it is optimized for analytical queries over large graphs, not for high-frequency transactional writes. This is a deliberate fit for its role. The Graph Daemon uses a two-layer write architecture: *SQLite* acts as the high-frequency write store, accepting the continuous stream of eBPF events and compositor events at full throughput. Structured, semantically valuable data – `UserAction` nodes, `Project` edges, `Session` nodes – is periodically promoted from SQLite into Ladybug, where it becomes available for graph traversal queries. Raw eBPF noise never reaches Ladybug; it is retained in SQLite for the anomaly detection window and then discarded. This means Ladybug operates on a clean, curated graph rather than a raw event log, which improves both query performance and the quality of AI context.

The two layers have different retention windows and query surfaces. SQLite answers time-range lookups over recent raw events; Ladybug answers multi-hop graph traversals over semantic history. The Graph Daemon presents a unified query interface to callers; the layer boundary is an implementation detail they never see.

Neo4j was evaluated and rejected. Neo4j is the most widely deployed graph database and has a large ecosystem, but it runs as a separate server process, its community edition has usage restrictions, and its JVM-based runtime carries memory overhead inappropriate for a component embedded in a user session.

Ladybug uses *Cypher* as its query language, an open graph query standard (openCypher) originally developed for Neo4j. It reads close to natural language:

```
MATCH (f:File)-[:OPENED_BY]->(a:App)
WHERE f.last_modified > datetime('2026-03-01')
RETURN f.path, a.name
ORDER BY f.last_modified DESC
```

This query finds all files modified after a given date, which applications opened them, and returns the results sorted by modification time.

2.9 D-Bus Coexistence

D-Bus is the standard inter-process communication system on Linux desktops, in use since 2003. NetworkManager uses it to report network state changes, BlueZ uses it for Bluetooth events, `systemd-logind` uses it for session management, and the *XDG Desktop Portal* uses it to give sandboxed applications access to OS-level dialogs. It cannot be removed without breaking the wider Linux software ecosystem.

D-Bus works as a message broker: a central `dbus-daemon` process routes messages between senders and registered receivers identified by well-known bus names such as `org.freedesktop.NetworkManager`. This model works well for the ecosystem-level events it was designed for. The central broker and its overhead make it a poor fit for the high-frequency structured events between project-internal components, which is why the project uses its own Event Bus for internal communication.

This system does not replace D-Bus. It runs alongside it. D-Bus remains responsible for third-party application notifications via `org.freedesktop.Notifications`, hardware events, portal interfaces, and session management. The internal Event Bus handles communication between project components only. The two systems are parallel and non-overlapping.

The rule for developers is simple: project-internal components communicate over the internal Event Bus via Unix sockets and protobuf. Standard Linux services and third-party applications communicate over D-Bus as they always have. No component spans both buses.

3 OS Foundation

Every operating system needs a foundation: a kernel that talks directly to hardware, an init system that starts services at boot, a package manager for installing software, and a base set of userspace tools. Building all of this from scratch would take decades and produce something inferior to what already exists. This project therefore builds on top of an existing Linux distribution that provides the entire foundation layer, and adds everything above it: the desktop environment, event system, knowledge graph, and AI layer.

3.1 What is a Linux Distribution

The *Linux kernel* is the core of the operating system: the program that manages hardware access, memory, processes, and the filesystem. It is developed by Linus Torvalds and thousands of contributors worldwide and is released under the GPL v2 license. The kernel alone, however, is not a usable system. A *Linux distribution* bundles the kernel together with an init system, a package manager, a C standard library, and a collection of userspace tools into a coherent, installable operating system.

Different distributions make different choices about package freshness, update cadence, governance, and default software selection. These choices have real consequences for a project building on top: a distribution with stale packages cannot support modern hardware or toolchains, and one with poor governance can change direction in ways that break downstream projects without warning.

3.2 Base Distribution

This project builds on *Fedora Linux* [38] as its base distribution. Fedora is a community distribution backed by Red Hat infrastructure but governed independently through the Fedora Council, which makes binding decisions on project direction without requiring Red Hat sign-off. Packages track upstream closely and a new major release ships every six months. Btrfs has been the default filesystem since Fedora 33 (2020), and the Flatpak and Wayland ecosystems receive stronger first-class support here than on most other distributions.

The selection criteria were deliberately narrow: packages current enough for modern toolchains and recent hardware without arriving unvalidated daily; governance not concentrated in a single commercial entity; Btrfs as a first-class filesystem; and broad enough hardware coverage to work on consumer machines without manual driver installation. Fedora satisfies all four.

RPM (RPM Package Manager) is the native package format. It is one of the two dominant Linux package formats alongside Debian's `.deb`. A package is a compressed archive containing the files to install, metadata about name and version, and dependency declarations that the package manager resolves automatically. The *Open Build Service* [86] (OBS), described in Section 3.4, handles building and signing RPMs for this project and is fully compatible with Fedora as a build target.

The six-month release cadence means periodic major upgrades are required. This is managed through the update strategy in Section 3.3: snapshots are taken before every upgrade, and the health check and rollback mechanism make a failed major upgrade recoverable without user intervention. Each Fedora release is supported for approximately thirteen months, giving a comfortable window for validation before an upgrade is required.

3.3 Update Strategy

Updates on most operating systems involve some degree of risk: something breaks, the user does not know why, and getting back to a working state requires either expertise or luck. This system addresses that with a layered approach that makes the worst-case outcome a single keypress rather than an afternoon in recovery mode.

The approach is a mutable filesystem with automatic rollback protection. *Mutable* means the filesystem works the way Linux filesystems always have: software can be installed freely, system files can be modified, and standard developer tools work without restrictions. This distinguishes it from *immutable* distributions such as Fedora Silverblue and NixOS, where the root filesystem is read-only and changes require going through a specific toolchain. Immutable systems have genuine safety benefits, but they break workflows that the target audience relies on: installing language runtimes with `pip` or `cargo`, building projects from source, or running a local development server. The safety benefits of immutability are achievable through snapshot tooling without those costs.

Figure 3 shows the full update flow.

Btrfs and copy-on-write. Most widely used Linux filesystems, including *ext4*, use *overwrite-in-place* semantics: when a file is modified, the old data is replaced at the same disk location. Once overwritten, the previous version is gone.

Btrfs [8] (B-Tree Filesystem) uses *copy-on-write* (CoW) semantics instead. When a file is modified, Btrfs writes the new data to a different location on disk and updates the metadata pointer. The old version remains intact until space is explicitly reclaimed. This has two important consequences: filesystem snapshots become nearly free in terms of disk space (a snapshot is a set of metadata pointers to existing blocks, not a copy of the data), and the filesystem remains self-consistent after a crash because incomplete writes never overwrite existing valid data.

Btrfs was merged into the Linux kernel in 2009 and has been the default filesystem on Fedora since 2020. Its CoW architecture is the technical foundation for this project's rollback strategy.

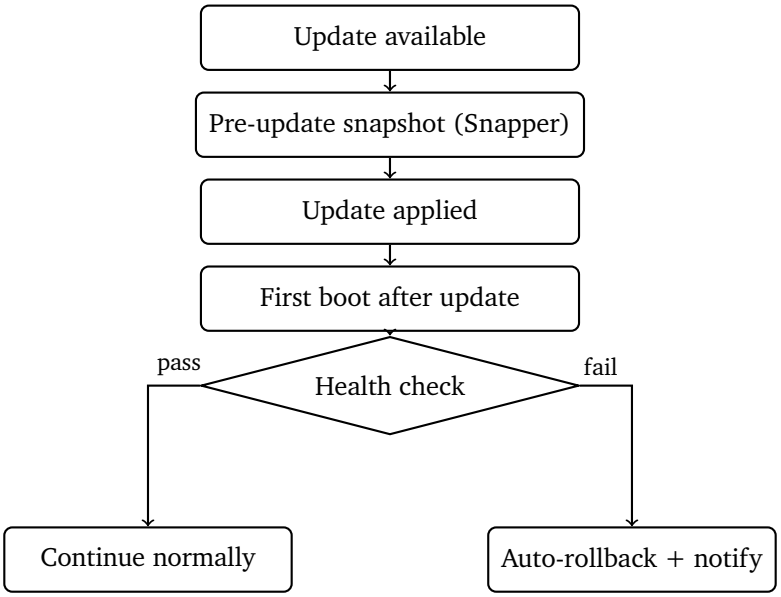

Figure 3 Update flow. A Snapper snapshot is created before every update. If the post-update health check fails, the system rolls back to the snapshot automatically and notifies the user.

Btrfs subvolumes. A *subvolume* is a named, independently snapshotable region of a Btrfs filesystem. It behaves like a directory but can be mounted separately, snapshotted independently, and assigned its own quota limits. This project uses the following layout, consistent with the convention established by Fedora:

```
@ -> mounted at / (system root)
@home -> mounted at /home (user data)
@snapshots -> mounted at /.snapshots
@log -> mounted at /var/log
@cache -> mounted at /var/cache
```

Keeping / and /home on separate subvolumes means a system rollback (restoring @ to a previous snapshot) never touches user data in @home. Log and cache data are excluded from system snapshots, preventing large log files from inflating snapshot sizes.

Snapper. *Snapper* [16] is the snapshot management tool that sits on top of Btrfs. It handles creating, listing, comparing, and restoring snapshots automatically. Snapper is configured per subvolume: the @ subvolume gets automatic pre/post snapshots around package manager operations and a timeline policy for background snapshots. The @home subvolume has its own independent daily snapshot schedule.

Before any update is applied, Snapper creates a tagged pre-update snapshot of the @ subvolume. This is a precondition for the update, not an optional step. A matching post-update snapshot is created after the package manager finishes, forming a before/after pair that Snapper can diff to show exactly what changed.

Bootloader integration. The bootloader is configured to list available snapshots as boot entries. If an update produces a system that fails to boot, the firmware presents the previous snapshot as a selectable entry on the next start. No terminal or recovery mode is required. On Fedora this is handled by `grub2-snapper-plugin`.

Post-update health check. On the first boot after an update, a systemd oneshot service runs a health check before the desktop session starts. It verifies that the display system initializes, that the network stack is reachable, that the Graph Daemon and Event Bus start successfully, and that no systemd services have entered a failed state. If any check fails, the service triggers an automatic rollback by setting the pre-update snapshot as the default boot entry and rebooting. The user receives a plain-language notification on the rolled-back system.

Snapshot retention. Snapper's retention policy for this project removes snapshots older than seven days while always retaining the five most recent. Both values are user-adjustable in Settings. The `@home` subvolume retains the three most recent daily snapshots; home snapshots are a safety net for accidental file deletion, not a system recovery mechanism.

3.4 System Infrastructure

systemd. *systemd* [17] is the init system and service manager on Fedora and on most modern Linux distributions. When the kernel finishes booting, the first userspace process it starts is `systemd` (PID 1), which reads *unit files* describing each service, its dependencies, and its startup environment, and starts them in dependency order. systemd also handles logging via the *journal*, socket activation, timer-based scheduling, and hardware event management.

All custom daemons in this project are deployed as systemd service units. systemd's unit file format provides sandboxing options that are enforced at the kernel level without any code changes in the service: `PrivateNetwork=true` to deny all network access, `PrivateTmp` for an isolated temporary directory, `MemoryDenyWriteExecute` to prevent writable executable memory, and `SystemCallFilter` to allowlist only the syscalls the service actually needs.

systemd-homed. *systemd-homed* [18] is a systemd component that manages user home directories as self-contained, portable, encrypted images. Unlike traditional Unix home directories, which are plain directories under `/home` identified by a numeric user ID, a systemd-homed home is a *LUKS-encrypted loopback image*: a file on disk that is mounted at login (after the encryption key is derived from the user's credentials) and unmounted at logout. When a user is not logged in, their home directory is an opaque encrypted file; even root cannot read its contents without the user's passphrase.

This is the technical foundation for the profile isolation system described in Section 8: each profile is a Linux system user with its own systemd-homed image. When a profile is not active, its image is unmounted and its key is discarded from memory.

LUKS (Linux Unified Key Setup [10]) is the standard disk encryption layer on Linux, built on top of the dm-crypt kernel module. It provides a standardized header format for encrypted block devices, support for multiple key slots so that several passphrases or key files can unlock the same volume, and memory-hard key derivation functions that make offline brute-force attacks expensive. LUKS is what systemd-homed uses internally to encrypt each home image.

Session manager and greeter. User sessions are managed by *greetd* [71], a minimal login daemon that handles authentication and session startup. greetd is intentionally small: it authenticates the user via PAM, launches the configured session command on success, and exits. All policy decisions, visual presentation, and user-facing interaction are delegated to a separate *greeter* process that communicates with greetd over a Unix socket using a simple JSON protocol.

This clean separation between authentication daemon and login UI is why greetd was chosen over display managers such as GDM or SDDM, which bundle both concerns into a single process with their own GUI frameworks. The login screen in this project is a first-party Tauri application that implements the greetd IPC protocol: it presents the user interface, collects credentials, and passes them to greetd for PAM verification. The Tauri greeter has access to nothing other than the greetd socket; it runs before any user session is active, with no access to the graph, the event bus, or any user data.

Package management. At the base level, dnf [97] handles dependency resolution, cryptographic package signature verification, and repository management.

First-party project packages are built and distributed through the *Open Build Service* (OBS [86]). OBS is a hosted build infrastructure originally developed by SUSE and used by many open-source projects across distribution families. Developers push package source files to OBS; OBS builds the resulting RPMs in a clean Fedora chroot for every configured release target and architecture, signs them with the project GPG key, and hosts the resulting repository. This keeps the build pipeline reproducible and independent of the developer's local environment.

Flatpak. Store applications are distributed as *Flatpak* [11] packages. Flatpak is a distribution-agnostic packaging format for desktop applications. A Flatpak bundles an application together with its specific library dependencies into a sandboxed unit that runs isolated from the rest of the system using Linux namespaces and *bubblewrap*, with controlled access to host resources via the *XDG Desktop Portal*.

The alternative formats are *Snap*, which routes all distribution through Canonical's servers and has faced criticism for this centralization, and *AppImage*, which is a self-contained executable with no sandboxing and no integrated update mechanism. Flatpak was chosen for its sandboxing model, distribution neutrality, and strong support in both Fedora and the broader Linux desktop ecosystem.

System-level components ship as native RPMs because they require tight integration with the system: daemons must run outside any sandbox, and systemd unit files must be installed to system paths. The Store handles both formats transparently.

3.5 Hardware Requirements

Three hard requirements determine the minimum supported hardware. They are not adjustable without significant changes to the security and graphics stack and are treated as hard constraints. In practice they exclude hardware older than roughly 2018–2019.

Table 4 Minimum hardware requirements

Requirement	Reason
Linux kernel 6.6 or newer	Required for the eBPF program types used by the event monitoring system (Section 4.3): `BPF_PROG_TYPE_TRACEPOINT` for syscall attachment, `BPF_PROG_TYPE_PERF_EVENT` for process lifecycle, and CO-RE (Compile Once, Run Everywhere) via BTF type information for portable eBPF binaries. Kernel 6.6 is the LTS release that ships with current Fedora.
Vulkan-capable GPU	Required for hardware-accelerated compositing and for Windows compatibility via DXVK and vkd3d-proton. Covered automatically on AMD and Intel via Mesa; Nvidia requires the proprietary driver, with installer guidance provided.
systemd 255 or newer	Required for full systemd-homed portable home image functionality used by the profile system.

eBPF program types and CO-RE. The event monitoring system uses three eBPF program types. Tracepoint programs attach to kernel syscall entry and exit points (`openat`, `execve`, `connect`, and others) and fire on every matching call across all processes. Perf event programs handle process lifecycle events including fork and exit. Ring buffer maps transfer collected events to userspace with low overhead and without copying data through the kernel I/O path.

All eBPF programs are compiled to eBPF bytecode at build time using the `aya` [23] framework and ship as pre-compiled binaries in the system image. No compiler is required on the end-user's machine. The programs use *CO-RE* (Compile Once, Run Everywhere) [81], a mechanism introduced in Linux 5.2 that uses BTF (BPF Type Format) kernel debug information to make eBPF programs portable across kernel versions without recompilation. Kernel 6.6 guarantees that the required BTF information is present and that the specific tracepoint hooks this system attaches to are stable. Kernels older than 5.8 lack ring buffer map support entirely [82]; kernels between 5.8 and 6.5 may work but are not tested and not supported.

Vulkan. *Vulkan* [66] is a low-level graphics and compute API by the Khronos Group, designed as the modern replacement for OpenGL. It provides explicit control over the GPU at the cost of a more verbose API. This project requires Vulkan for two reasons:

the compositor uses Vulkan for hardware-accelerated compositing, and the Windows compatibility layer (Section 8.3) translates Direct3D calls to Vulkan via DXVK and vkd3d-proton. On AMD and Intel hardware, Vulkan support is provided by *Mesa* [77], the open-source graphics driver stack included with Fedora. Nvidia hardware requires Nvidia's proprietary driver because the open-source `nouveau` driver does not provide Vulkan support on most hardware generations.

Hardware support scope. The design principle "works on first boot" requires an honest statement of scope. It means: on supported hardware classes, the system boots, the desktop starts, networking works, and audio works without manual configuration.

The supported hardware classes where this principle holds are mainstream x86-64 laptops and desktops with Intel or AMD CPUs released after 2018, with integrated or discrete Intel or AMD graphics. This covers the large majority of consumer hardware available today.

Two known categories require additional handling. Nvidia discrete GPUs require the proprietary driver [96]. RPM Fusion is configured as a repository source in the system image; on first boot, the installer detects Nvidia hardware and installs the proprietary driver automatically before the user reaches the desktop. No manual repository configuration is required. Broadcom Wi-Fi adapters, found in some Apple hardware and a smaller set of laptops, require a proprietary driver that is not included in the Fedora kernel [95]. The same first-boot detection logic applies where a driver package is available; hardware without a packaged driver is documented explicitly in the compatibility list rather than discovered by users after installation.

3.6 Configuration Format

All system components and first-party applications use *TOML* [93] (Tom's Obvious Minimal Language) as the configuration format. TOML was designed by Tom Preston-Werner (co-founder of GitHub) to be unambiguous and easy to edit without special tooling.

The two most common alternatives are *YAML* and *JSON*. YAML is human-readable and widely used, but its indentation-sensitive syntax is a consistent source of subtle errors: a single misaligned space changes the meaning of the document, and certain values parse unexpectedly (bare `NO` is a boolean in YAML 1.1, not a string; YAML 1.2 fixes this, but library support for 1.2 is inconsistent in practice). JSON is unambiguous but has no comment support and is more verbose than necessary for human-edited files.

TOML has clear, explicit syntax with no indentation sensitivity, supports inline comments, and has an unambiguous type system: strings, integers, floats, booleans, datetimes, arrays, and tables. Using one format consistently means a developer who reads one configuration file can read all of them, and validation and migration tooling only needs to be written once.

Personal settings live at `~/.config/<app-id>/config.toml` and system-wide settings at `/etc/<service-name>/config.toml`, following the *XDG Base Directory Specification* [43].

Some third-party applications expect other formats; a small helper translates without moving the source of truth:

```
# Export as environment variables for a legacy app
eval $(os-config-helper export \
    --toml ~/.config/myapp/config.toml --format env)

# Export as JSON for tools that require it
os-config-helper export \
    --toml ~/.config/myapp/config.toml --format json
```

The TOML file is always the authoritative source; the helper is a read-only translation layer with no business logic of its own.

4 Event System and Knowledge Graph

4.1 Overview

For the Knowledge Graph to build a picture of what is happening on the machine, something has to watch and report. The Event System is that something: it observes activity across the entire OS and feeds a continuous stream of structured events into the graph, without any visible effect on the user and without requiring applications to be rewritten for this platform.

Three sources feed into it, each covering a different slice of system activity.

Table 5 Event sources and their coverage

Source	App cooperation	What it sees
Kernel monitoring (eBPF)	None required	File opens, network connections, process lifecycle
Wayland compositor	Built-in (compositor is part of this project)	Window focus, app open/close, clipboard operations
First-party app events	Required; apps emit events directly	Rich semantic actions: document saved, search performed, tab opened

The design goal is baseline coverage for every app on the system without requiring anything from third-party developers, combined with richer semantic data where apps support it. eBPF provides the floor; structured app events raise the ceiling.

4.2 Wayland as an Event Source

The Wayland compositor sits at the intersection of every user interaction with the graphical desktop. Because the compositor is a first-party component of this project,

it can emit structured events to the Event Bus directly without any protocol extension or third-party cooperation.

Wayland provides the following event categories natively, without requiring applications to do anything:

Window lifecycle. The compositor knows when a Wayland surface is created, mapped (becomes visible), unmapped, and destroyed. This translates directly to app open and close events carrying the app ID, window title, and timestamp. The `xdg-shell` protocol provides this for all standard application windows; `wlr-foreign-toplevel-management` extends it to allow the shell to observe all open windows including those from other clients.

Focus transitions. The compositor controls which surface receives keyboard input at any given moment. Every focus change (the user switching windows, clicking into a different app, or the active window closing) is an event the compositor observes before any application does. Focus events carry the incoming and outgoing app IDs and the timestamp, giving the graph an accurate record of how attention moves across applications.

Clipboard operations. The Wayland clipboard protocol (`wl_data_device`) routes all clipboard interactions through the compositor. The compositor observes when content is placed on the clipboard and when a paste request is made, including which application is the source and which is the destination. Clipboard *content* is never logged; only the operation metadata (source app, destination app, MIME type, timestamp) reaches the Event Bus.

Output events. Monitor connect, disconnect, and configuration changes are compositor-level events. These are logged as session context, useful for queries like "which monitor was I on when I worked on this file."

What Wayland does *not* provide, and what therefore requires eBPF or app-level instrumentation: filesystem access, network connections, what the user typed, the content of windows, or any in-application state beyond the window title. The compositor sees surfaces, not their contents. This is by design: it is the same isolation property that makes Wayland more secure than X11, and it defines the boundary between what the compositor can contribute and what other event sources must cover.

4.3 Kernel Monitoring with eBPF

eBPF (extended Berkeley Packet Filter) is a Linux kernel subsystem that allows small, sandboxed programs to run inside the kernel without modifying kernel source code or loading a kernel module [34]. The name is historical: the original Berkeley Packet Filter from 1992 was a mechanism for filtering network packets; the *extended* version, introduced in Linux 3.18 (2014) and substantially expanded in subsequent releases, is a general-purpose in-kernel execution environment with applications far beyond networking.

Before eBPF, the two main approaches to system monitoring were kernel modules and `ptrace`. A kernel module runs with full kernel privileges and no sandboxing; a bug in a module crashes the system. `ptrace` is a syscall that allows one process to intercept the syscalls of another, but it introduces substantial overhead and can only

observe one process at a time. eBPF solves both problems: programs are verified by the kernel's eBPF verifier before loading (the verifier rejects any program that could loop infinitely, access out-of-bounds memory, or crash the kernel), run at near-native speed via JIT compilation, and can attach to hooks covering all processes on the system simultaneously.

An eBPF program attaches to a specific kernel hook, called a *tracepoint* or *kprobe*, and is executed each time the corresponding event fires: a file being opened (`openat` syscall), a new process being created (`clone`/`execve`), a network connection being established, and so on. The program runs in kernel context, collects the relevant data (file path, process ID, return value), and passes it to a user-space daemon via a shared *ring buffer* without the overhead of standard kernel I/O paths. The result is a read-only window into every process on the machine with overhead typically below 2% of CPU time under normal desktop load.

At this layer, the system knows *that* a process opened a file but not *why* or what it means to the user. That is intentional: eBPF provides the universal baseline, and higher-level sources fill in the semantics where they are available.

Why aya. Three libraries exist for writing eBPF programs from user space: `libbpf` [72], `BCC` (BPF Compiler Collection), and `aya` [23].

`libbpf` is the reference C library maintained in the Linux kernel tree. It is stable and widely used, but integrating it into a Rust codebase requires C FFI (Foreign Function Interface) bindings. Every call across the Rust/C boundary requires `unsafe` blocks that the Rust compiler cannot verify, partially defeating the purpose of writing the daemon in Rust.

`BCC` takes a different approach: eBPF programs are written as C strings embedded in Python or Lua and compiled at runtime by LLVM on the target machine. This makes BCC inappropriate for a production daemon: it requires LLVM to be present on the user's system, compilation adds startup latency of several seconds, and the Python runtime overhead is significant for a background service.

`aya` is a Rust-native eBPF library [23]. Both the eBPF programs (which run inside the kernel) and the user-space daemon are written in Rust. Programs are compiled to eBPF bytecode at build time using `cargo`, not at runtime, so no compiler is required on the end-user's machine. `aya` supports *CO-RE* (Compile Once, Run Everywhere) [81], which uses BTF type information embedded in the kernel to make eBPF programs portable across kernel versions without recompilation. The result is a fully Rust codebase with no unsafe FFI boundaries, no runtime compilation, and no additional system dependencies beyond what the kernel already provides.

eBPF Normalizer. Raw eBPF events arrive at high frequency from the kernel ring buffer. Before they reach the Event Bus they pass through the *eBPF Normalizer*, a user-space component that performs three operations in sequence.

First, *deduplication*: identical operations on the same file by the same process within 100 milliseconds are collapsed into a single event with an updated timestamp. A text editor that calls `read()` on a file hundreds of times per second produces one graph-relevant event, not hundreds.

Second, *path filtering*: events on paths with no semantic value for a graph of user activity are discarded entirely before reaching the bus. Discarded prefixes include `/proc`, `/sys`, `/dev`, `/tmp`, shared libraries under `/usr/lib`, and kernel internal paths. An event on `/proc/1234/status` carries no information the Knowledge Graph needs; dropping it at the normalizer saves the bus, the Graph Writer, and SQLite from processing it at all.

Third, *UUID assignment*: each surviving event receives a UUID v7 identifier. UUID v7 embeds a millisecond-precision timestamp in its most significant bits, making events generated close together sort together lexicographically, a property that improves write performance and time-range query efficiency in SQLite.

The normalizer runs in the same user-space process as the eBPF ring buffer consumer. It introduces no IPC hop; filtered events pass directly to the Event Bus client in the same process. Drop rates and deduplication ratios are exposed as metrics and monitored by the Anomaly Detector.

4.4 Event Bus

The Event Bus is a background daemon that receives events from all sources, validates them against the event schema, and forwards them to registered consumers. Its only job is to move and validate data; it does not interpret or store anything. Figure 4 shows the full pipeline.

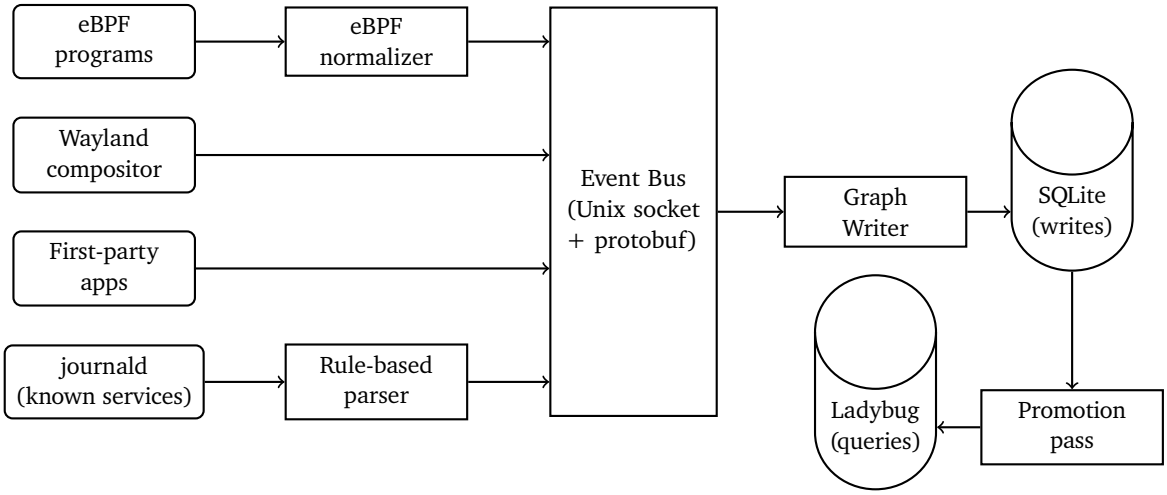

Figure 4 Event pipeline. Raw kernel events pass through the eBPF normalizer before reaching the bus. journald output from known system services passes through a rule-based parser. First-party app events and compositor events arrive already structured. All sources converge on the Event Bus, which validates and forwards to the Graph Writer Daemon. The Graph Writer writes to SQLite at full event throughput; a background promotion pass periodically moves semantically valuable nodes into Ladybug for graph traversal queries.

Event schema. Every event, regardless of source, shares a common envelope. The identifier uses *UUID v7* [31], a variant of the universally unique identifier standard

introduced in RFC 9562 (2024). Unlike UUID v4, which is purely random, UUID v7 embeds a millisecond-precision Unix timestamp in the most significant bits. This makes UUIDs generated close together sort together lexicographically, which matters for a time-series workload: inserting events into the graph in roughly time order avoids the random-access write pattern that UUID v4 would produce and improves query performance for time-range lookups. The envelope also carries a microsecond timestamp, a type label, a source identifier, the originating process ID, and a session reference. The type-specific data travels as a protobuf payload inside the envelope.

```
message Event {
  string id = 1; // UUID v7 (time-sortable)
  string type = 2; // "file.opened", "window.focused", etc.
  int64 timestamp = 3; // microseconds since epoch
  string source = 4; // "ebpf", "wayland", "app:<id>"
  uint32 pid = 5; // originating process
  string session_id = 6; // user session reference
  bytes payload = 7; // type-specific protobuf message
}
```

Listing 1 Common event envelope (protobuf)

A file.opened payload carries the file path, the application name, and the operation flags. A window.focused payload carries the app ID and window title. App events use a generic structure with a category, an action, a subject, and an optional metadata map, which allows any first-party application to emit meaningful events without requiring a dedicated payload type per application.

Clipboard content is never logged. The bus records that a clipboard operation occurred and its MIME type, not the content itself.

D-Bus coexistence. The custom Event Bus handles communication between project-internal components only. D-Bus remains in place for everything it already handles: desktop notifications, NetworkManager, Bluetooth, session management via systemd-logind, sandboxed app portals, and standard third-party applications. The two systems run in parallel with no overlap; nothing that communicates over D-Bus today is changed.

4.5 Write Architecture

The Knowledge Graph has two structurally different workloads: a high-frequency write path that must absorb thousands of eBPF events per second without dropping data, and a query path that must traverse multi-hop graph relationships efficiently for AI context and user queries. These two workloads have conflicting requirements. A database optimized for one performs poorly at the other. The write architecture addresses this by separating the two into dedicated stores.

SQLite as the write store. All incoming events are written to *SQLite* [56], a battle-tested embedded relational database with excellent write throughput for append-heavy

workloads. SQLite is the write store: every event from every source lands here first, at full throughput, with no analytical overhead.

The Graph Writer Daemon maintains a ring buffer of 10,000 event slots. A batch write to SQLite is triggered by whichever threshold is reached first: 500 milliseconds have elapsed, or 1,000 events have accumulated. Both values are configurable. When the buffer is full, the system works through three tiers before discarding anything. First, it checks whether an identical event (same type, source, and subject) already exists in the buffer and updates its timestamp rather than inserting a duplicate. If no duplicate exists, it drops the lowest-value event in the buffer, prioritizing raw eBPF read and write events with no app-level context. Only if neither approach frees a slot is the incoming event discarded, which should not occur under normal operating conditions. Drop rates per tier are exposed as metrics and monitored by the Anomaly Detector.

Raw eBPF events are retained in SQLite for 30 days and then discarded. They serve the anomaly detection window and nothing else; they are never promoted to Ladybug.

Ladybug as the query store. *Ladybug* [69] is the query store: an embedded graph database written in C++ with first-class Rust bindings, released under the MIT license. Embedded means it runs inside the Graph Daemon process with no separate server process, no network port, and no additional service to manage. Ladybug uses *Cypher* [88] as its query language, which is the interface the AI daemon, first-party applications, and the FUSE timeline daemon all use for graph traversal.

Ladybug is a community fork of *Kuzu* [68], an embedded graph database originally developed by Kùzu Inc. and based on research published at VLDB [62]. In October 2025, Kùzu Inc. was acquired by Apple and archived the open-source Kuzu repository without a migration path. Ladybug emerged as the primary community-maintained continuation: it carries the full Kuzu codebase, Cypher API, and Rust bindings forward under the same MIT license, with active releases continuing from where Kuzu left off.

This dependency carries a known risk that is acknowledged rather than minimised. Ladybug's contributor base is small and its long-term governance is not yet as established as projects like SQLite or DuckDB. The mitigation is architectural: the Graph Daemon is the sole component that opens the Ladybug database files directly. No other part of the system has a compile-time or runtime dependency on Ladybug's API. A migration to an alternative embedded graph database (whether DuckDB with its `duckpgq` graph extension, a future stable fork, or a custom SQLite-based traversal layer) would be contained entirely within the Graph Daemon and require no changes to any consumer of the graph query interface. This isolation is a deliberate design choice that applies the same dependency containment principle used throughout the rest of the system.

Ladybug operates on a curated graph, not a raw event log. Raw eBPF noise never reaches it. Only semantically valuable nodes are promoted here: `File`, `App`, `Session`, `Project`, `UserAction`, and their relationships. This keeps the Ladybug graph clean and traversal queries fast.

Promotion pipeline. A background promotion pass runs periodically during idle periods, reading from SQLite and writing promoted nodes and edges to Ladybug. The pass is incremental: it tracks a high-water mark in SQLite and processes only new records since the last run. Promotion criteria are deterministic: Event nodes that have been correlated into UserAction clusters, File nodes that have been accessed more than once, and all Session and Project nodes are promoted. Single-occurrence raw eBPF events that have not been correlated are not promoted.

The two stores have different query surfaces. SQLite answers time-range lookups over recent raw events; Ladybug answers multi-hop graph traversals over semantic history. The Graph Daemon presents a unified query interface to all callers; the layer boundary is an implementation detail they never see.

```
MATCH (f:File)-[:OPENED_BY]->(a:App)
WHERE f.last_modified > datetime('2026-03-01')
RETURN f.path, a.name
ORDER BY f.last_modified DESC
```

Listing 2 Example Cypher query against Ladybug: files accessed last Tuesday

4.6 Knowledge Graph

Why a graph. A relational database stores relationships as foreign keys across tables. A graph database stores them as first-class edges. The difference is felt most when querying across multiple entity types simultaneously: finding which files a user worked on last Tuesday, which applications touched them, and whether any of those applications are still running requires three joins and a subquery in SQL and a single traversal in a graph. For an operating system where everything is connected (processes open files, files belong to projects, projects span sessions, sessions overlap with other applications), a graph models the domain naturally.

The write architecture that feeds the graph is described in Section 4.5: SQLite absorbs the high-frequency event stream, and a promotion pipeline moves semantically valuable data into Ladybug for graph traversal. This section describes the Ladybug graph itself: its schema, ingestion tiers, and the data that lives in it.

Why not RDF and SPARQL. *RDF* (Resource Description Framework) and *SPARQL* are W3C standards for the *Semantic Web*: a vision of interlinked, publicly queryable knowledge graphs on the open internet. They are designed for schema-flexible, distributed, open data: a fundamentally different problem than this system's.

This system's Knowledge Graph is local, private, schema-fixed, and performance-sensitive. RDF's triple-store model (subject predicate object) is optimized for schema flexibility at the cost of query expressiveness and traversal performance. SPARQL is significantly less expressive than Cypher for multi-hop graph traversals and pattern matching, which are the primary query patterns this system needs. The reference implementations of RDF stores (rdf4j, Apache Jena) run on the JVM, which is incompatible with the project's Rust-first system stack and introduces runtime overhead inappropriate for an embedded component.

W3C interoperability is not a design goal. The Knowledge Graph is an internal, local data store; it does not need to federate with external systems or conform to open web standards. If data export is needed, the export format is a separate concern from the storage format and can be addressed at the export layer without constraining the internal representation.

Schema. Figure 5 shows the core node types and their relationships. The schema is intentionally minimal at this stage; it will grow as the system matures. `File`, `App`, `Session`, `Project`, `Event`, and `UserAction` are the primary entities. Every raw occurrence from the event pipeline lands in SQLite first; promotion to Ladybug happens after correlation into higher-level nodes. `UserAction` nodes are derived abstractions built from clusters of related events and represent meaningful activity rather than individual syscalls.

Event ingestion tiers. Not all event sources are equal in structure or volume, and the ingestion strategy reflects that. Three tiers operate independently. All three write to SQLite first via the write architecture described in Section 4.5; promotion to Ladybug follows asynchronously.

Tier 1: structured sources. eBPF events, compositor events, and first-party app events arrive already typed and schema-validated. They are written to SQLite with no intermediate processing step. This tier handles the large majority of writes and is the only tier with hard latency requirements.

Tier 2: rule-based parsing. journald output from known system services (NetworkManager, bluetoothd, systemd-logind, and others) carries useful structured information in a line-oriented format. A rule-based parser with per-service patterns extracts this information synchronously. No machine learning is involved; the patterns are deterministic and fast. Unknown services and unmatched lines are discarded rather than approximated.

Tier 3: semantic enrichment (opt-in). For document content specifically — PDFs opened in the viewer, browser page titles, text file contents — a small local model running at idle can extract topic labels and named entities and attach them as properties on the corresponding `File` or `Event` node. This is explicitly opt-in, runs only during idle periods, makes no real-time promises, and is the only tier that involves a language model. The appropriate model class is a sub-2B parameter instruction model (Qwen2.5-1.5B or similar) that can run on CPU without saturating it; the workload is low-frequency enrichment of user documents, not stream processing. The graph is fully functional without this tier enabled. The graph is fully functional without this tier enabled.

4.7 Project System

A project is not a folder. It is a named context node in the Knowledge Graph that groups related files, sessions, and activity across time and application boundaries. The grouping is expressed as typed edges; a file does not carry a project property, it participates in a `PART_OF` relationship. This means a file can belong to multiple projects

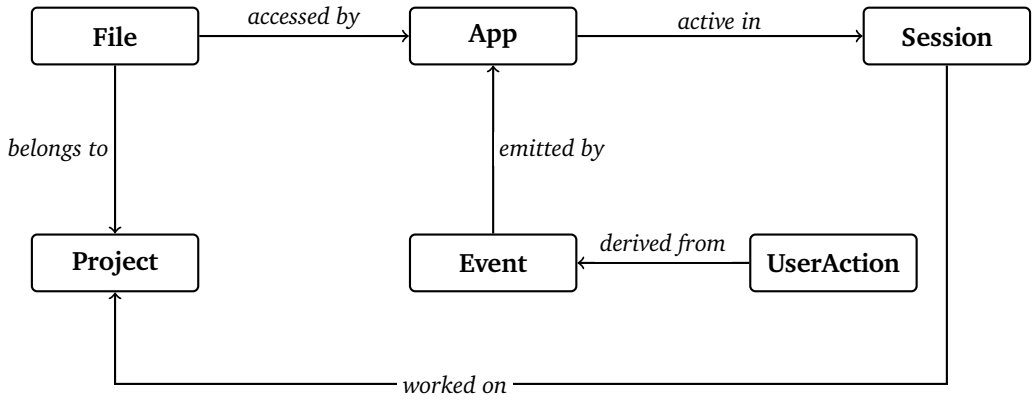

Figure 5 Knowledge Graph core schema. Nodes represent entities; edges represent relationships. Event nodes are the lowest-level entries, created directly from the event pipeline. UserAction nodes are derived abstractions built from clusters of raw events. A File can belong to multiple Project nodes simultaneously; membership is an edge, not a property of the file.

simultaneously, and project membership can be established, changed, or removed without touching the file itself.

Philosophy: emergent, not prescribed. Projects are designed to arise from observed behavior and be confirmed by the user, not to be created upfront and maintained manually. The graph accumulates enough signal to infer that a coherent context exists; the user names and confirms it. Both paths are available: explicit creation is supported for users who prefer structure from the start, but it is not required.

The .project file. Every project can optionally carry a .project file at its root directory. This is a TOML file that acts as the authoritative configuration for that project and is the primary structured data source the Graph Daemon uses when building the Project node. It is designed to be committed alongside source code: a developer who clones a repository gets the project context with it.

```
[project]
name = "CoffeeShop"
description = "Wii␣U␣homebrew␣mod␣manager"
status = "active" # active | paused | archived
created = 2025-11-01
due = 2026-06-01 # optional

[paths]
root = "~/Repositories/coffeeshop"
include = [
    "~/Documents/coffeeshop-notes",
    "~/Downloads/wiiu-sdk-reference.pdf",
]
exclude = ["target/", "*.log"]

[git]
```

```toml
remote = "https://github.com/timkicker/coffeeshop"
default_branch = "main"

[accounts]
github = "timkicker"

[links]
docs = "https://wiiu.hacks.guide"

[tracker]
provider = "github-issues"
url = "https://github.com/timkicker/coffeeshop"
# provider options: github-issues | gitlab-issues | openproject |
# plane | jira

[tags]
values = ["cpp", "wiiu", "homebrew"]

[appearance]
accent_color = "#6aaa64" # optional; overrides shell accent when focused

[focus]
suppress_notifications_from = ["steam", "discord"]

[modules]
active_when_focused = ["com.example.rust-waypointer"]

[ai]
context_scope = "project" # minimal | project | time-scoped

[collaboration]
# Planned for a future release. The data model supports shared projects
# from the start; this field is reserved and currently ignored.
enabled = false
```

Listing 3 Example .project file

The .project file is the source of truth; the Graph Daemon reads it at session start and whenever it changes on disk. Projects without a .project file are fully supported: the node exists in the graph and all features work, but configuration defaults apply and the project cannot be distributed as a template.

Project node properties. The Project node in the graph carries the following properties, sourced from the .project file where present and inferred or defaulted otherwise: name, description, status, root path, creation timestamp, optional due date, tags, and an accent color. The status field drives visibility across the system: active projects appear prominently in Waypointer, the File Manager sidebar, and AI context suggestions; paused projects are searchable but not proactively surfaced; archived projects are read-only in the graph and only reachable via explicit search. Archiving is a transition, not a deletion: graph data is retained under the normal retention policy.

File membership and the PART_OF edge. A file becomes part of a project through one of four mechanisms, in decreasing order of specificity. First, explicit declaration in the [paths] block of the .project file: any file under a declared include path is a member. Second, rootdir heuristic: any file under the project root directory is a candidate member, confirmed once the user has acknowledged the project. Third, co-occurrence clustering: files that consistently appear together in the same sessions, accessed by the same applications and terminal working directories, are grouped as project candidates by a background inference pass. Fourth, manual assignment via the File Manager or Waypointer: the user explicitly adds a file to a project, which writes a PART_OF edge directly.

A file can carry PART_OF edges to multiple projects simultaneously. The graph enforces no uniqueness constraint on this relationship.

Project-scoped AI context. When the AI access level is set to project-scoped (Level 2, described in Section 5), the .project file serves as the context seed for the AI daemon. Name, description, tags, linked accounts, and recent project activity from the graph are included in the context window before the user's query. Waypointer surfaces project-specific quick actions when a project is active: recent files, last commit message, open issues, and time spent this week. These are graph queries, not hardcoded integrations.

Git integration. The presence of a .git directory in the project root is treated as a strong project signal: the directory is automatically proposed as a project root if no .project file exists. The current branch name is attached as a property on the active Session node, providing context for queries like "what was I working on during the refactor branch last week." The [git] block in the .project file links the remote and default branch for use in Waypointer quick actions and AI context.

Full git integration, covering commits as graph nodes, branch switches as session context events, and pull request status as linked resources, is planned for a subsequent release. The graph schema includes Commit and Branch node types from the start so that this integration does not require a schema migration when it arrives.

Issue tracker integration. The [tracker] block in the .project file connects the project to an external issue tracker. The Graph Daemon reads this block and exposes the configuration to the corresponding tracker Integration Package, which registers a Waypointer provider that surfaces issues as typed issue results (Section 6.7).

When a project is focused, typing # in Waypointer opens issue search against the configured tracker. Selecting an issue opens an inline expansion showing title, description, status, assignee, and labels without leaving Waypointer. From there the user can activate issue-scoped Focus Mode, open the issue in the browser, or create a related issue with pre-filled context.

Issue-scoped Focus Mode adds the issue as a sub-context layer on top of the active project. The AI daemon receives the issue title, description, and labels in its context window. When the focus session ends, a UserAction node is written to the graph recording which files were touched and the session duration, giving automatic time tracking at the issue level without any manual input.

Supported tracker providers, each delivered as a separate Integration Package: GitHub Issues, GitLab Issues, OpenProject, Plane, and Jira. The provider field in [tracker] selects which package handles the connection; the Install Daemon installs the matching Integration Package automatically when it detects the .project file declares a provider.

Project-aware system integrations. Projects are a cross-cutting concept and several system components integrate with them directly. These integrations are described in detail in the sections covering each component; the connections are summarized here for reference.

The *Focus Mode* (Section 6) is activated when the user switches to a project context. It coordinates the shell accent color, Waypointer search scope, File Manager root, terminal working directory, and notification priority rules simultaneously.

The *Health Daemon* (Section 9.2) tracks active focus time per project from session data. This enables queries about work distribution across projects and provides the signal for wellbeing observations such as sustained single-project concentration over extended periods.

The *Companion App* (Section 9.1) uses the active project as the Handoff context: switching away from the desktop surfaces the project's linked resources, recent files, and notes on the phone.

The *Module System* (Section **??**) supports the active_when_focused manifest field, which activates a module only when a specific project is in focus. A Rust-specific Waypointer provider that surfaces cargo commands is the canonical example: it runs only when the focused project carries the rust tag.

The *Audit Log* (Section 8.4.7) supports project-scoped export: all activity records associated with a project within a specified date range can be exported as a structured report. This is useful for time tracking, client billing, and project handover.

Signed Profile Packages (Section 8.2) can bundle .project template files. An organization deploying a work profile can pre-configure project structure, required accounts, AI scope restrictions, and notification rules as part of the profile installation.

Automatic project inference. Project nodes in the graph are created explicitly when the user names a project or a .project file is detected, but the graph also accumulates enough signal to infer project membership without either. Files that consistently appear together in the same sessions, terminal working directories that recur alongside a fixed set of applications, and browser tabs that co-occur with specific local files are all indicators that a coherent project exists even without a name. A background inference pass groups these co-occurrence clusters into candidate project nodes and presents them to the user as suggestions rather than applying them automatically. The user can confirm, rename, or dismiss each suggestion. Automatic inference is opt-in and runs locally.

4.8 Retention Policy

The graph accumulates data indefinitely if left unchecked, which creates two problems: storage grows, and the privacy surface area expands. The retention policy defines how long data lives, with user control at every tier.

Table 6 Knowledge Graph retention tiers

Tier	Default	Content
Raw eBPF events (SQLite)	30 days	High-volume, low-semantic-value syscall events. Retained for the anomaly detection window; never promoted to Ladybug and discarded after expiry.
Semantic nodes (Ladybug)	12 months rolling	File, App, Session, and Network-Connection nodes promoted from SQLite. This is where AI query value lives. Nodes past the window are summarized rather than deleted.
Pinned nodes (Ladybug)	Permanent	Nodes explicitly marked by the user, active Project nodes, and nodes recently referenced in AI queries. Only manual deletion removes these.

Summarization. Hard deletion of aged nodes loses information abruptly. Instead, nodes past the retention window are compacted into summary nodes that retain aggregate statistics (access count, primary application, active period) while discarding the individual event edges. The node still exists in the graph and AI queries still find it; the granularity is reduced but the semantic content is preserved. Summarization is triggered by access frequency rather than age: a node that has not been queried in six months is a candidate regardless of when it was created, while a node from two years ago that is queried regularly remains granular. A background daemon identifies candidates nightly and processes them during idle periods.

User controls. Settings exposes the retention window per tier with documented minimums, a manual summarize-now option for specific time ranges, node and subgraph deletion, full graph export for data portability, and storage usage broken down by tier and time range. In managed deployments, administrators can set a minimum retention for compliance-relevant audit events in a separate audit log, but they cannot shorten the user's configured retention window for personal graph data.

4.9 Temporal Filesystem View

The Knowledge Graph contains a complete record of which files were accessed, created, and modified, when, and by which application. This makes it possible to answer queries like "show me everything I worked on last Tuesday" without relying on file modification timestamps, which applications regularly reset, or on search indices, which do not capture access history.

The temporal filesystem view exposes this as a navigable virtual directory tree via a *FUSE* (Filesystem in Userspace) mount. FUSE is a Linux kernel mechanism that allows a userspace process to implement a filesystem: the kernel forwards directory listing and file open calls to the FUSE daemon, which answers them from its own data source rather than from disk. The temporal view mounts at ~/.timeline/ and is populated on demand by a small daemon that translates directory traversal into Graph Daemon queries.

A path like ~/.timeline/2026-03-04/ lists all files the user touched on that date, grouped by project where the graph has enough context to determine project membership. The project grouping is deterministic when a .project file is present and heuristic otherwise. Opening a file through this path opens the actual file at its real location; the timeline view is an index, not a copy. No additional storage is used.

```
# List files worked on a specific date
ls ~/.timeline/2026-03-04/

# Browse by project across all time
ls ~/.timeline/projects/CoffeeShop/

# Files touched in the last week, grouped by project
ls ~/.timeline/last-7-days/

# Open a file as it appears in the timeline (opens real file)
xdg-open ~/.timeline/2026-03-04/CoffeeShop/report-draft.md
```

Listing 4 Temporal filesystem view examples

The FUSE daemon holds no persistent state of its own; every listing is a live graph query. The mount is available only while the user session is active and the Graph Daemon is running. It is read-only: files cannot be created, renamed, or deleted through the timeline path.

A natural extension of the temporal view is causal chain visualization: because the graph records not only which files were accessed but the sequence and application context of those accesses, it is possible to reconstruct the workflow that produced a given output. A query asking "what did I do to produce this file?" can trace the chain of files opened, commands run, and applications active in the session where the file was last modified. This is planned for a subsequent release, built on the same FUSE and graph infrastructure described here.

5 AI Layer

5.1 Overview

The AI layer sits on top of the Knowledge Graph and provides both the user and the system with access to an AI model that has structured context about what is happening on the machine. It is not a chatbot grafted onto an existing OS; it is an interface that can query the graph, invoke application capabilities through a standardized protocol, and optionally act autonomously when the user explicitly enables that.

The AI layer is not the primary reason this system exists. The desktop, the permission system, the profile isolation, and the store all stand on their own. AI is an enhancement that becomes more useful as the Knowledge Graph fills in over time, and users who want none of it can disable it completely without losing any other functionality.

Three principles govern the design of this layer.

Query-first. The AI responds when asked. Nothing happens in the background without an explicit trigger. Autonomous features exist but are separate opt-ins, not defaults that the user has to find and disable.

Provider-agnostic. The system does not care whether the model runs locally or in the cloud. Every provider implements the same interface; the rest of the codebase never speaks to a specific backend. The user decides what runs where, including routing sensitive requests to a local model only.

Opt-out at any point. The entire layer can be disabled in Settings under AI. When disabled, no local model runs, no graph queries originate from the AI layer, Waypointer falls back to standard search, and no MCP servers are active. The setting is not hidden five menus deep.

5.2 Cold-Start Behavior

The Knowledge Graph is empty on a fresh install. An AI feature that promises contextual awareness but has nothing in the graph yet is worse than no feature at all: it creates an expectation and immediately fails to meet it. The system handles this with straightforward communication rather than by faking usefulness.

During onboarding, AI features are presented with an honest explanation of what to expect: the assistant learns context gradually as the system is used, starts with nothing, and becomes more useful over days and weeks. There are no demo interactions, no placeholder context, and no fabricated examples of what it will eventually be able to do.

At onboarding the user can optionally seed the graph with existing data from their current system: browser history, calendar events, and documents. This is presented as a trade-off, not a recommendation. Each source can be imported independently, so someone who is comfortable sharing their calendar but not their browsing history can

import only the former. Importing gives the AI immediately useful context without a multi-week waiting period.

The AI assistant does not surface proactively until a threshold is met: seven days of usage or a completed manual import. Before that threshold the feature is accessible but does not advertise itself. The non-AI parts of the system are functional and complete from the first boot, so there is no gap in usefulness during this period.

5.3 Provider Abstraction

Different AI providers have different APIs, different latency profiles, and significantly different privacy implications. Sending the contents of a private file to a cloud API is appropriate in some contexts and completely unacceptable in others. The system needs to handle both without the rest of the codebase caring which provider is active.

The solution is a Rust trait that every AI backend implements. The AI layer constructs a `CompletionRequest`, passes it to whichever provider is configured, and receives a `CompletionResponse`. No provider-specific code is visible outside the provider module. The `#[async_trait]` macro [113] is a Rust community crate that enables async methods in traits, which the stable Rust compiler does not yet support natively; it desugars the async methods into an equivalent form that compiles correctly.

```rust
#[async_trait]
pub trait AiProvider: Send + Sync {
    async fn complete(
        &self,
        request: CompletionRequest,
    ) -> Result<CompletionResponse>;
    async fn available(&self) -> bool;
    fn name(&self) -> &str;
}
```

Listing 5 AI provider trait

Figure 6 shows how the AI layer relates to the available backends.

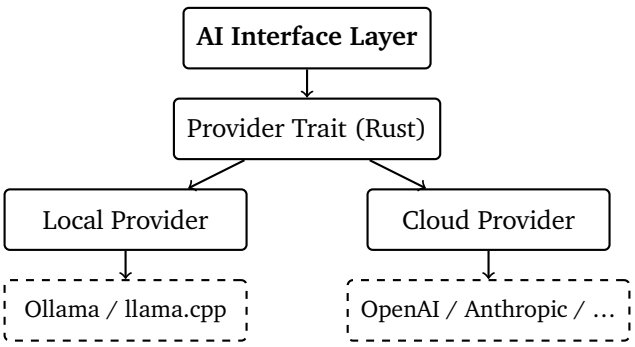

Figure 6 Provider abstraction. The AI layer speaks only to the trait; the concrete backend is a configuration choice. Local and cloud providers are interchangeable from the perspective of anything above the trait boundary.

Local providers. A local provider runs an AI model entirely on the user's machine. No data leaves the device. The two primary options are *Ollama* [85] and *llama.cpp* [46].

llama.cpp is a C++ inference engine for running large language models on consumer hardware, originally developed by Georgi Gerganov in 2023. It achieves viability on consumer hardware through *quantization*: a technique that reduces the numerical precision of a model's weight values (for example from 32-bit floats to 4-bit integers), shrinking both the memory footprint and computation cost at a modest quality trade-off [47]. A 7-billion parameter model that would require roughly 28 GB of RAM at full 32-bit precision runs in approximately 4 GB after 4-bit quantization, which falls within the RAM budget of a typical consumer laptop. llama.cpp exposes an HTTP API compatible with the OpenAI request format, so any client written for the OpenAI API works against a local llama.cpp instance by changing the base URL.

Ollama wraps llama.cpp and other backends in a higher-level tool that adds model management (download, version tracking, storage), a simpler CLI, and a REST API. For most users, Ollama is the practical entry point. llama.cpp is available as a direct option for cases where fine-grained control over quantization levels and inference parameters is needed.

Both are supported as local provider backends. The specific model is a user configuration choice. As of 2025, models in the 7B–13B parameter range with 4-bit quantization offer the best balance of quality, memory footprint, and inference speed on consumer hardware without a dedicated GPU.

Cloud providers. Cloud providers send the request to a remote API over HTTPS and return the response. The system ships with built-in support for the OpenAI [87] and Anthropic [1] APIs, both of which follow similar request/response formats. Additional providers can be added by implementing the provider trait; the interface is intentionally minimal. API keys are stored in the system keyring (not in a config file) and are never written to the Knowledge Graph.

Routing policy. Multiple providers can be active simultaneously under a routing policy the user configures in TOML. The baseline: use the local provider if one is configured and available; fall back to cloud if the local model is unavailable or if the request exceeds the local model's context window. Additional rules can be layered on top: route all code-related requests to a specific provider; prevent any request that includes personal file contents or graph data from leaving the machine; route requests originating from a specific application to a designated provider. Rules are evaluated in order; the first match wins.

5.4 MCP Integration

MCP (Model Context Protocol) is an open protocol published by Anthropic in November 2024 that standardizes how AI models interact with external tools and data sources [2]. The core idea is a client/server split: the AI model talks to an *MCP client*, which in turn dispatches to one or more *MCP servers*. Each server exposes a list of *tools* (functions the AI can call) and *resources* (data the AI can read). The protocol

runs over JSON-RPC 2.0 [64], a lightweight remote procedure call format, typically over a Unix socket or stdio pipe for local servers and over HTTP/SSE for remote ones.

The key property of MCP is that the AI model does not need to know anything about the application it is talking to. A model that has never seen this OS can still call list_directory on the file manager MCP server, because the tool definition (name, parameters, description) is discovered at runtime. This makes MCP servers independently deployable and the AI layer application-agnostic.

In this system, first-party applications expose MCP servers as part of their standard implementation via the *os-sdk* boilerplate described in Section 2: the file manager exposes list_directory, open_file, and move_file; the terminal exposes run_command and get_output; the Knowledge Graph exposes a query interface. The AI can coordinate across multiple applications in a single interaction by calling their respective servers in sequence. Figure 7 shows the topology.

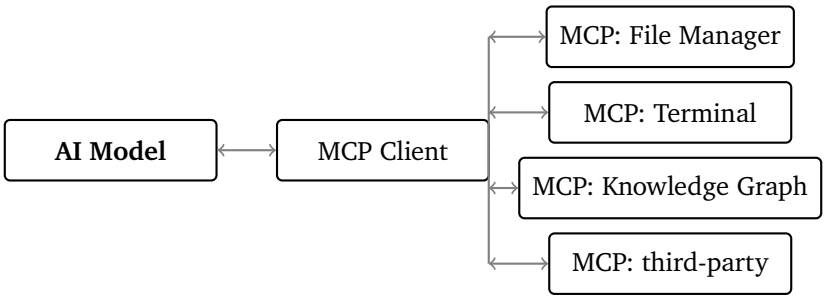

Figure 7 MCP topology. The AI model communicates through a single MCP client that dispatches to individual application servers. Third-party applications from the Store can ship their own MCP servers, subject to the permission model described in Section 5.7.

CLI tools do not need MCP servers. If a tool has a –help flag and produces sensible output, the AI can invoke it as a subprocess and read stdout. MCP is reserved for GUI applications and stateful interfaces where a subprocess call is insufficient.

5.5 Interaction Model

Query-based (default). The user asks a question, the AI responds. No background activity occurs without an explicit trigger. The AI has access to the Knowledge Graph and whatever MCP servers are permitted under the current configuration. Typical interactions include querying session history, finding files related to a project, correlating system events with observed behaviour, and composing multi-step actions across applications.

When a question is asked, the AI layer translates the natural language input into a Cypher query, executes it against the Knowledge Graph via the Graph Daemon, and passes the structured results to the model as context. Relative time expressions are resolved to absolute UTC timestamps before the generation step, so the model always operates on concrete values rather than ambiguous phrases. The Cypher generation and the result formatting are two separate model calls, since combining them causes

the model to bias the query toward a desired-sounding answer rather than accurately querying the schema.

The query validator sits between generation and execution: it checks that the generated Cypher uses only known node types and relations, contains no write operations, and includes a LIMIT clause. If validation fails, the error is fed back for a retry; after two failed attempts the user is told the question could not be answered, with no silent fallback.

Autonomous agent (explicit opt-in). A background agent that observes the event stream and acts proactively is available as an opt-in capability. It is not enabled by default and each autonomous behaviour is a separate toggle: suggesting related files when a document is opened is independent of automatically tagging new files by project. The agent uses the same provider abstraction and MCP layer as the query interface; the difference is that it initiates actions rather than responding to them.

5.6 Project Context

When a project is active in Focus Mode (Section 6.8), the AI layer adjusts its behavior to match. This happens automatically when Focus Mode activates and is reversed when it deactivates; no separate AI configuration is required.

Context seed. The .project file (Section 4.7) is loaded as a structured context seed at the start of every AI query while a project is focused. The model receives the project name, description, tags, linked accounts, relevant links, and a summary of recent project activity from the graph before processing the user's question. This is the minimum context needed for the AI to give useful answers without the user having to re-explain their situation every time.

Graph scope. If the project's [ai] block declares context_scope = "project", the Graph Daemon restricts the AI daemon's traversal to nodes carrying a PART_OF edge to the active project. The AI sees a narrower but more relevant slice of the graph: recent project files, sessions where the project was active, applications used in those sessions, and notifications received during focused work. Nodes outside the project are not visible to the model for the duration of the focused session.

This is Access Level 2 from the table in Section 8.4.6, applied automatically by Focus Mode rather than requiring the user to configure it manually per session.

Waypointer integration. Inside Waypointer's AI mode (Tab), queries are answered with project context active. Asking "which files did I touch yesterday?" returns results filtered to the project rather than the full graph. Project-specific quick actions — recent files, last commit message, open issues link, focus time this week — are derived from live graph queries against the active Project node and its edges, not from cached or hardcoded data.

5.7 MCP Permission Model

MCP servers are the interface through which the AI acts on applications. The permission model distinguishes between reading information, which the AI can do by default within the user's configured AI permission level, and modifying state, which requires explicit per-session authorization.

Table 7 Read-only MCP servers (permitted by default)

Server	Exposes
File Manager	Directory listings, file metadata
Calendar	Upcoming and past events
Browser	History and open tabs (no passwords, no form data)
System Monitor	Process list and resource usage
Notes	Note titles and content

Reading to answer a question is what the user expects when they ask the AI something. Answering "what meetings do I have today?" without a confirmation dialog is the correct behaviour. Reading does not change state, and the AI permission level configured in Settings still gates what data each server exposes regardless of this default.

Table 8 Write and action MCP servers (per-session authorization required)

Server	Can do
Terminal	Execute shell commands
File Manager	Create, move, rename, delete files
Calendar	Create and modify events
Email	Compose and send messages
Store	Install and remove packages

Authorization is granted once per session and covers the scope the user specifies: approving file management does not imply approval for email. It expires at session end and is not persisted. Within an authorized scope, a separate hardcoded list of actions always requires explicit confirmation regardless of session permissions, because their consequences are irreversible or high-impact: permanent file deletion, sending any external message, installing or removing system packages, modifying system configuration outside ~/.config, any network request to a host outside the application's declared permissions, and running commands with elevated privileges. This list is not configurable.

Third-party MCP servers shipped by Store applications are always treated as action-requiring by default. A community server that claims to be read-only receives the same treatment until it carries a verified security audit covering its MCP implementation.

5.8 System Explanation Mode

A running operating system is largely opaque to its user. Background processes run, network connections open and close, disk activity spikes, and the user has no practical way to understand what is happening without opening a terminal and running several diagnostic commands. This is a gap this system is positioned to close, because it already has the data.

The AI daemon can answer the question *"What is my computer doing right now?"* by querying the live event stream and the Knowledge Graph simultaneously. The event stream provides the current moment: which processes are active, what they are reading or writing, which network connections are open, what the CPU and memory usage is and which process owns it. The Knowledge Graph provides context: whether this activity is normal for this time of day on this machine, which project the files being accessed belong to, whether this process has behaved unusually in the past.

The result is a dynamically generated plain-language summary, not a set of predefined status strings. A typical response might note that the package manager is fetching updates in the background, that a first-party application is indexing a recently imported folder, and that network activity is within the normal range for this time of day. If something is unusual, the summary says so: a process accessing node types it has never previously accessed, or a network connection to a destination outside the application's declared permissions, is flagged in the same natural language response rather than silently logged.

The explanation mode is available on demand via the AI query interface and as a persistent ambient view in System Monitor. It does not run continuously in the background; it is a query like any other, executed when the user asks. Everything it uses is local: the event stream and the Knowledge Graph never leave the machine, and the model generating the summary runs locally unless the user has configured a cloud provider. The output can be shared by the user if they choose, but nothing is reported automatically.

6 Desktop Environment and Design Language

The desktop environment is the most visible part of the system and the primary differentiator for users switching from other platforms. The goal is a fast, coherent desktop that does not feel like a collection of independent projects assembled by different teams in different decades. The stack reflects this: performance-critical and system-close code is Rust, everything visual is TypeScript with Tailwind CSS, and the two halves communicate over well-defined interfaces that neither side reaches past.

6.1 Architecture and IPC

Three communication channels connect the compositor core to the shell and its frontend, each carrying a different class of data.

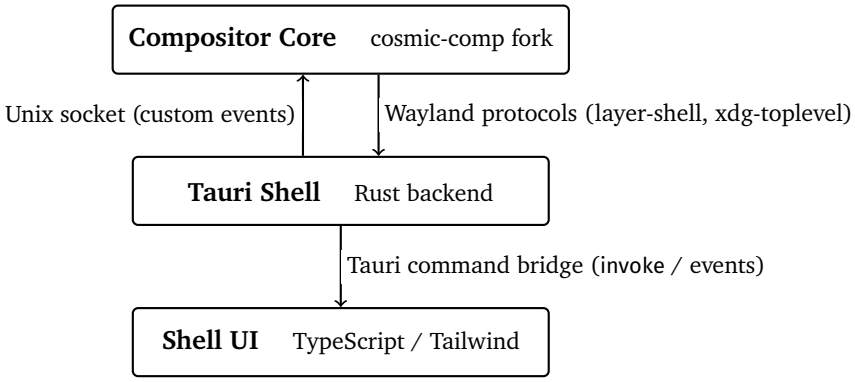

Figure 8 Shell IPC architecture. Wayland protocols carry window management data (open windows, focus, workspace state). The Unix socket carries system-specific events with no Wayland equivalent: theme updates, module lifecycle events, graph notifications. The Tauri command bridge connects the Rust backend to the TypeScript frontend within the shell process.

Wayland protocol extensions are the correct channel for anything the Wayland ecosystem has standardized. The Unix socket exists for everything that does not have a protocol equivalent and where a custom solution is both simpler and more direct. Messages on the socket use JSON during development, with migration to protobuf planned once the schema stabilizes.

Within the Tauri shell, the Rust backend and TypeScript frontend communicate through Tauri's typed command system: the frontend calls Rust via `invoke()`, Rust pushes events to the frontend via Tauri's event emitter. No ad-hoc serialization; the bridge handles it.

6.2 Compositor and Window Manager

The Wayland compositor is a fork of *cosmic-comp* [103], as described in Section 2.6. This section covers what that means in practice for window management and the shell boundary.

The fork retains the full window management engine from cosmic-comp: floating and tiling layouts switchable per workspace independently, focus rules, workspace management, and all the low-level Wayland plumbing that cosmic-comp inherits from *Smithay* [100].

DRM/KMS (Direct Rendering Manager / Kernel Mode Setting) is the Linux kernel subsystem responsible for talking directly to display hardware. DRM manages GPU memory and command submission; KMS handles setting display modes, resolutions, refresh rates, and which GPU output drives which physical connector. Smithay, which underlies cosmic-comp, abstracts this into a safe Rust API [9].

libinput [73] is the standard Linux input processing library. It reads raw events from kernel input devices (keyboards, mice, touchpads, touchscreens), applies device-specific quirks and acceleration curves, and emits normalized events. Using libinput means the compositor gets correct two-finger scrolling, palm rejection, and tap-to-click on every supported touchpad without implementing any of that per-device logic itself.

Buffer management in Wayland means handling *wl_buffer* objects: the shared memory regions or GPU buffers that applications render their window contents into. The compositor receives these buffers, composites them together (applying transforms, clipping, and blending), and presents the result to the display hardware via DRM/KMS. Smithay handles the buffer lifecycle including GPU-side zero-copy paths via dmabuf.

What the fork does not retain is the COSMIC shell integration. The custom IPC protocols that cosmic-comp uses to communicate with cosmic-panel and cosmic-launcher are removed. The shell boundary is defined entirely by standard Wayland protocol extensions, described below, plus the minimal custom Unix socket channel described in Section 2.6.

Wayland protocol extensions. The core Wayland protocol covers only the basics: surfaces (drawable regions), input routing, and output (monitor) management. Everything a desktop actually needs – panels anchored to screen edges, a taskbar that knows about open windows, workspace switching – requires *protocol extensions*: additional interface definitions that the compositor implements and clients can use if present. Table 9 lists the extensions this compositor implements and what each enables.

The wlr- prefixed protocols are part of the *wlroots* protocol collection [127], originally developed alongside the wlroots compositor library and now widely implemented across compositors including Sway, Hyprland, and KWin.

XWayland. X11 compatibility is provided via *XWayland* [22], an X11 server that runs as a Wayland client. From the compositor's perspective it is just another Wayland client; from legacy X11 applications' perspective it is a regular X11 display. It starts on demand when the first X11 application launches and shuts down when none are running. Users who do not need X11 compatibility can disable it in Settings.

XWayland has structural limitations that cannot be solved at the compositor level; they are documented rather than hidden. Copy-paste between X11 and Wayland applications requires a clipboard synchronization bridge. Drag-and-drop across protocol boundaries is unreliable. X11 applications cannot use xdg-desktop-portal for screen-sharing. Most significantly, X11 has no inter-application isolation: one X11 application can read the input events of any other X11 application within the XWayland session, which is a keylogger-class capability contained within the XWayland boundary. This is recorded in the audit log whenever X11 applications are active.

6.3 Shell and Application Framework

The shell – global menu bar, taskbar, app launcher, notification center, quick settings panel, and lock screen – is built with Tauri (introduced in Section 2.5) using TypeScript and *Tailwind CSS* for the interface layer. Tauri renders the UI in a *WebKitGTK* WebView, which provides full modern web rendering capabilities without the memory cost of an Electron-style approach. The shell is a Wayland client that uses wlr-layer-shell to anchor its surfaces to screen edges.

Table 9 Implemented Wayland protocol extensions

Protocol	What it enables
wlr-layer-shell [127]	Allows a client (the taskbar) to claim a fixed strip at a screen edge and stay above all application windows. Without this, the taskbar would be just another window the user could accidentally cover.
wlr-foreign-toplevel-management [127]	Lets a client outside the compositor query and control all open windows: title, app ID, minimized/maximized state, and focus. This is what makes the taskbar able to list running applications and switch to them.
wlr-workspace-management [127]	Exposes workspace state (current workspace, window-to-workspace mapping) to clients. Enables the workspace switcher in the taskbar.
xdg-shell	The standard protocol for normal application windows: creation, resize, maximize, fullscreen, and popup menus. Every Wayland application uses this.
xdg-output	Extends monitor information beyond the core protocol: logical position and size for multi-monitor layouts, human-readable names.
xdg-desktop-portal [44]	A D-Bus interface (not a Wayland protocol) that gives sandboxed applications controlled access to OS capabilities: file pickers, screensharing, printing, and camera access, each with a user-consent dialog. Flatpak applications use this exclusively; they have no direct filesystem or device access.
wp-fractional-scale-v1	Allows the compositor to tell an application its exact fractional scale factor (for example 1.5 on a 1440p display), so it can render at the correct pixel density rather than relying on integer scaling with blurring.

WebKitGTK on Fedora. WebKitGTK [125] is the GTK port of the WebKit rendering engine and is the WebView backend Tauri uses on Linux. It is worth addressing its known weaknesses directly, because the argument "WebKitGTK is already present on any modern Linux desktop" can sound more reassuring than the reality warrants.

WebKitGTK has historically lagged behind WebKit upstream in JavaScript engine performance and in adopting newer web platform features [107]. It has also shown rendering inconsistencies between distributions when the installed version differs. On some distributions, the packaged version trails upstream by several months, creating a fragmented target for applications that rely on specific CSS or JavaScript behaviour.

Fedora as the base distribution is the concrete mitigation for these problems. Fedora packages a recent WebKitGTK version promptly after upstream releases and carries security patches quickly. The project targets only Fedora, which means the WebKitGTK version in the OBS repository and in user systems is known and controlled. Applications are tested against this specific version. The fragmentation problem that affects multi-distribution projects – where the same application must run correctly on WebKitGTK versions that differ by a year or more – does not apply here.

The remaining risk is JavaScript performance for complex UI interactions. This is a real trade-off relative to Chromium-based WebViews. For the use cases in this project – settings panels, file browsers, launchers, and notification centers – the performance difference is not perceptible. The trade-off would matter for applications with continuous animation at 60 frames per second or with heavy computational JavaScript; those use cases are not part of this system's first-party application set.

What GTK is. *GTK* (GIMP Toolkit, now branded simply as GTK) is the most widely used UI toolkit on Linux [49]. It provides the standard set of interface components – buttons, text inputs, dialogs, menus, scrollable lists – and is the foundation for applications such as GNOME, Nautilus, and most of the Linux application ecosystem. *GTK4* is the current major version, a significant rewrite from GTK3 with a new rendering pipeline (using GPU shaders via OpenGL/Vulkan rather than Cairo), improved accessibility, and a better layout engine.

Qt is the main alternative to GTK and is used by the KDE ecosystem and many cross-platform applications [110]. The first-party applications in this project do not use GTK or Qt directly; they use Tauri with WebKitGTK. However, GTK4 and Qt applications from the ecosystem still run on this desktop and receive theming coverage, which is why both frameworks appear in the theming section.

Tailwind CSS. *Tailwind CSS* [105] is a utility-first CSS framework. In conventional CSS, developers write named classes that each describe a component (`.notification-card { padding: 1rem; border-radius: 0.5rem; }`). In Tailwind, styling is applied directly with utility classes in the HTML: `<div class="p-4 rounded-lg bg-surface">`. Each class does exactly one thing. The result is that the visual design and the structure live together in the component file; there is no separate stylesheet to maintain or keep in sync.

For a project where multiple developers will build different applications, this matters: consistency comes from shared utility classes and design tokens, not from trusting that each developer remembers to use the right custom class name. A button looks

the same across applications because it uses the same token values, not because two people independently wrote matching CSS.

Tailwind is a build-time tool. Only the utility classes actually used in the project are included in the final CSS output; unused classes are removed.

ui-kit. A shared TypeScript package provides the visual foundation for every first-party application. It is based on *shadcn/ui* [99], a collection of accessible, unstyled UI components built on *Radix UI* [128] primitives and styled with Tailwind.

The key design decision behind shadcn/ui is its distribution model: components are not installed as an npm dependency. Instead, component source files are *copied into the project* and become part of the codebase. This means there is no upstream version to break things silently, no peer-dependency conflicts, and full ownership of every component. When a component needs to deviate from the default (different animation, extra accessibility attribute), the source file is there to edit directly.

ui-kit is organized in three layers. The first layer is the *design token system*: values defined as CSS custom properties that establish the entire visual language. A *CSS custom property* (also called a CSS variable) is a named value defined in a CSS rule and referenced throughout the stylesheet: `--color-accent: #3b82f6` defined once and used as `color: var(--color-accent)` everywhere. When the accent color changes, one value changes and every element using it updates instantly, without reloading any component. This is the mechanism behind live theme switching. The second layer contains base components: Button, Input, Dialog, Card, and similar primitives built from shadcn/ui and styled with the token system. The shadcn/ui component set covers the full range of shell needs directly: Popover and DropdownMenu for the global menu bar and its context popup, Command for Waypointer's search interface, Carousel for the workspace switcher, Toast for notifications, and Progress for toolbar status indicators. Because shadcn/ui distributes component source rather than a compiled library, each of these can be adapted to shell-specific behaviour without fighting an upstream API. The third layer contains OS-specific components: `WindowChrome`, `AppShell`, `GlobalMenuBar`, `AppMenuPopup`, and `NotificationToast`, which implement patterns specific to this desktop environment.

os-sdk. A shared Rust crate provides system integration to every first-party application. It bundles a typed Graph Daemon client (a Rust struct that opens the daemon's Unix socket and sends Cypher queries), an Event Bus client, MCP server boilerplate (the scaffolding to register tools and handle invocations), and the TOML-based config system with automatic schema validation. A new first-party application adds Tauri, ui-kit, and os-sdk and immediately has graph access, event emission, an MCP server, config handling, and a consistent UI without writing any infrastructure code. The standard application layout reflects this:

```
app-files/
  src-tauri/
    Cargo.toml # includes os-sdk
    src/
      main.rs # Tauri setup
      commands.rs # Tauri command handlers
```

```
      mcp.rs # this app's MCP server
  src/
    main.tsx
    components/ # app-specific components (uses ui-kit)
    pages/
```

Listing 6 Standard first-party app structure

6.4 Shell API

Beyond graph access, event emission, and MCP integration, os-sdk exposes a shell namespace that gives first-party applications structured access to desktop-level capabilities. Each surface is permission-gated: the app declares which shell.* capabilities it needs in its manifest, and the system enforces this at runtime. What is not declared is not accessible.

The shell API is TypeScript-first: all surfaces are available to the Tauri frontend via the same invoke bridge used for other Tauri commands. Rust equivalents exist for capabilities that the backend needs directly.

Clipboard. Applications interact with the clipboard using explicit sensitivity labels. The system enforces these labels at read time: an application without clipboard.read.sensitive permission receives null when requesting clipboard contents that carry a sensitive label.

```
// Write with explicit sensitivity label
shell.clipboard.write({
  content: "sk-ant-api03-...",
  mime: "text/plain",
  label: "sensitive" // "normal" is default
})

// Read -- returns null if content is sensitive and app lacks permission
const entry = await shell.clipboard.read()

// React to clipboard changes
shell.clipboard.onChanged((entry) => {
  // entry.mime and entry.label always available
  // entry.content only if not sensitive or app has permission
})

// History (requires clipboard.history permission)
const history = await shell.clipboard.getHistory({ limit: 10 })
```

Listing 7 shell.clipboard

Presence. Applications describe what the user is currently doing. This produces semantically precise UserAction nodes in the Knowledge Graph without relying on eBPF inference, and provides structured context for the Companion App's Handoff feature. Presence data never leaves the device without an explicit user action.

```
shell.presence.set({
  activity: "editing", // "editing"|"reading"|"reviewing"|"building"|custom
  subject: "report-draft.md",
  project: "coffeeshop", // optional; inherits Focus Mode if omitted
  metadata: { language: "rust", line: 247 },
  autoClear: "on-blur" // clear automatically when app loses focus
})

shell.presence.clear()
```

Listing 8 shell.presence

Search. Applications open Waypointer programmatically and register as handlers for specific query patterns or file types, making Waypointer an extensible action surface rather than only a launcher.

```
// Open Waypointer with a pre-filled query
shell.search.open({ query: "files␣in␣this␣project", mode: "ai" })

// Register as a handler for query patterns
shell.search.registerHandler({
  pattern: /^calc\s+(.+)/i,
  label: "Calculate",
  icon: "calculator",
  handle: async (match) => ({
    result: evaluate(match[1]),
    label: `= ${evaluate(match[1])}`,
    action: "copy-to-clipboard"
  })
})

// Register as a handler for file types
shell.search.registerFileHandler({
  extensions: [".rs", ".toml"],
  label: "Open␣in␣My␣Editor",
  handle: async (file) => openFile(file.path)
})
```

Listing 9 shell.search

Annotations. Applications attach structured metadata to existing Knowledge Graph nodes without requiring arbitrary_query graph permissions. Each application writes only to its own namespace; reading another application's annotations requires an explicit permission declaration.

```
shell.annotations.set({
  target: { type: "File", path: "/home/tim/report.md" },
  namespace: "com.example.editor",
  data: { word_count: 1240, last_cursor_position: 847 }
})

```

```
const mine = await shell.annotations.get({
  target: { type: "File", path: "/home/tim/report.md" },
  namespace: "com.example.editor"
})

// Read another app's annotations (requires declared permission)
const gitInfo = await shell.annotations.get({
  target:     {  type: "File", path: "/home/tim/report.md" },
  namespace: "com.example.git-integration"
})

shell.annotations.onChanged({
  target:     {  type: "File", path: "/home/tim/report.md" },
  namespace: "com.example.git-integration",
  handle:     (data) => updateGitStatusDisplay(data)
})
```

Listing 10 shell.annotations

The filter callback in context menu and annotation queries receives graph metadata about the target – project membership, last accessed timestamp, which application last modified the file – enabling precise conditional logic without a separate graph query.

Intents. A typed intent system decouples applications from each other. Applications dispatch intents describing what they want to happen; the system routes to the best registered handler. When multiple handlers match, the user chooses once and the preference is remembered.

```
// Dispatch an intent
shell.intents.dispatch({
  action: "view",
  type: "url",
  data: "https://github.com/timkicker/coffeeshop/issues/42",
  fallback: "open-in-default-app"
})

// Register as a handler
shell.intents.register({
  action: "view",
  type: "url",
  pattern: /github\.com\/.*\/issues\/\d+/,
  label: "Open in GitHub Integration",
  handle: async (intent) => openIssue(intent.data)
})
```

Listing 11 shell.intents

Intents differ from xdg-open in that they are typed and pattern-aware. A handler can match type: "url" with a specific hostname pattern, routing only GitHub issue URLs to the GitHub integration while all other URLs go to the browser. This is not expressible with MIME types alone.

Timeline. Applications write explicit semantic entries to the Knowledge Graph timeline. These produce UserAction nodes that are more precise than eBPF inference and are immediately available in ~/.timeline/ and AI queries.

```
// Point-in-time event
shell.timeline.record({
  label: "Exported␣PDF",
  subject: "/home/tim/report.pdf",
  type: "export",
  metadata: { pages: 12, format: "pdf/a" }
})

// Duration event
shell.timeline.record({
  label: "Build␣succeeded",
  subject: "coffeeshop",
  type: "build",
  started_at: buildStart,
  ended_at: new Date(),
  metadata: { duration_ms: 4230, target: "release" }
})
```

Listing 12 shell.timeline

Timeline differs from Presence: Presence describes the current state and is ephemeral; Timeline describes completed events and is persistent. Both produce graph nodes but with different semantics.

Shortcuts. Applications register contextual quick actions that appear in Waypointer and the global menu bar when the application is focused. Unlike menu items, shortcuts are designed for immediate single-action access without navigation. Context tags connect shortcuts to Focus Mode: a shortcut tagged "git" appears only when the active project carries that tag.

```
shell.shortcuts.register([
  {
    label: "New␣Issue",
    icon: "plus-circle",
    action: "issue.new",
    context: ["project", "git"],
    confirm: "Create␣new␣issue␣in␣current␣project?"
  },
  {
    label: "Run␣Tests",
    icon: "play",
    action: "test.run",
    context: [] // always visible when app is focused
  }
])

// Update shortcut state dynamically
shell.shortcuts.setState("git.push", {
  enabled: hasUnpushedCommits,
```

```
  badge: unpushedCount > 0 ? `${unpushedCount}` : undefined
})
```

Listing 13 shell.shortcuts

Badges. Applications set status indicators on their taskbar icon. The system renders all badges consistently; applications cannot define custom badge styles or colors beyond the four status types. Error and warning badges are recorded in the Knowledge Graph; count-only badges are not.

```
shell.badges.set({ count: 3 })
shell.badges.set({ dot: true })
shell.badges.set({ status: "warning" }) // "success"|"warning"|"error"|"progress"
shell.badges.set({ status: "progress", value: 0.67 })
shell.badges.set({ count: 2, status: "warning" })
shell.badges.clear()
```

Listing 14 shell.badges

Context menu. Applications add entries to system context menus – right-click on files, directories, or Waypointer results. Filter callbacks receive graph metadata about the target, enabling precise conditional entries.

```
shell.contextmenu.register({
  target: "File",
  filter: (file) => file.extension === ".md",
  label: "Open␣in␣My␣Editor",
  icon: "file-text",
  action: "open.markdown",
})

shell.contextmenu.register({
  target: "File",
  filter: (file) => file.extension === ".md",
  label: "Export␣as...",
  icon: "export",
  children: [
    { label: "PDF", action: "export.pdf" },
    { label: "HTML", action: "export.html" },
    { label: "EPUB", action: "export.epub" },
  ]
})

shell.contextmenu.register({
  target: "Directory",
  filter: (dir) => dir.hasFile(".git"),
  label: "Open␣Git␣Dashboard",
  icon: "git-branch",
  action: "git.open",
})
```

Listing 15 shell.contextmenu

Onboarding. Applications register setup steps that are woven into the system onboarding wizard at first login. All steps use ui-kit components and appear as a single coherent flow rather than per-app dialogs.

```
shell.onboarding.register({
  id: "com.example.editor.import",
  label: "Import␣your␣notes",
  subtitle: "Bring␣your␣existing␣notes␣from␣Obsidian␣or␣Notion",
  icon: "import",
  optional: true,
  order: "after:accounts",

  component: OnboardingImportStep,

  shouldShow: async () => checkForExistingNotes(),

  onComplete: async (result) => {
    if (result.imported) {
      shell.timeline.record({
        label: "Imported␣notes",
        type: "import",
        metadata: { count: result.count, source: result.source }
      })
    }
  }
})
```

Listing 16 shell.onboarding

shouldShow runs before the wizard renders. Steps that return false are omitted entirely; the user never sees an empty or irrelevant step.

Ambient. Applications influence the visual state of the desktop subtly – a slow pulse of the accent color while a build runs, a warning tint when tests fail. Ambient effects are peripheral rather than interruptive: they are perceptible without demanding attention. The user can disable all ambient effects globally in Settings. Only colors from the token system are permitted; arbitrary hex values are rejected. Maximum intensity is capped at 0.5.

```
shell.ambient.set({
  effect: "pulse",
  intensity: 0.3,
  speed: "slow", // "slow" | "medium" | "fast"
  reason: "build-running",
})

shell.ambient.set({
  effect: "tint",
  color: "warning", // token system color, not a hex value
  intensity: 0.15,
  reason: "test-failing",
  autoClear: 5000 // ms
})
```

```

shell.ambient.clear()
```

Listing 17 shell.ambient

Spatial. Applications provide window placement hints to the compositor. Hints are suggestions, not commands: the compositor applies them when possible and ignores them when the current layout does not permit it. User-initiated window moves always take precedence.

Full spatial hint support requires a compositor-side extension of the cosmic-comp fork and is planned for a subsequent release. The SDK surface is defined now so that applications can declare hints without breaking changes when the compositor support arrives. Until then, shell.spatial calls are accepted and silently ignored.

```
shell.spatial.suggest({
  relation: "beside",
  target: "com.example.terminal",
  side: "right",
  size: 0.4
})

shell.spatial.suggest({
  relation: "float",
  anchor: "cursor",
  size: { width: 600, height: 400 }
})

// Remember layout preference for a Focus Mode context
shell.spatial.remember({ context: "focus-mode" })
```

Listing 18 shell.spatial

Voice. Applications access speech recognition through the AI daemon's local inference pipeline. No application needs to implement its own speech recognition. Cloud fallback is available only if the user has explicitly configured a cloud AI provider in Settings.

When any application is actively listening, the system displays a microphone indicator in the global menu bar. This indicator cannot be suppressed by the application. It is a system-level affordance that the user always knows when the microphone is in use.

```
// One-shot transcription
const text = await shell.voice.transcribe({
  language: "de-AT",
  local: true
})

// Streaming transcription
shell.voice.stream({
  language: "en",
  onPartial: (text) => showLivePreview(text),
  onFinal: (text) => processCommand(text),
```

```
  onSilence: () => stopListening()
})

// Command grammar
shell.voice.listenForCommand({
  grammar: [
    { phrase: "open␣$project", action: "project.open" },
    { phrase: "new␣file", action: "file.new" },
  ],
  onMatch: (action, params) => handleCommand(action, params)
})

shell.voice.setIndicator({ visible: true, label: "Listening..." })
shell.voice.setIndicator({ visible: false })
```

Listing 19 shell.voice

The shell includes a persistent narrow bar anchored to the top edge of each monitor. Its design is informed by macOS and GNOME: a single shared bar rather than per-window title bars carrying duplicate controls. The left side shows the active application's name followed by its registered menu structure. The right side carries system status indicators: clock, network, audio, battery, and the quick settings toggle. The center displays the active project name when Focus Mode is active, providing a persistent ambient signal of the current context.

App menu registration. First-party applications register their menu structure once via os-sdk at startup. The shell reads the registration and renders it in the bar; the application does not manage its own menu rendering at all. The registration declares the full menu tree, item types, default keyboard shortcuts, and initial enabled states:

```
shell.menu.register({
  items: [
    {
      label: "File",
      children: [
        { label: "New", shortcut: "Ctrl+N", action: "file.new" },
        { label: "Open", shortcut: "Ctrl+O", action: "file.open" },
        { label: "Save", shortcut: "Ctrl+S", action: "file.save",
          enabled: false },
        { separator: true },
        { label: "Export␣as␣PDF", action: "file.export.pdf",
          context: ["project"] },
      ]
    },
    {
      label: "View",
      children: [
        { label: "Show␣Sidebar", action: "view.sidebar",
          type: "toggle", checked: true },
        { separator: true },
        { label: "Small", action: "view.size.small",
          type: "radio", group: "view.size" },
```

```
        { label: "Large", action: "view.size.large",
          type: "radio", group: "view.size" },
      ]
    },
    {
      label: "File",
      children: [
        { label: "Open␣Recent", type: "recent",
          node_type: "File", limit: 8 },
      ]
    }
  ]
})
```

Listing 20 App menu registration via os-sdk (TypeScript)

The action field is an opaque string identifier. When the user selects an item, the shell dispatches it back to the application via the Tauri command bridge; the application handles it and never touches menu rendering directly. Shortcuts declared in the menu are registered automatically with the Global Input Manager (Section **??**), ensuring the displayed binding always reflects the user's actual configured binding rather than a hardcoded default.

Dynamic state updates. Menu items change state at runtime: Save becomes enabled when the document is modified, Undo carries the name of the last operation, toggles reflect the current view state. The application pushes these updates through os-sdk without re-registering the full menu:

```
// Enable Save when document is modified
shell.menu.setState("file.save", { enabled: true })

// Update Undo label to reflect the last action
shell.menu.setState("edit.undo", {
  enabled: true,
  label: "Undo␣Typing"
})

// Reflect sidebar toggle state
shell.menu.setState("view.sidebar", { checked: false })
```

Listing 21 Runtime menu state updates

Recent items. The type: "recent" item type is a declarative slot that os-sdk fills automatically from the Knowledge Graph. An application declares which node type it wants recent entries for and the maximum count; the shell queries the graph for the most recently accessed nodes of that type opened by this application and populates the submenu. No application-level recent-files tracking is needed. The list updates on every menu open, so it always reflects the current graph state without the application maintaining its own history.

Context tags. Menu items can carry optional context tags. When Focus Mode is active and the active project carries matching tags, items tagged accordingly receive a subtle visual marker in the menu – a small indicator, not a structural change. This surfaces project-relevant actions without reordering or hiding anything. An export action tagged "project" is marked when a project is focused; an action tagged "git" is marked when the project carries the git tag. Context marking is opt-in per item and off by default.

Third-party app coverage. GTK applications expose their menu structure via the org.gtk.Menus and org.gtk.Actions D-Bus interfaces, which GNOME originally designed for global menu support. The shell implements a bridge that reads these interfaces and renders GTK app menus in the bar alongside first-party menus. Coverage depends on the application implementing the interface; most modern GTK4 applications do. Qt applications do not have an equivalent standard interface; Qt apps retain their in-window menus. This asymmetry is documented rather than hidden. XWayland applications are not covered by the global menu bar regardless of toolkit.

Toolbar extensions. Beyond the menu, applications can push three additional elements into the bar via os-sdk. These are additive slots that appear between the menu items and the system status indicators when the application is focused.

Quick Actions are a small set of icon buttons for primary operations. The limit is three; exceeding it is rejected by the SDK. Icons come from the ui-kit icon set to guarantee visual consistency:

```
shell.toolbar.setQuickActions([
  { icon: "share", action: "file.share", tooltip: "Share" },
  { icon: "star", action: "file.bookmark", tooltip: "Bookmark",
    toggle: true, active: false },
])
```

Listing 22 Quick Actions registration

A Breadcrumb replaces the quick actions slot when the application is in a navigable hierarchy – a file manager in a deep directory, a settings panel several levels in. The shell renders it as a compact inline path with each segment clickable:

```
shell.toolbar.setBreadcrumb([
  { label: "Home", action: "nav.home" },
  { label: "Projects", action: "nav.projects" },
  { label: "CoffeeShop", action: "nav.current" },
])
```

Listing 23 Breadcrumb registration

A Progress indicator covers long-running background operations. It appears as a subtle fill on the bar itself and disappears automatically when cleared:

```
shell.toolbar.setProgress({ value: 0.67, label: "Compiling..." })
shell.toolbar.clearProgress()
```

Listing 24 Progress indicator

Quick Actions, Breadcrumb, and Progress are mutually exclusive in the same slot; setting one clears the others. All three are cleared automatically when the application loses focus.

Audit log integration. Every menu action executed by the user is written to the Knowledge Graph as a `UserAction` node via os-sdk, without the application emitting a separate event. The node carries the application identity, the action string, and the timestamp. This gives the AI layer a structured record of what the user did inside applications, not just which files they opened, enabling queries like "what did I export last Tuesday" without any extra instrumentation work from the application developer.

6.5 App Menu Context Popup

Accessing the global menu bar requires moving the pointer to the top of the screen. For users working in the center or bottom of the display this is a deliberate interruption. The App Menu Context Popup addresses this: a keyboard shortcut opens the active application's full menu as a floating popup at the current pointer position, without touching the bar.

The popup renders the identical menu structure registered in the bar – the same items, the same enabled states, the same labels, the same submenu hierarchy. It is not a separate registration or a summarized version; it is the same data rendered in a different location. A user who knows the menu structure from the bar will find the popup immediately familiar.

Positioning. The popup anchors to the current pointer position with space-awareness: it opens in whichever direction has more room, capping at the screen boundary. Near the right edge it opens to the left; near the bottom it opens upward. Submenus follow the same rule independently of the parent popup's direction. The positioning logic is centralized in the shell and applied uniformly; no application code is involved.

Keyboard navigation. The popup opens in a keyboard-ready state: the first top-level item is focused, arrow keys navigate, Enter selects, Escape closes. The pointer remains where it was; the user does not need to move it. Mouse hover works in parallel – moving the pointer over items highlights them normally. The two input modes are fully composable within a single popup session.

The bar highlights the active application's menu section while the popup is open, providing a visual connection between the two surfaces.

Keybinding. The specific binding is configured in Settings as part of the unified keybinding system (Section **??**). There is no hardcoded default beyond a suggested starting point. It is treated as a first-class system shortcut, not an application-level binding, because it operates on any focused application regardless of what that application registers.

Third-party app coverage. The popup is available for any application whose menu is visible in the bar: first-party Tauri apps and GTK apps that expose the D-Bus menu interface. Applications without a registered menu show an empty popup with a single disabled item noting that no menu is available. Qt and XWayland apps are not covered for the same reasons as the bar.

6.6 Default Applications

The system ships a curated set of default applications. The selection criteria are consistent: Wayland-native where possible, open source, and either first-party Tauri applications or well-maintained projects with a clear separation between backend logic and visual presentation.

Terminal emulator. The default terminal is *Ghostty* [55], a GPU-accelerated terminal emulator written in Zig by Mitchell Hashimoto. Ghostty is Wayland-native, carries an MIT license, and ships its own terminfo entry. The base system image includes the `xterm-ghostty` terminfo entry so that SSH sessions to remote hosts do not require a separate setup step. The primary alternative, *kitty*, was considered but Ghostty's Wayland-first design and more permissive license are better fits for this project.

Default shell. The default interactive shell is *Zsh* [130]. Zsh is POSIX-compatible, shares the broad shell scripting ecosystem with Bash, and has a mature plugin and completion infrastructure. It has been the default shell on macOS since 2019, which reduces friction for users arriving from that platform. The system ships a minimal Zsh configuration that enables syntax highlighting, history-based autosuggestions, and a clean prompt without pulling in a framework such as oh-my-zsh. Fish was evaluated for its superior out-of-the-box UX but rejected for its POSIX incompatibility, which would make a significant fraction of shell snippets and scripts from the wider ecosystem fail silently.

Web browser. The default browser is *Zen Browser* [129], a Firefox-based browser focused on privacy and a clean UI. It inherits Firefox's rendering engine and extension ecosystem while adding a more opinionated interface. Using a Firefox-based browser aligns with the project's European digital autonomy goals: Mozilla is a non-profit foundation, and the Gecko engine is the only major browser engine not controlled by a US technology company.

File manager. The file manager is a first-party Tauri application. The Rust backend handles filesystem operations using the standard library and the `notify` crate for filesystem event watching; the TypeScript frontend provides the directory view, file previews, and bulk operations. The *conaticus/FileExplorer* [30] project, a Tauri v2 file explorer with a similar backend architecture, was reviewed as a reference implementation. The first-party file manager integrates with the Knowledge Graph via os-sdk: opening, moving, and deleting files emit events that become graph nodes, enabling the temporal filesystem view described in Section 4.9. One known Tauri limitation is that

drag-and-drop of files to native (non-Tauri) applications is not currently supported by the framework; this is documented and will be resolved when upstream Tauri adds the capability.

The file manager sidebar displays active projects alongside the standard directory tree. Selecting a project in the sidebar sets the file manager root to the project's configured root path and shows a recent-files view filtered to that project's graph context. Files can be assigned to a project via right-click context menu, which writes a PART_OF edge directly to the Knowledge Graph without moving the file on disk.

Text editor. The default text editor is a first-party Tauri application embedding *CodeMirror 6* [29] as the editing component. CodeMirror provides syntax highlighting for over 150 languages, a language server protocol (LSP) client interface, and accessible keyboard navigation out of the box. The Tauri backend handles file I/O and integrates with the graph for recent-file tracking.

PDF viewer, image viewer, archive manager. These three applications are first-party Tauri applications with minimal backends. The PDF viewer embeds *PDF.js* [79] in the WebView. The image viewer renders directly in a <canvas> element with a Rust backend handling format decoding via the image crate [59]. The archive manager calls libarchive from the Rust backend for read and write operations across zip, tar, and common compression formats. All three write open-file events to the Knowledge Graph.

6.7 Waypointer

Waypointer is the system-wide launcher and command palette: the primary keyboard-driven entry point to the entire desktop, triggered by the Super key by default and accessible from any context as a centered overlay. It handles app launching, file and web search, inline calculation, unit conversion, timezone lookup, and contextual actions derived from the current selection or active application. It is not a file manager; deep folder navigation and bulk operations belong in the File Manager. Waypointer surfaces the result the user wants immediately or hands off to the right application.

When the AI layer is active, Waypointer gains an AI mode toggled by Tab. Natural language queries are interpreted by the AI layer, resolved against the Knowledge Graph and available MCP actions, and returned as structured results in the same interface. Asking "show me what I worked on this week" is a Waypointer query, not a file search.

When a project is active in Focus Mode (Section 6.8), Waypointer's default search scope narrows to that project's graph context. Typing a filename searches within project files first before widening to the full graph. Project-specific quick actions are surfaced at the top of the result list: recent files, last git commit message, open issues, and time logged this week. These are live graph queries, not cached data.

Third-party modules can register as waypointer.provider to add result types or as waypointer.action to add commands, as described in Section **??**.

Typed result system. Waypointer results carry an explicit type that determines how they are rendered. Every type has a defined data structure and a native render component in ui-kit; Integration Packages and modules deliver data in the correct format and Waypointer renders consistently regardless of which provider supplied the result. A GitHub Integration Package and a GitLab Integration Package both deliver issue type results – the user sees the same layout independent of the underlying provider.

Table 10 Built-in Waypointer result types

Type	Rendered fields
app	Icon, name, description, launch action
file	Path, last modified, project membership
action	Label, icon, registered keybinding
issue	Number, title, status, assignee, labels, inline expansion
commit	Hash, message, author, date, changed files
contact	Name, avatar, last interaction timestamp
result	Generic label and subtitle for custom providers

Inline expansion. Typed results support inline expansion: pressing the right arrow key or a dedicated expand shortcut opens a detail panel inside the Waypointer overlay without opening a new window or leaving the current context. The expansion renders the full content of the result – for an issue type this means title, full description, status, assignee, labels, and available actions.

Actions available from the inline expansion depend on the result type and the provider. For issue results the standard actions are: focus on this issue (activates Focus Mode with the issue as sub-context), open in browser (opens the issue URL in the default browser), create related issue, and copy URL. The provider declares which actions are available; Waypointer renders them uniformly.

Issue tracker integration. When a project declares a [tracker] block in its .project file (Section 4.7), Waypointer surfaces issues from that tracker as first-class results when the project is focused. Typing # triggers issue search; typing #42 jumps directly to that issue number.

```
shell.waypointer.register({
  id: "com.lunaris.github-issues",
  trigger: "#",
  placeholder: "Search issues...",
  onQuery: async (q) => {
    const issues = await fetchIssues(q)
    return issues.map(issue => ({
      id: `issue-${issue.number}`,
      title: `#${issue.number} ${issue.title}`,
      subtitle: issue.status,
      icon: issueIcon(issue.status),
```

```
      type: "issue",
      data: issue,
      actions: [
        { label: "Focus␣on␣this", action: "issue.focus" },
        { label: "Open␣in␣Browser", action: "issue.open-browser" },
        { label: "Copy␣URL", action: "issue.copy-url" },
      ]
    }))
  }
})
```

Listing 25 Issue provider registration via module SDK

Issue tracker Integration Packages are available for GitHub Issues, GitLab Issues, OpenProject, Plane, and Jira. Each is a separate Store package that implements the waypointer.provider interface and declares the [tracker] provider type it handles. The .project file specifies which provider to use; Waypointer routes to the correct Integration Package automatically.

Issue Focus Mode sub-context. When the user selects "Focus on this issue" from a Waypointer issue result, Focus Mode activates with the issue as an additional sub-context layer on top of the active project. The AI daemon receives the issue title, description, and labels as part of its context window. Waypointer surfaces files that have been recently modified in sessions where this issue was active. The terminal opens in the project root. A timeline entry is created when the focus session ends, recording which files were touched and how long the session lasted – giving the user a traceable record of work done per issue without any manual time tracking.

Creating issues from context. From any file's context menu or from Waypointer directly, the user can create a new issue in the active project's tracker. The creation form is pre-filled with context: project name, the currently focused file if relevant, and the active branch name if the project has a git integration. The form is minimal – title and description only – with a "Open full editor in browser" option for everything else. The created issue is immediately available as a Waypointer result without a manual refresh.

6.8 Focus Mode

Focus Mode is the coordinated state the desktop enters when the user switches to a project context. It is not a distraction blocker and does not restrict what the user can do. It adjusts the environment to match the active project so that switching context is a single action rather than a series of manual steps across multiple applications.

Activating a project — via Waypointer, the File Manager sidebar, or the project quick-switch keyboard shortcut — triggers the following changes simultaneously:

The shell's accent color updates to the project's configured appearance.accent_color value if one is declared, giving an immediate, ambient visual signal of the active context. If no project color is declared, the user's default accent remains.

Waypointer's search scope narrows to the active project as described in Section 6.7.

The File Manager navigates to the project's root directory and switches its sidebar view to the project context.

A new terminal window opened while a project is active launches with the working directory set to the project root.

Notification priority rules declared in the project's [focus] block take effect. Applications listed under suppress_notifications_from have their notifications demoted to Low priority for the duration of the focused session, sending them silently to the Notification Center without a toast. Critical-priority system alerts are never suppressed.

Modules declared in active_when_focused are started by the Module Runtime if not already running. When the project is deactivated, those modules are stopped.

The AI daemon's read scope is updated to reflect the project context. If the project's [ai] block specifies context_scope = "project", the daemon limits graph traversal to nodes associated with the active project.

All of these changes are reversed when the user deactivates the project or switches to a different one. Focus Mode state is not persisted across sessions; the project is not automatically re-activated on login, though this is configurable in Settings.

6.9 Settings

Settings is the primary control center for the entire system. It is a first-party Tauri application built on ui-kit and os-sdk like any other first-party app, but with privileged write access to the configuration files of other system components. It is not a data store: every component owns its own TOML configuration file and Settings is a UI layer that reads and writes those files. Settings can crash, restart, or never be opened – the system functions regardless.

Architecture. Settings writes to component configuration files via the filesystem and signals changes via the Event Bus. Components observe their configuration files through inotify and reload automatically. For changes that must take effect immediately – theme switches, AI access level, permission updates – Settings additionally emits an Event Bus message and waits for a confirmation from the affected daemon before marking the change as applied. For changes that take effect only after a restart – keybinding profiles, profile type changes – Settings shows a "Wirksam nach Neustart" notice without forcing an immediate restart.

Structure. Settings is organized as a two-level sidebar: a small set of top-level categories on the left, each expanding into specific sections. A global search field at the top of the sidebar finds settings by name, synonym, or keyword across all categories and all depths. "Dark mode", "night mode", and "appearance" all reach the Theme section; "shortcut" and "hotkey" both reach Keybindings.

App pages. Every installed application has a dedicated page in Settings, reachable from the Privacy category or directly via deep link. The page aggregates everything the system knows about that application in one place: permission profile (filesystem, network, device, graph), Integration Package status and adapter configuration if

Table 11 Settings top-level categories

Category	Contents
Appearance	Themes, accent colors, font scale, dark/light mode, animations
Privacy	Permissions per app, graph retention, audit log, clipboard sensitivity
AI	Provider configuration, access level, routing policy, MCP servers
Connections	External service connections, OAuth tokens, outbound event configuration
Accounts	Online accounts, login token handling, account resolution order
System	Updates, snapshots, startup apps, hardware info
Input	Keybindings (single source of truth), mouse, touchpad, keyboard layout
Display	Monitor layout, resolution, scale, refresh rate, rotation
Profiles	Profile creation and management, transfer rules, signed packages
Health	Data sources, graph integration level, retention, export

installed, audit log entry count with export option, and ephemeral mode toggle. For applications that have registered native Settings via os-sdk, their configuration sections appear at the top of the page above the system sections, matching the macOS model where app-specific preferences precede system-managed permissions.

Managed environment display. When a setting is controlled by a Signed Profile Package, it is displayed normally but carries a visible "Managed by organization" badge. A tap on the badge explains which profile package controls it and what the current enforced value is. The setting is not hidden and not greyed out without explanation. This follows the system's general principle of transparency: the user always understands what is happening on their device and why.

Keybindings. The Input category hosts the unified keybinding editor: a searchable list of every registered action from every application, module, and system component. Each entry shows the application or component name, the action label, the current binding, and whether it is a user override or the registered default. Conflicts – two actions sharing the same binding – are highlighted in red with both conflicting entries visible simultaneously. The Global Input Manager is the authoritative source; Settings reads from and writes to it directly. No application manages its own bindings independently.

Connections. Connections is the Settings category for external service integrations: GitHub, GitLab, OpenProject, n8n, OpenClaw, and any other third-party service the system or a project needs to communicate with. It is the single place where these integrations are created, configured, and revoked. No other component stores connection credentials or creates integrations independently.

A connection has an identifier, a provider type, authentication credentials stored in the system keyring via the Account Daemon (never in plaintext), and a data flow classification that describes where data goes. The classification is one of four values:

local for services running on the same machine, self-hosted for services the user controls on their own infrastructure, encrypted for end-to-end encrypted channels such as Signal or Matrix, and third-party for external services whose servers process the data. The classification is displayed prominently on the connection list and on each connection's detail page. It is not a warning – it is information the user needs to make an informed decision about what to connect.

Multiple connections to the same provider are fully supported. A user with two GitHub accounts configures two separate connections with different identifiers; Settings distinguishes them by showing the account name as a subtitle on each entry. When creating a second connection for a provider that already has one, Settings shows a notice but does not block the action.

Each connection detail page shows the provider and account, the data flow classification with a plain-language explanation, which system components are permitted to use the connection, which projects reference it, the last successful contact timestamp from the audit log, and controls to edit or revoke it. The permitted components list is editable: by default only the components that need a connection can use it, and the user can restrict this further.

When a connection's token expires, the Account Daemon attempts a silent refresh using the stored refresh token if one exists. If the refresh fails or no refresh token is available, the connection is marked as expired. All components that request the token receive an access-denied response with the reason token_expired rather than a silent failure. Settings surfaces a warning badge on the connection entry and sends a single High-priority notification with a deep link to the connection detail page.

Deleting a connection that is still referenced by projects or components requires confirmation. Settings presents a list of every dependency before proceeding. It does not automatically update the referencing files; the user cleans up the references after deletion. Settings highlights any .project files that still reference the deleted connection identifier after the deletion completes, each with a deep link to the project's configuration.

When a .project file references a connection identifier that does not exist locally, the system treats this as an onboarding moment rather than an error. Waypointer shows a one-time prompt offering to create the missing connection or assign an existing one to that identifier. A local override file at ~/.config/lunaris/connection-overrides.toml records any identifier remappings; this file is never committed to version control.

```toml
# ~/.config/lunaris/connections.toml

[[connection]]
id = "github-personal"
provider = "github"
account = "timkicker"
scopes = ["repo", "issues"]
token = { keyring = "lunaris/github-personal" }
data_flow = "third-party"
allowed = ["tracker", "ai-daemon"]

[[connection]]
id = "n8n-local"
```

```
provider = "n8n"
endpoint = "http://localhost:5678/mcp"
token = { keyring = "lunaris/n8n-token" }
data_flow = "local"
allowed = ["ai-daemon"]
workflows = ["wf_abc123", "wf_def456"]

[[connection]]
id = "openclaw-local"
provider = "openclaw"
endpoint = "ws://127.0.0.1:18789"
data_flow = "local"
allowed = ["ai-daemon"]
```

Listing 26 Example connection configuration

For connections that support outbound events, such as n8n, the detail page includes an Outbound Events section. The user selects which event types to forward from a fixed list derived from the Event Bus schema. The endpoint receiving these events is bound to the connection and cannot be changed to an arbitrary URL outside of it. The Event Bus enforces a whitelist of permitted outbound endpoints that it receives from the Account Daemon; any event targeting an endpoint not on the whitelist is silently discarded. Outbound events are rate-limited to 60 per minute per connection to prevent high-frequency desktop activity from generating uncontrolled external traffic. When the limit is reached the connection receives a warning badge in Settings.

For n8n connections specifically, the detail page also shows an Allowed Workflows section. The AI daemon may only invoke workflows explicitly listed here. The default is an empty list, meaning no workflows are callable until the user explicitly permits them. A fetch button retrieves the available workflows directly from the configured n8n instance and presents them as a selectable list, so the user never needs to type workflow identifiers manually.

Deep links. Settings registers a settings:// URL schema at startup. Any application, notification, or module can open a specific Settings location directly:

```
settings://apps/org.mozilla.firefox
settings://apps/org.mozilla.firefox/permissions
settings://apps/org.mozilla.firefox/adapter
settings://privacy/audit-log
settings://ai/providers
settings://input/keybindings
settings://appearance/themes
settings://connections
settings://connections/github-personal
settings://connections/new?provider=github
settings://connections/new?provider=n8n
```

Listing 27 Settings deep link examples

If Settings is already open, incoming deep links navigate without opening a new window. If Settings is closed, it opens directly at the target location. The Store uses

a complementary store:// schema for links in the other direction, described in Section 7.6.

Reset. Every setting has a reset path. Individual settings show a reset icon that restores the system default. Each section header offers a "Reset this section" option. Keybindings can be reset per action or for all actions at once. Resets never require confirmation for individual settings – the action is immediately reversible by setting the value again. Section resets show a single confirmation step.

6.10 Notification System

Desktop notifications on Linux follow the *freedesktop.org Desktop Notifications Specification* [42], a D-Bus protocol that any application can use via libnotify [50] without knowing anything about the specific notification implementation on the system. An application calls notify_send() or the equivalent library function; the notification daemon receives it over D-Bus, decides how to present it, and handles persistence, grouping, and dismissal.

This project ships a first-party notification daemon written in Rust. Its architecture follows the pattern established by *swaync* [35] (SwayNotificationCenter), which demonstrated that the daemon logic and the visual presentation can be cleanly separated: the daemon handles the D-Bus interface, queuing, persistence, and routing policy, while the visual layer is a separate process that connects to the daemon. Rather than adopting swaync's GTK-based frontend, this project replaces the visual layer with Tauri components that integrate with the design language and the Event Bus.

The notification system has two visual components. The *Toast Renderer* is a layer-shell surface anchored to the top-right corner of each monitor; it displays transient notifications for a configurable duration and dismisses them automatically. The *Notification Center* is a persistent panel surface that shows the full notification history, grouped by application, with actions and dismissal controls. Both are Tauri windows that receive notification data from the daemon over a Unix socket.

Every notification that passes through the daemon is written to the Knowledge Graph as a Notification node with timestamp, source application, summary, and whether it was interacted with. This gives the AI layer visibility into what the user was interrupted by and when, and enables queries like "what notifications did I miss while in the meeting?" without any changes to the applications that sent them.

6.11 Screen Lock

Screen locking on Wayland uses the ext-session-lock-v1 [124] protocol, a standard Wayland extension that allows a client to take exclusive control of all outputs and receive input until it explicitly unlocks. Once the lock surface is committed, the compositor freezes all other clients' rendering and routes all keyboard and pointer input to the lock client only. No other application can render to the screen or receive input events; this is enforced by the compositor, not by the lock client.

The lock screen is a first-party Tauri application that implements this protocol. On activation it acquires the session lock, renders its surface on every connected monitor, and communicates with PAM via a small Rust backend for credential verification. The visual design is part of the standard theme system and follows the same token architecture as the rest of the shell.

Lock activation is triggered by three paths: an explicit keyboard shortcut, the power menu, and idle timeout. Idle detection uses the ext-idle-notify-v1 Wayland protocol, which allows a client to register a timeout and receive a notification from the compositor when no input has been received for that duration. The idle-notify client is a minimal system daemon that triggers the lock screen on timeout and optionally suspends the system after a second, longer timeout.

The lock screen has no access to the graph, the event bus, or any user data beyond what the compositor provides (monitor layout and scale factors). It authenticates via PAM only. A failed unlock attempt is logged to the Audit Log as a security event; the lock screen itself does not write to the Knowledge Graph.

6.12 Multi-Monitor and HiDPI

Wayland treats multi-monitor and HiDPI as first-class concepts: each output carries its own scale factor, every Wayland-native application queries its output properties and renders at the correct scale automatically, and WebKitGTK respects the Wayland scale factor transparently. This means Tailwind layouts render identically on 4K and 1080p without extra work in the application.

The taskbar runs as a separate layer-shell surface per monitor, each correctly sized and scaled for its output. Monitor configuration, covering resolution, refresh rate, scale factor, position, and rotation per display, is stored in ~/.config/display/config.toml and applied at compositor startup. When a new monitor is connected, the compositor checks for a saved configuration and applies it immediately if found; for an unknown monitor it applies sensible defaults and surfaces a notification offering to open the display settings.

Fractional scaling at 1.5x (common on 1440p displays) uses the wp-fractional-scale-v1 Wayland protocol. Applications that implement the protocol render crisply at fractional scales; those that do not are rendered at the nearest integer scale, which results in slightly soft output but is acceptable and documented. XWayland has no per-monitor DPI concept and uses a single global DPI matched to the primary monitor; X11 applications do not re-scale when moved between monitors, which is a structural X11 limitation with no compositor-level fix.

6.13 Design Language and Theming

The specific visual design language, exact color palettes, and motion design are finalized in a separate design phase. This section documents the technical theming architecture: how themes are structured, how they propagate, and how community themes work. A theme is not limited to colors: it can cover cursor, sounds, wallpa-

per, typography, motion, shadows, icons, the login screen, and the terminal prompt alongside the visual layer.

Design principles. Seven principles govern the visual design language. They are constraints on all design decisions, not aspirations.

Content comes first. The shell recedes visually. Colors, shadows, and animations exist to frame content, not to be noticed. A user who is working should forget the shell is there.

Consistency comes from the system, not from discipline. No developer decides an arbitrary spacing value. All visual decisions come from the token system. What is not in the token system does not exist.

Depth is used sparingly and always communicates meaning. A popup floats above content; a sidebar sits behind an active window. Shadows are not decoration. Elevation always means something.

Motion is informative, not decorative. No animation exists that does not convey direction, feedback, or causality. Elements that appear come from somewhere. Elements that disappear go somewhere.

Typography is the primary visual element. Strong typographic hierarchy reduces the need for decorative elements. Bold for primary labels, regular for content, light for metadata. Italic is reserved for figure labels in diagrams.

Color communicates state, not style. The accent color marks interactive elements and active state. Semantic colors – red, amber, green – mark error, warning, and success respectively. No color exists purely for decoration.

Accessibility is not a feature. High Contrast is a fully designed built-in theme, not an afterthought. Reduce Motion is always respected. Font size is scalable system-wide.

Surface token hierarchy. Themes use a surface-aware token hierarchy rather than flat global tokens. This is what makes the Panda theme possible: different surfaces declare different background colors while sharing accent, border, and semantic colors. The hierarchy is:

```
color.bg.shell # topbar, sidebar, dock
color.bg.app # application window background
color.bg.card # cards, elevated elements within apps
color.bg.overlay # popups, modals, dropdowns
color.bg.input # text fields, search inputs
color.fg.primary # primary text
color.fg.secondary # secondary and metadata text
color.fg.disabled # disabled state
color.accent # interactive elements, active state
color.border # dividers and outlines
```

Listing 28 Surface token hierarchy

ui-kit components reference these surface-specific tokens rather than a single `–color-bg`. A button inside a card uses `–color-bg-card`; a dropdown uses `–color-bg-overlay`. Theme authors set these values per surface; the shell renders each surface with the correct background automatically.

Built-in themes. Four themes ship with the system, all generated from the same token structure.

Panda is the default. It uses a split-surface design: shell surfaces (topbar, sidebar, dock) use a dark background, while application surfaces use a light background. This creates a high-contrast frame around the working area – the shell is the tool, the application is the focus. When the user is in full-screen, the distinction disappears gracefully because only the application surface is visible. The Panda theme works because the surface token hierarchy makes it a first-class design choice rather than a hack: color.bg.shell is set dark and color.bg.app is set light, and every component that uses these tokens renders correctly without any per-component special casing.

Light sets all surfaces to light variants. *Dark* sets all surfaces to dark variants. Both are fully supported and receive the same design attention as Panda. A user who prefers a uniform desktop has an equally polished option.

High Contrast targets *WCAG AAA* [120] accessibility compliance, which requires a minimum contrast ratio of 7:1 between text and its background. It uses no transparency effects, no gradients, and conveys no information through color alone. It has both a dark and light variant and is generated from the same token system as the others. blur.enabled is set to false automatically for this theme.

Token system. All themes are defined in TOML and compiled to platform-specific outputs by the build system. The TOML file is the single source of truth; no theme value exists anywhere else. From one tokens.toml, the build generates CSS custom properties for the Tauri shell and all first-party applications applied via a root stylesheet loaded at startup; a GTK4 CSS theme applied via gtk-theme-name in ~/.config/gtk-4.0/settings.ini; a Wine .msstyles file so that Windows applications running under Wine use the same accent color and surface colors as the rest of the desktop; and base dark and light palettes for GTK3 and Qt.

Changing the accent color in the TOML file propagates to every output on the next build; in a live switch (see below), the CSS custom property update is applied without a rebuild.

Project accent colors. A project can declare an optional accent color in its .project file (Section 4.7). When Focus Mode activates that project, the declared color is applied as a temporary token override on top of the active theme, replacing only –color-accent and its derived values. All other theme tokens remain unchanged. The override is applied via the same CSS custom property mechanism used for live theme switching and takes effect immediately without a reload. When Focus Mode deactivates, the user's configured theme accent is restored. Project accent colors do not affect GTK4, Qt, or Wine theming; they apply to the Tauri shell and first-party applications only. This scope is disclosed in Settings.

GTK and Qt coverage. GTK4 receives a full custom theme generated from tokens. GTK3 and Qt receive fixed base dark and light themes only; the split-surface Panda design requires component-level CSS control that neither framework exposes cleanly, so they fall back to whichever base variant is closer to the active theme.

Store themes. Community themes are distributed through the Store. A theme package declares which layers it covers in its manifest and can inherit from any built-in theme, overriding only the values it changes. The Store displays coverage before installation so users know which layers remain at the base theme. Two authoring formats are supported: token overrides for authors who want to change colors and radius values without touching CSS, and full compiled CSS for themes that need structural changes beyond what the token system exposes.

Cursor. A theme package can include a complete cursor set in the standard *xcursor* format. xcursor supports frame-based animation natively, so animated cursors for states such as loading or resize require no additional mechanism. The cursor files are installed to ~/.local/share/icons/<theme-name>/cursors/ on theme activation; the Settings daemon sets XCURSOR_THEME and XCURSOR_SIZE in the session environment. This applies automatically to all Wayland applications and XWayland. A theme that does not declare a [cursor] block leaves the system cursor unchanged.

```
[cursor]
theme = "bundled"
size = 24
```

Listing 29 Cursor theme declaration

Greeter. The login screen is a first-party Tauri application and therefore fully covered by the token system like any other Tauri surface. It is listed as an explicit coverage layer in the theme manifest so that users can see whether a theme extends to the login screen or leaves it at the system default. The greeter reads the active theme from a system-wide path at startup since it runs before any user session is active; live theme switching does not apply to it.

Sounds. A theme can declare system sounds and per-application sound overrides. System sounds cover the standard events: notification, error, warning, and action completion. Applications declare their own sound slots in their manifest; a theme can override these by namespace.

```
[sounds]
system.notification = "sounds/notification.ogg"
system.error = "sounds/error.ogg"
system.warning = "sounds/warning.ogg"

# Per-app overrides by namespace
"org.lunaris.calendar.notification.reminder" = "sounds/soft-chime.ogg"
```

Listing 30 Sound theme declaration

All sounds are played by the Notification Daemon, never by applications directly. This means Do Not Disturb, Focus Mode suppression, and the global volume setting apply automatically to every sound regardless of which theme declared it. OGG/Vorbis is the required format. A theme that omits [sounds] leaves system sounds unchanged.

Motion. Animation timing and easing curves are theme tokens like colors. A theme declares duration and easing values that apply to every transition across the shell and all first-party applications.

```
[motion]
duration.short = "120ms"
duration.medium = "200ms"
duration.long = "350ms"
easing.default = "cubic-bezier(0.4, 0, 0.2, 1)"
easing.enter = "cubic-bezier(0.0, 0.0, 0.2, 1)"
easing.exit = "cubic-bezier(0.4, 0.0, 1.0, 1)"
easing.spring = "cubic-bezier(0.34, 1.56, 0.64, 1)"
```

Listing 31 Motion token declaration

These compile to CSS custom properties (–duration-short, –easing-default, and so on) consumed by ui-kit components and the shell. When the user enables *Reduce Motion* in Settings, the system overrides all duration tokens to 0ms regardless of the active theme. This override is not suppressible by a theme package; accessibility takes precedence over aesthetic preference.

Shadows and depth. In-application shadows and blur effects are CSS custom properties and work within the Tauri WebView layer without compositor involvement.

```
[depth]
shadow.active = "0 8px 32px rgba(0,0,0,0.4)"
shadow.inactive = "0 2px 8px rgba(0,0,0,0.15)"
shadow.popup = "0 4px 16px rgba(0,0,0,0.3)"
blur.panels = "12px"
blur.popups = "8px"
blur.enabled = true
```

Listing 32 Depth token declaration

blur.enabled is set to false automatically for High Contrast themes; blur reduces text legibility and is incompatible with WCAG AAA compliance. Compositor-rendered window shadows (outside the Tauri window boundary) require a compositor-side extension and are planned for a subsequent release; the token declarations are defined now so that themes can declare values without breaking changes when support arrives.

Typography. A theme can bundle font files and declare the typefaces used across all surfaces. Bundled fonts are installed to ~/.local/share/fonts/ on theme activation and removed on deinstallation.

```
[typography]
font.sans = "Inter"
font.mono = "JetBrains Mono"
font.size = "14px"
font.weight.regular = 400
font.weight.bold = 600
font.lineheight = 1.5

[typography.bundled]
```

```
files = [
  "fonts/Inter-Regular.woff2",
  "fonts/Inter-Bold.woff2",
  "fonts/JetBrainsMono-Regular.woff2",
]
```

Listing 33 Typography token declaration

Only fonts licensed for redistribution may be bundled in a Store theme. The Open Font License (OFL) and equivalents are acceptable; proprietary fonts are not. Store submission validates the declared license field against the bundled files.

Wallpaper. A theme declares a wallpaper in one of three modes. Static delivers a single image. Dynamic delivers multiple images selected by time of day, matching the ambient light cycle. Video delivers an animated wallpaper rendered as a looping video via *mpv* [80] as a background Wayland client.

```
# Static
[wallpaper]
type = "static"
file = "wallpapers/wallpaper.jpg"

# Dynamic (time-of-day)
[wallpaper]
type = "dynamic"
dawn = "wallpapers/dawn.jpg"
morning = "wallpapers/morning.jpg"
day = "wallpapers/day.jpg"
evening = "wallpapers/evening.jpg"
night = "wallpapers/night.jpg"

# Animated video
[wallpaper]
type = "video"
file = "wallpapers/wallpaper.mp4"
loop = true
fps = 30
fallback = "wallpapers/wallpaper.jpg"
```

Listing 34 Wallpaper declaration

Video wallpapers are opt-in: the system uses the static fallback by default and switches to the video only when the user explicitly enables animated wallpapers in Settings. A clear notice explains the GPU cost. The `fallback` field is required for video declarations.

Icons. A theme can supply a complete icon set in the Freedesktop icon theme format, installed to `~/.local/share/icons/`. This covers application icons in Waypointer, the File Manager, and the taskbar, as well as GTK application icons. First-party Tauri application UIs use the ui-kit SVG icon set internally; Store themes can supply overrides for these separately as SVG files declared in the manifest.

The Store listing shows two icon coverage indicators independently: *System icons* (Freedesktop) and *UI icons* (ui-kit). A theme that covers only one of the two is valid and common; full icon coverage requires both.

Terminal prompt style. A theme can supply a complete *Starship* [102] configuration that defines the prompt layout, information density, symbols, and structure alongside the color tokens already covered by CLI theming. The prompt configuration is delivered as a separate file and merged via Starship's include mechanism, leaving the user's own Starship configuration untouched.

```
[terminal.prompt]
type = "bundled"
file = "starship/prompt.toml"
```

Listing 35 Terminal prompt style declaration

The Settings daemon writes the bundled file to ~/.config/starship-theme-prompt.toml on theme activation. The user opts in by adding includes = ["starship-theme-prompt.toml"] to their own starship.toml. As with all generated files, the user's own configuration is never modified directly.

CLI and terminal theming. The token system is not limited to GUI layers. The same color tokens that drive the Tauri shell and GTK4 can be projected onto terminal emulators, shells, and CLI tools, giving a user who installs a theme coherent colors across every context from the desktop to the command line.

This is implemented as two complementary mechanisms. The first is *generated configuration files*: for applications with a known, stable configuration format, the Settings daemon generates the appropriate file and writes it to the correct location on every theme switch. The second is *environment variables*: for tools that accept color configuration via environment variables, the session environment is updated at login from the active theme's token values.

Table 12 lists the supported applications, their integration mechanism, and what the user must do to opt in.

The generated-file mechanism follows a strict non-destructive policy. The Settings daemon never writes to a file the user created manually. Each application has a dedicated *theme-managed* file at a predictable path; the user's main configuration file is never touched. The user opts in by adding a single include or require line to their own configuration, and can remove it at any time to stop receiving theme updates for that application. This means a user with a heavily customized Neovim configuration loses nothing by enabling theme management: only the colors defined in the managed file change, and every other setting in the user's own files remains intact.

Store themes can declare CLI coverage in their manifest alongside GUI coverage. A theme that covers Ghostty and Neovim alongside the shell and GTK4 shows all four coverage indicators in the Store listing. CLI coverage entries in a theme package contain only color token overrides in the same TOML format used for GUI tokens; they cannot contain arbitrary configuration directives or executable code. The Settings daemon generates the final output files from these token values, not by copying the

Table 12 CLI and terminal theme integration

Application	Mechanism	User action required
Ghostty	Generated config	Enable in Settings; Settings daemon writes ~/.config/ghostty/theme-managed.conf, included from main config.
Zsh prompt (Starship)	Generated config	Enable in Settings; generates ~/.config/starship-theme.toml, merged by Starship at startup.
Neovim / Vim	Generated Lua/Vimscript	Enable in Settings; generates ~/.config/nvim/theme-managed.lua. User adds require('theme-managed') to init.lua.
bat	Environment variable	Automatic; BAT_THEME set in session environment.
fzf	Environment variable	Automatic; FZF_DEFAULT_OPTS color flags set in session environment.
delta / difftastic	Environment variable	Automatic; syntax theme exported via DELTA_FEATURES.

theme's content verbatim. This constraint prevents a malicious Store theme from injecting shell aliases or editor keybindings under the guise of color configuration.

Live switching. Figure 9 shows the theme inheritance chain. When the user applies a theme, the Settings daemon loads the package, merges its token overrides with the base theme, and broadcasts updated CSS custom property values to all running Tauri processes via a Unix socket message. Each Tauri WebView has a JavaScript bridge that receives the update and writes the new values to the document's :root CSS rule; all elements that reference those properties via var() update immediately without a component reload or page refresh. This is what makes theme switching feel instantaneous for the Tauri layer. The GTK4 theme is regenerated and re-applied (which requires relaunching affected GTK4 applications to pick up the change), and Wine prefix themes are updated if applicable. Token-based switches complete in under a second; GTK4 regeneration may take one to two seconds on slower hardware.

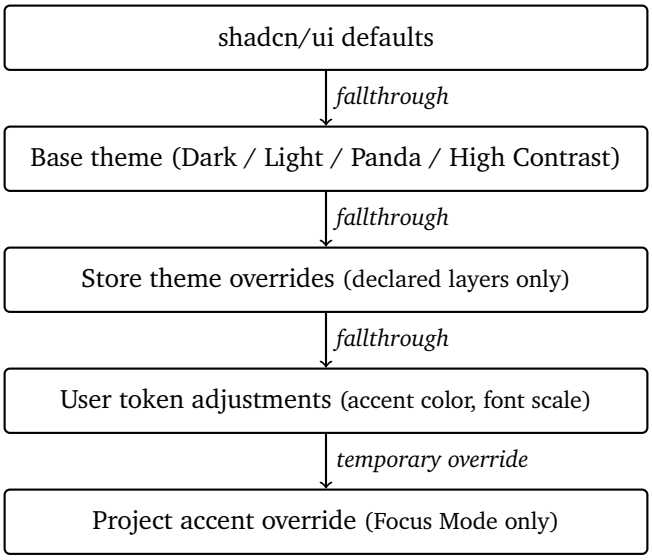

Figure 9 Theme inheritance chain. Each layer overrides only what it declares; everything else falls through to the layer below. The project accent override is the outermost layer and applies only while Focus Mode is active; it is removed when the project is deactivated, restoring the user's configured accent. A Store theme that covers only the shell and GTK4 leaves Wine and Qt at the base theme.

7 Apps, Modules, and the Store

7.1 App Ecosystem

The application layer divides cleanly into two categories: first-party applications built as part of this project, and the broader ecosystem of third-party Linux software. Both operate under the same permission system and benefit from the same infrastructure, but they differ in how deeply they integrate with the platform.

First-party applications. All core system applications are built with the standard stack: Tauri, ui-kit, and os-sdk. Each one follows the shared design language, implements an MCP server for AI integration, emits structured events to the Event Bus, and uses the shared TOML configuration system. Table 13 lists the planned core applications.

Third-party integration layers. Third-party applications are not rewritten. Integration happens in four layers of increasing depth, each building on the one below.

At the baseline, every application on the system receives passive tracking via eBPF regardless of whether it does anything to support this platform: file access, network connections, and process lifecycle are recorded automatically. No changes to the application are required.

For applications that produce useful log output, a wrapper captures stdout and stderr and passes them through asynchronous LLM extraction to produce graph entities. These wrappers are shipped alongside packages in the package manager where practical, and the extraction runs in the background without blocking anything.

Table 13 First-party system applications

Application	Notes
File Manager	Graph-aware features: recent files by project, related files sidebar
Terminal	MCP server exposing command history and session output
Settings	System-wide configuration: theming, permissions, connections, AI level
Store	App and module discovery, installation, and updates
Text Editor	Basic editor primarily for configuration files
System Monitor	Process list, resource usage, and Anomaly Detector alerts
Wine Manager	Prefix management, Winetricks access, per-prefix version selection

For popular applications, dedicated MCP server wrappers expose structured interfaces to the AI layer. These are maintained as separate packages and where possible contributed back to the community as independently useful tools.

Finally, applications that explicitly support the platform APIs emit structured typed events directly to the Event Bus and implement native MCP servers. This is the long-term goal for major open-source applications but is not a precondition for any of the layers above it.

Shared standards. Applications that choose to integrate with the platform work with a common set of conventions: TOML configuration files under ~/.config/<app-id>/, CSS custom properties from the token system for applications that render via WebView, the standard notification daemon via the Freedesktop D-Bus interface or the Event Bus for richer notifications, and a published MCP server interface specification.

7.2 Notification System

The Notification Daemon is the single entry point for all notifications from all sources. Figure 10 shows the pipeline.

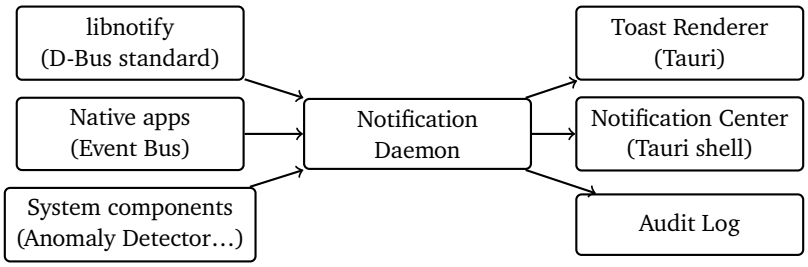

Figure 10 Notification pipeline. All sources converge on the Notification Daemon, which normalizes, groups, and applies priority logic before forwarding to the toast renderer and notification center. Every notification is logged in the Audit Log by sender and timestamp, never by content.

The daemon receives from two sources: the standard D-Bus interface at org.freedesktop.Notifications, which means every third-party application using libnotify or notify-send works without changes; and the Event Bus, through which native system components send richer notifications with priority metadata and graph context. Both are normalized to a common internal schema before any further processing.

Table 14 Notification priority levels

Level	Behavior	Do Not Disturb
Critical	Persistent toast, no auto-close	Always shown, never silenced
High	Toast auto-closes after approximately 8 seconds	Delayed, not blocked
Normal	Toast auto-closes after approximately 4 seconds	Silenced
Low	No toast; goes silently to Notification Center only	Silenced

Critical priority is reserved for system-level alerts only: Anomaly Detector warnings, component compatibility failures, and security events. Applications cannot self-assign Critical. Notifications from the same application are grouped in the Notification Center; as a toast, only the most recent notification from an application is shown, with a counter if a previous one from the same application is still visible. Do Not Disturb has three modes (off, on, and scheduled by time of day) and is togglable from the taskbar quick panel.

Focus Mode (Section 6.8) interacts with the notification priority system directly. When a project declares a [focus] block with suppress_notifications_from, the daemon demotes matching application notifications to Low priority for the duration of the focused session. This demotion is applied by the daemon at routing time, not by modifying the application's permanent permission profile. Critical-priority alerts from system components are exempt and always pass through regardless of Focus Mode state.

7.3 Permission System

Every application on the system, whether a first-party app, a Flatpak, or an RPM package, operates within a permission profile that defines what it can access. The system is unified: one file format, one enforcement layer, one place for the user to review and adjust everything.

Permission file format. Each application has a TOML permission file at ~/.config/permissions/<app-id>.toml that serves as the sole source of truth for what that application is allowed to do. The file is human-readable and user-editable; the Settings app provides a UI for those who prefer not to edit it directly. Permissions are enforced via Linux namespaces, AppArmor, and seccomp, described in Section 8.4.

```
# ~/.config/permissions/org.mozilla.firefox.toml
```

```

[filesystem]
allow = ["~/Downloads", "~/Documents", "~/.mozilla"]

[network]
allow = ["full"]

[devices]
allow = ["gpu"]

[graph]
# no entry = no access, never prompted
```

Listing 36 Example permission file

Graph access is never inferred or learned. An application with no [graph] section has no graph access, with no exceptions and no automatic prompting. Graph permissions must be explicitly declared.

Store-supplied profiles. The Store ships a curated permission profile alongside known applications. Common software such as Firefox, LibreOffice, and VLC has community-maintained profiles that are installed automatically and shown to the user at install time for review. Community members can submit profiles for applications not yet covered; submitted profiles go through Store review before publication, since an overly permissive profile is a meaningful risk. Permission profiles are one component of the broader Integration Package system described in Section 7.4.

Flatpak adapter. Flatpak applications carry their own permission declarations. When a Flatpak is installed, the system automatically generates a native permission file by mapping Flatpak permissions to the native format: filesystem=home becomes filesystem.allow = ["~/"], and so on. The generated file is shown to the user at install time and can be tightened before confirming. The Flatpak manifest is the floor; the system never grants more than what the Flatpak declared.

First-run learning mode. If no Store profile exists and the application is not a Flatpak, the first launch runs in learning mode: all accesses are permitted and logged, nothing is blocked. The learning window closes automatically ten minutes after first launch, or when the session ends, whichever comes first. The user then receives a summary of what the application accessed, with options to review and confirm, edit first, or start with all permissions blocked. From the second launch onward, the confirmed profile is enforced. Learning mode applies to filesystem, network, and device access only; graph access is always denied until explicitly granted and cannot be learned. During the learning window, at most five new permission grants are recorded per minute; exceeding this rate suspends learning immediately and generates an anomaly alert. This limits the damage from a compromised application that attempts to accumulate permissions in bulk during its first session.

Runtime behavior. When an application attempts an access outside its profile, the access is denied and the user receives a notification offering to extend the profile. If the user allows, the profile is updated and the application retries the action; if the user denies, the block stands. Repeated denials for the same path are collapsed: after three denials for the same access, the system asks once whether to always block it rather than continuing to prompt.

Ephemeral app mode. Any application can be launched in ephemeral mode, either from the right-click context menu in the launcher or via a command-line flag. Ephemeral mode combines several existing sandbox mechanisms into a single preset with no configuration required.

In ephemeral mode, a private Mount Namespace is created with no write access to the home directory; the application sees a temporary overlay that is discarded on exit. Graph access is entirely disabled regardless of the application's normal permission profile. Network activity is permitted if declared in the application manifest, but is not logged to the Audit Log. On exit, the temporary overlay is deleted, no entries are written to the Knowledge Graph, and no permission changes persist.

The intended use case is opening untrusted or one-off content: a PDF from an unknown sender, a binary downloaded for a single use, or any application the user wants to run without leaving a trace. Ephemeral mode is not a replacement for the full profile isolation that separate Linux users provide; it does not protect against a compromised application exploiting a kernel vulnerability. What it provides is a low-friction way to reduce the footprint of routine untrusted content without requiring a separate session or a virtual machine.

7.4 Integration Packages

An Integration Package is a Store package that describes how the system integrates with a specific third-party application. It is not a bundle: it contains no application binaries. It is a declaration that references an existing application and provides three optional components alongside it: a permission profile, a Settings adapter, and native Settings registrations for os-sdk applications.

The application itself and its Integration Package are two independent Store entries with separate versions, separate maintainers, and separate update cycles. A package published by Mozilla ships Firefox; an Integration Package published by the community ships the permission profile and Settings adapter for Firefox.

Package format. An Integration Package declares which application it targets and what it provides:

```
[package]
id = "community.firefox.integration"
name = "Firefox␣Integration"
version = "2.1.0"
conflicts = ["community.firefox.privacy"]

[app]
```

```
# At least one identifier required; all three are optional
store_id = "org.mozilla.firefox"
flatpak_id = "org.mozilla.Firefox"
package_name = "firefox"

# Compatible with which app versions
compatible_with = ">=120.0, <130.0"

[includes]
permissions = "permissions.toml" # optional
adapter = "firefox-adapter.toml" # optional
```

Listing 37 Integration Package manifest

The conflicts field lists Integration Packages that are incompatible with this one. When the Store detects a conflict at install time, it presents the user with a choice rather than silently merging or overriding. If both packages are installed simultaneously (for example via a package manager), the Settings page for the affected application shows a conflict warning and the permissions of the first-installed package remain active until the user resolves the conflict.

Settings Adapter. The Settings Adapter is a declarative TOML document that describes how a third-party application's native configuration file maps to Settings UI elements. It contains no executable code; Settings interprets the declaration and performs all reads and writes itself.

```
[adapter]
schema_version = "1.0"
write_strategy = "requires_app_closed" # or "anytime"

[sources]
prefs = {
    path = "~/.mozilla/firefox/*/prefs.js",
    format = "firefox-prefs",
    instance_strategy = "last_used" # or "all", "ask"
}

[[settings]]
key = "browser.startup.homepage"
source = "prefs"
label = "Homepage"
type = "string"
default = "about:home"
section = "General"
verify = true

[[settings]]
key = "browser.tabs.tabMinWidth"
source = "prefs"
label = "Minimum tab width"
type = "integer"
min = 50
max = 250
```

```
default = 76
section = "Tabs"
readonly = false

[[settings]]
key = "browser.cache.disk.capacity"
source = "prefs"
label = "Disk␣cache␣size␣(internal)"
type = "integer"
section = "Advanced"
readonly = true
```

Listing 38 Settings Adapter declaration

write_strategy controls when Settings may write to the configuration file. anytime writes immediately; requires_app_closed disables all edit controls while the application is running and writes only when it is not, because some applications overwrite their configuration on exit and would discard changes made while running.

instance_strategy applies when the source path glob matches multiple files. last_used selects the file with the most recent modification timestamp, appropriate for applications that maintain separate profiles. all writes to every matched file simultaneously. ask prompts the user once on the first opening of the Settings page and remembers the selection.

verify causes Settings to read the field back after writing and compare it to the value it attempted to write. If the values do not match, Settings displays an error rather than silently succeeding. This catches cases where an application has renamed or removed a configuration key in a newer version.

readonly fields are displayed but not editable. They carry a lock indicator in the Settings UI.

The adapter supports the following configuration formats via built-in format handlers: TOML, JSON, INI and .conf files, Firefox prefs.js, .env key-value files, and flat key=value files. Binary and proprietary formats are not supported; the application page in Settings shows only the Permissions section when no compatible adapter is available.

Adapter source files are restricted to paths within known user configuration directories. An adapter cannot declare a source outside ~/.config/, ~/.mozilla/, ~/.local/, and similar user-owned paths. System paths are never writable through an adapter.

Compatibility and updates. The compatible_with field declares which application versions the Integration Package supports. When the application updates beyond the declared range, Settings displays a compatibility warning on the application page with a direct link to the Store listing of the Integration Package. The adapter remains active – most configuration keys do not change between application versions – but the warning signals that the package maintainer should review and update the package.

Integration Package updates are independent of application updates. The Install Daemon (Section 7.5) monitors both streams and keeps them consistent.

Native app Settings registration. First-party applications built with os-sdk do not use the adapter mechanism. They register their Settings interface directly at startup via os-sdk, using the same TOML configuration system the application already uses internally. There is no configuration file bridge and no format handler involved:

```
shell.settings.register({
  sections: [
    {
      label: "General",
      items: [
        {
          key: "startup.restore_session",
          label: "Restore␣last␣session␣on␣startup",
          type: "boolean",
          default: true,
        },
        {
          key: "editor.font_size",
          label: "Font␣size",
          type: "integer",
          min: 8,
          max: 32,
          default: 14,
        },
        {
          key: "editor.theme",
          label: "Editor␣color␣scheme",
          type: "enum",
          options: ["system", "dark", "light"],
          default: "system",
        },
      ]
    }
  ]
})
```

Listing 39 Native Settings registration via os-sdk (TypeScript)

Settings reads this registration and renders the application's configuration section automatically. Values are read from and written to the application's own ~/.config/<app-id>/config.toml via the os-sdk configuration system. The application observes changes via the same inotify mechanism all other components use. No adapter, no format handler, no intermediate layer.

The distinction between the two paths is clean: the adapter is for existing third-party applications that the system does not control; native registration is for applications built for this platform that already use os-sdk.

7.5 Install Daemon

The Install Daemon is a background service that sits as an abstraction layer over all package managers active on the system. Its role is to ensure that every application

installation, regardless of how it was performed, results in a consistent system state: permissions enforced, Integration Package loaded if available, Settings page populated.

A user who types dnf install firefox in a terminal has the same expectation as a user who finds Firefox in the Store: the application works, its permissions are set to sensible defaults, and its Settings page is ready. The Install Daemon makes that expectation true regardless of installation path.

Package manager coverage. The daemon registers hooks with all package managers it finds on the system. For systemd-integrated package managers (dnf, Flatpak), it uses systemd unit hooks that emit an Event Bus message after every install, update, or removal. For non-systemd package managers (Homebrew, Nix, Cargo, pip with –user), it monitors known installation paths via inotify:

```
~/.homebrew/ ~/.nix-profile/
~/.cargo/bin/ ~/.local/bin/
/opt/homebrew/
```

Listing 40 Monitored installation paths

New package managers installed after first login are detected when their binary appears in a known package manager location. From that point the daemon begins monitoring their output paths as well.

When the daemon detects a new application it cannot associate with a known app-ID, it takes no action. Only positively identified applications trigger Integration Package lookup.

Integration Package matching. After detecting a new application installation, the daemon queries the Store catalogue for Integration Packages that reference the installed app-ID via any of the three reference fields (store_id, flatpak_id, package_name). Matching packages are evaluated against a three-tier rating policy:

Table 15 Install Daemon Integration Package rating policy

Condition	Behavior
Rating $\geq$ 4.0 and installs $\geq$ 1,000	Installed automatically. Brief notification after completion.
Rating 3.0–3.9 or installs $<$ 1,000	Notification with Install button. Not automatic.
Rating $<$ 3.0 or installs $<$ 100	Not surfaced. Available in the Store via manual search.

If only one Integration Package exists for an application and it does not meet the automatic threshold, the minimum rating for automatic installation is lowered to 3.5. No Integration Package is worse than a completely absent one from a user experience perspective.

Dual installation handling. When the same application exists as both an RPM and a Flatpak, the daemon treats them as two separate entries. Settings shows both

with distinct labels (for example “Firefox (System)” and “Firefox (Flatpak)”) and separate application pages. The user designates one as primary in Settings; the primary installation is used for default application associations and Waypointer suggestions.

Offline queue. When an application is installed without network access, the daemon stores a pending lookup entry locally. The next time the network becomes available – detected via a NetworkManager event on the Event Bus – the daemon processes queued entries and applies the normal matching policy. Queue entries expire after 30 days without resolution.

Update compatibility. Application updates are never blocked by Integration Package compatibility status. When an application updates beyond an Integration Package’s declared `compatible_with` range, the daemon checks the Store for a newer version of the Integration Package. If one exists it is installed automatically. If none exists, Settings displays a compatibility warning. The application update proceeds regardless.

Deinstallation. When an application is removed, its Integration Package is set to inactive rather than uninstalled. It remains visible in a dedicated “Inactive Integrations” section of Settings. If the application is reinstalled, the Integration Package reactivates automatically without prompting. Explicit removal of the Integration Package is a separate user action available from the inactive list or the Store.

Audit log. Every Install Daemon action is recorded in the Audit Log via the Audit Daemon: automatic Integration Package installations, manual installations triggered by notification, deactivations, reactivations, and failed queue resolutions. Each entry carries the app-ID, Integration Package ID and version, trigger source, rating and install count at decision time, and whether the action was automatic or user-initiated.

The system is extensible at multiple layers through a unified module mechanism. All module types use the same installation channel (the Store) and the same manifest format, but they operate as independent subsystems with separate APIs. A single Store package can bundle multiple module types together.

Module categories. Table 16 lists the available extension points. Theme modules are a special case: they consist of CSS and token overrides only, contain no executable code, and require no sandbox. All other categories run sandboxed with declared permissions.

Manifest format. Every module package has a single manifest that declares what it provides, its dependencies, conflicts, and permissions. The `[permissions]` block uses the same schema as application permission files; at install time the Module Runtime generates the permission file from the manifest and the user can inspect and adjust it in Settings like any other application.

```
[package]
id = "com.example.ai-waypointer"
name = "AI␣Waypointer␣Provider"
```

Table 16 Module categories and extension points

Category	Extension point	Example
theme	Design language layer	Community color schemes
waypointer.provider	Waypointer search results	Web search, stock prices
waypointer.action	Waypointer command actions	Create task, send message
shell.topbar	Topbar element slot	Custom status widget
shell.widget	Desktop widget area	Clock, weather, system stats
mcp.server	AI MCP integration	External service connector
keybinding.profile	Global input manager	Alternative layout profiles

```
version = "1.0.0"

[provides]
waypointer.provider = true

[activation]
# Optional: only run this module when a focused project carries these tags.
# The Module Runtime starts and stops the module as Focus Mode changes.
# Omit this block for modules that run unconditionally.
active_when_focused = { require_tags = ["rust"] }

[dependencies]
shell = ">=1.2"
modules = ["com.example.ai-core"]

[permissions]
network.allow = ["api.openai.com"]
graph.allow = ["read"]
```

Listing 41 Example module manifest

The [activation] block is optional. When present, the Module Runtime evaluates it against the active Project node each time Focus Mode changes. A module with require_tags = ["rust"] is started when the focused project carries the rust tag and stopped when Focus Mode deactivates or switches to a project that does not. The module's declared permissions apply for the duration it is running; they are not expanded or restricted by the activation condition. A module started via project activation follows the same crash recovery policy as any other module.

Modules cannot declare permissions beyond what the manifest specifies, and manifests are reviewed during Store submission. Unlike applications, modules cannot expand their permissions at runtime through a learning mode; their profile is set at install time and any undeclared access returns EACCES.

Module Runtime. A dedicated Module Runtime daemon manages lifecycle, sandboxing, and crash recovery for all active modules. Each module runs in its own isolated WebView context, separate from the shell's WebView. A module crash does not affect the shell or other modules.

The runtime uses exponential backoff per module instance for crash recovery, shown in Table 17. The 60-second window resets on a clean run, so a module that runs for several minutes and then crashes once gets an immediate restart.

Table 17 Module crash recovery policy

Crash count (within 60 s)	Behavior
1st	Immediate restart
2nd	Restart after 5 seconds
3rd	Restart after 30 seconds
4th or more	Marked as failed; no further automatic retries

When a module is marked failed, Settings shows a subtle error indicator with the last crash log. The user can trigger a manual retry at any time. There is no notification spam.

Module updates are downloaded in the background and applied at different points depending on category: theme updates apply immediately as a token and CSS swap; Waypointer modules update on the next Waypointer open; shell widgets and topbar modules update on the next session start by default, with a manual "Restart now" option available in Settings; MCP server modules update on the next session start to avoid interrupting active AI interactions.

Shell API. Modules interact with the system through a restricted TypeScript SDK. No direct access to the compositor or Graph Daemon is available; what is not in the SDK is not accessible.

```
import { shell } from "@os/module-sdk"

// Window management
const windows = await shell.getWindows()
const active = await shell.getActiveWindow()

// Graph (requires graph = "read" in manifest)
const recent = await shell.graph.query(
  "MATCH␣(f:File)␣RETURN␣f␣LIMIT␣5"
)

// Waypointer provider registration
shell.waypointer.register({
  id: "com.example.ai-waypointer",
  placeholder: "Ask␣AI...",
  onQuery: async (q) => [ /* result objects */ ]
})

// Focus Mode: react to project context changes
shell.focus.onActivate((project) => {
  // project.name, project.tags, project.rootPath available
})
shell.focus.onDeactivate(() => { /* cleanup */ })
```

```

// Badges (always available, error/warning types logged to graph)
shell.badges.set({ count: 3 })
shell.badges.clear()

// Ambient effects (respects user's global ambient preference)
shell.ambient.set({ effect: "pulse", intensity: 0.2, reason: "syncing" })
shell.ambient.clear()
```

Listing 42 Module SDK surface (TypeScript)

The shell.focus API allows modules to react to Focus Mode transitions without polling. A Waypointer provider that wants to narrow its result set to the active project subscribes to onActivate and adjusts its query scope accordingly; onDeactivate signals that the module should return to its default behavior. These callbacks fire after all system-level Focus Mode changes have been applied, so the project context is already consistent when the module receives the event.

Global Input Manager. Keybindings and gestures are configured in a single place in Settings, not per application or per module. Modules register named actions with default bindings; the user overrides them centrally. The module never receives the binding string directly; the Input Manager calls the registered action handler when the binding fires. This means a user can rebind everything from one place and no module can silently capture input outside its declared actions.

Developer experience. A CLI scaffolds a complete module project with one command: npx create-os-module <n>, producing a Vite, TypeScript, Tailwind, and module-sdk setup with a manifest template. In dev mode, the module runs unsigned in a dedicated dev sandbox, all declared permissions are granted without Store review, WebView DevTools are accessible, and hot-reload via Vite is active. Dev mode is only available on developer builds; production builds reject unsigned modules.

For distributing a finished module outside the Store, a developer generates a self-signed key pair with the CLI, signs the module, and distributes it as an .osmod file. Installation uses TOFU: the first install prompts the user to trust the key fingerprint; subsequent updates from the same key are silent. Sideloaded modules receive no expanded permissions and are marked “Not from Store” in Settings.

7.6 Store

The Store is the primary distribution channel for applications, themes, modules, and profile packages. It is designed to be financially sustainable for the project without becoming a gatekeeper for the ecosystem.

Publishing. Publishing an application to the Store is free and remains free indefinitely. There is no annual developer fee. Every upload runs automated technical checks: valid permission declarations, correct package signing, sandbox policy compatibility, and resolvable declared dependencies. These checks are automated; no manual review is

required for packages that pass them. A failed check returns a specific error. A manual Security Review is available optionally and is the basis for the Security Audit Badge described below.

Revenue model. Supported pricing models include free, one-time purchase, subscription, in-app purchases, pay-what-you-want, and freemium. Developers can also direct users to external payment links; the Store payment infrastructure is a convenience, not a monopoly.

Commission is calculated on total annual revenue across all applications in a developer account, using the tiered structure in Table 18. When a threshold is crossed mid-year, the new rate applies to all subsequent transactions in that year, not retroactively.

Table 18 Store commission tiers

Annual revenue (account total)	Commission
Up to 10,000€	0%
10,000€ to 50,000€	5%
50,000€ to 500,000€	15%
500,000€ to 5,000,000€	20%
Over 5,000,000€	25%

User-facing features. Purchases are tied to the user account, not the device. Trials are declared in the app manifest and enforced by the Store; no credit card is required to start a trial, and when it ends the user receives a single notification with a purchase option and no automatic charges. Family Sharing is opt-in per application: developers can allow up to six to eight family members to use an application under one purchase. Every application's Store page shows a full price history including every price change with its date; this is automatic and cannot be disabled by the developer. All active subscriptions across all applications are visible and cancellable in one place in Settings.

Developer dashboard. Developers see install counts, active user numbers, revenue and payout history, ratings and written reviews, crash reports from users who opted into crash reporting at the system level, and version adoption numbers. All data is aggregated and anonymized; the developer sees counts, never identities. No third-party analytics are required or supported.

Security Audit Badge. Applications can display a Security Audit badge on their Store page by uploading an independent security audit report. The Store verifies that the report originates from the stated auditing organization and links it from the application page. Users can read the full report and judge the auditing organization's credibility themselves. The badge means an audit exists and is accessible, not that the Store has endorsed the application's security.

Long-term support. Developers can declare a Long Term Support commitment in the application manifest by setting a support end date. The Store displays this date publicly on the application page. It is a voluntary public commitment with no automated enforcement, but the public visibility creates accountability. LTS applications are filterable in Store search for users who prioritize stability.

Profile package distribution. Signed environment profile packages (.lenv files, described in Section 8) are distributed through the Store using the same infrastructure as applications. Profile packages are free to publish and carry no commission.

Integration Package distribution. Integration Packages (described in Section 7.4) are distributed through the Store as a distinct package type. They are free to publish and carry no commission. Publishing runs the same automated technical checks as applications: valid signing, resolvable app references, and schema validation of any included adapter declaration. A manual Security Review is available and recommended for packages that include permission profiles, since an overly permissive profile represents a meaningful risk.

Store listings for Integration Packages display the target application, the package components included (permissions, adapter, or both), the compatible_with version range, and the standard rating and install count. The Install Daemon uses rating and install count to determine automatic installation eligibility as described in Section 7.5.

Deep links between Integration Package Store listings and the corresponding application page in Settings use the store:// and settings:// URL schemas described in Section 6.9.

7.7 MCP Safety and External Connections

The MCP Server Layer enables powerful integrations between the AI daemon, first-party applications, and external tools such as n8n and OpenClaw. Two mechanisms exist to prevent this power from causing runaway or uncontrolled behavior: a call depth limit for MCP chains, and a controlled outbound webhook surface for the Event Bus.

Call depth limit. Every MCP call chain carries a call_chain_id and a depth counter. When a component initiates an MCP call the depth starts at one. If that call triggers a further MCP call, the depth increments and the chain identifier is forwarded. A call that would exceed depth five is rejected immediately with the error mcp_depth_exceeded rather than executing. The full call chain is written to the Audit Log on rejection so the user can diagnose what triggered the loop.

The default limit of five is intentionally conservative. Legitimate nested MCP calls rarely exceed two or three levels. Users who build complex multi-step workflows can raise the limit to a maximum of ten in Settings under AI; no configuration can exceed ten regardless of how it is set. The limit applies globally across all MCP callers including the AI daemon, modules, and external connections.

Outbound Event Bus webhooks. The Event Bus can forward selected events to external endpoints configured as Connections in Settings (Section 6.9). This enables tools like n8n to react to desktop events without polling. The mechanism is intentionally restrictive to prevent uncontrolled data leakage.

Outbound webhook endpoints are bound to a Connection. There is no mechanism for arbitrary URL registration outside of a Connection; a module or a .project file cannot register a webhook endpoint on its own. The Event Bus receives its permitted endpoint whitelist directly from the Account Daemon at startup and after any Connection change. Any event targeting an endpoint not on the whitelist is silently discarded without error or retry.

The user selects which event types to forward per Connection from a fixed list derived from the Event Bus schema. Raw eBPF events are not available as outbound event types regardless of configuration; only semantic events such as focus.activated, timeline.record, and issue.status_changed can be forwarded. This prevents low-level system telemetry from leaking to external services even if the user misconfigures a connection.

Outbound events are rate-limited to 60 per minute per Connection. When the limit is reached the remaining events in that minute are dropped, not queued, and a warning badge appears on the Connection in Settings. The limit resets at the start of the next minute. This prevents high-frequency desktop activity from generating uncontrolled external traffic volume.

8 Platform, Profiles, and Security

8.1 Account System

The Account System is the single place where the user manages all online accounts and external service connections. Set up once, available everywhere, with full visibility into which application accesses what. The design principle is straightforward: no application ever directly handles OAuth flows, stores tokens, or manages credentials. All credential logic lives in a dedicated *Account Daemon*.

The daemon serves two related but distinct purposes. First, it manages user-facing online accounts such as Google, Microsoft, and Nextcloud, including OAuth flows, token refresh, and the GNOME Online Accounts bridge. Second, it manages the credentials for external service Connections configured in Settings (Section 6.9): GitHub, GitLab, n8n, OpenClaw, and any other Connection the user creates. Both credential types live in the Linux Kernel Keyring and are accessed exclusively through the daemon. No other component reads credentials from the Keyring directly.

OAuth 2.0 [54] is the standard authorization framework used by cloud services to grant third-party applications access to user data without sharing the user's password. The flow works by redirecting the user to the service's login page, where they authenticate and approve a specific set of *scopes* (declared permissions such as "read email" or "write calendar"), after which the service issues a short-lived *access token* and a long-lived *refresh token*. The access token is sent with each API request; the refresh

token is used to obtain new access tokens when the current one expires. An application that holds both tokens has persistent access to the user's account; protecting them is therefore critical.

Architecture. Applications declare what they need; the daemon handles the rest. When an application needs access to a service, it sends a request to the Account Daemon, which checks the application's permission profile, retrieves or derives the appropriate token, logs the access in the Audit Log, and returns what the application needs. The application then communicates with the provider's API directly using that token. The daemon is the sole keeper of all credentials; tokens never live in application memory longer than necessary.

Service-level permissions. Applications request access at the service level, not the OAuth scope level. The user sees human-readable service names and never raw OAuth scope strings. When an application requests account access for the first time, the user sees exactly which service is being requested, grants or denies it, and the decision is stored in the application's unified permission profile as described in Section 7.3. Account permissions are part of the same system as filesystem, network, and graph permissions.

```
Google Account (personal@example.com)
  Mail -> allowed for: Thunderbird
  Calendar -> allowed for: Calendar App, AI daemon (read-only)
  Drive -> allowed for: File Manager
  Contacts -> not granted to anyone
```

Listing 43 Service-level permission view (Settings)

Multiple accounts and resolution order. When multiple accounts of the same type exist, the system resolves which to use via a three-tier model. The system default applies globally; an Isolated Desktop context can declare its own defaults, so that switching to the Work desktop automatically switches the active account context; and individual applications can be manually pointed at a specific account regardless of the desktop context. Resolution proceeds from most specific to least specific: per-app override, then desktop default, then system default.

```
# resolved in order: per-app override -> desktop default -> system default
[defaults]
google = "personal@example.com"

[desktop.work.defaults]
google = "work@example.com"

[app."org.mozilla.thunderbird"]
# always uses personal account regardless of active desktop
google = "personal@example.com"
```

Listing 44 Account resolution configuration

Token security. Credentials never leave the Account Daemon as-is. Two paths are used depending on provider support. For providers that support OAuth scope restriction, the daemon issues a short-lived derived token scoped only to what the application declared, with a default lifetime of five minutes. The full OAuth token and refresh token never leave the daemon at all. For providers that do not support downscoping, the full token is handed to the application but with two compensating controls: every token handoff is logged in the Audit Log, and tokens are actively revoked when the application process exits. All credentials at rest are stored in the Linux Kernel Keyring [12]: a kernel-managed storage area for cryptographic keys, tokens, and passwords that lives in kernel memory rather than userspace files, making it inaccessible to other processes and not visible in `/proc` or on disk. The daemon loads them into memory only when actively needed and discards them immediately after.

Offline behavior. Two failure modes are handled distinctly. When the provider rejects the token because the user revoked access or the account password changed, the daemon emits a single system notification directing the user to Settings and does not repeat it; applications receive a clear `AccountAuthRequired` error code and surface a gentle in-app hint. When the network itself is unavailable, the daemon serves cached data from the last successful sync, clearly marked with the timestamp of the last update. A calendar application shows last-known events labeled with their age rather than an empty state. When connectivity is restored, the daemon refreshes silently in the background and notifies applications of updated data via the Event Bus.

Connection credential management. External service Connections configured in Settings are stored as named entries in the Linux Kernel Keyring under the `lunaris/connections/` namespace. The Account Daemon is the sole process that reads and writes these entries. When a component requests the credential for a Connection it provides the Connection identifier and its own component identity; the daemon checks whether the requesting component is listed in the Connection's `allowed` field before returning anything. Unauthorized requests receive `EACCES` with no credential data.

Token refresh for Connections follows a two-path model. For providers that issue refresh tokens alongside access tokens, the daemon attempts a silent refresh in the background when the access token approaches expiry. If the refresh succeeds the new token is stored in the Keyring and the Connection continues working without interruption. If the refresh fails, or if the provider does not issue refresh tokens at all, the daemon marks the Connection as `expired`, returns `token_expired` to requesting components rather than a stale or invalid token, and sends a single High-priority notification to the user with a deep link to `settings://connections/<id>`. The notification is sent once and not repeated until the user interacts with it or the Connection is renewed.

The daemon also maintains the outbound webhook endpoint whitelist for the Event Bus. Whenever a Connection with outbound events is created, modified, or deleted, the daemon pushes the updated whitelist to the Event Bus over a dedicated internal channel. The Event Bus never queries the daemon at send time; it uses the cached whitelist to avoid any latency in the hot event path.

GNOME Online Accounts bridge. *GNOME Online Accounts* [48] (GOA) is a GNOME framework that centralizes cloud service credentials (Google, Microsoft, Nextcloud, and others) and exposes them to applications via a D-Bus interface. Applications built for GNOME that use GOA expect a D-Bus interface at org.gnome.OnlineAccounts. The system provides a GOA-compatible bridge that implements this interface exactly but sources all data from the Account Daemon internally. Evolution, GNOME Calendar, and other GOA-dependent applications work without modification. Internally every request goes through the Account Daemon, through the permission system, and through the Audit Log. The bridge translates; it does not add logic.

8.2 Profile System

A profile is a complete, isolated user environment backed by a separate Linux system user. Each profile has its own home directory, its own Knowledge Graph, its own account configuration, and its own set of running system daemons. Isolation is enforced at the kernel level, not by convention.

The profile system replaces the concept of isolated desktops within a session. Rather than trying to isolate individual desktops within a single session, the entire session is the isolation boundary. Switching profiles is a session switch, not a window toggle.

Personal profiles. The primary profile, created at system setup, cannot be deleted. It has no restrictions: full access to all installed applications, all accounts, and all system features. This is the default context the user logs into.

Work and custom profiles. A user-created or organization-deployed profile for a specific context such as work, a project, or a secondary identity. Backed by a separate Linux user with its own encrypted home image via systemd-homed. Virtual desktops, accounts, Knowledge Graph, and application configurations are all independent from the Personal profile. Multiple custom profiles can coexist. Each is a full Linux user; switching to one is a standard session switch with saved and restored window state.

Locked profiles. A profile deployed by an institution for controlled sessions: exams, regulated work environments. Backed by a temporary Linux user created at session start and deleted when the session ends. The user cannot switch to another profile while a Locked session is active; this is enforced at the compositor level, not by UI alone. After the session ends, the Linux user and all associated data are removed. No data from the Locked session survives into the Personal profile.

Guest mode. A lightweight temporary context for short-term use by another person. Unlike other profile types, Guest does not create a second Linux user or a second compositor session. It runs as a sandboxed environment on top of the current session: a separate namespace, an empty home directory, no access to the host user's data. The host session is suspended in the background and resumed instantly when Guest ends. All Guest data is discarded on exit.

Technical foundation. Each non-Guest profile is a Linux system user. Profile creation is useradd plus systemd-homed home image setup; profile switching is a standard Linux user session switch; profile deletion is userdel plus home image removal. Isolation is kernel-enforced. systemd-homed provides per-profile LUKS-encrypted home images. When a profile is not active, its home image is unmounted and its encryption key is discarded from memory. Each profile runs its own instance of all system daemons: Graph Daemon, Account Daemon, Event Bus. There is no shared daemon state between profiles.

Data transfer between profiles. By default profiles are fully sealed from each other. Controlled data transfer is configured per profile pair in Settings, with independent directional control over clipboard, drag-and-drop, and a shared folder:

```
[profile.work.transfer]
clipboard = "in-only" # "off" | "in-only" | "out-only" | "both"
drag_drop = "in-only"
shared_folder = "~/Transfer/Work"
shared_folder_permission = "read-write" # or "read-only" | "off"
```

Listing 45 Cross-profile transfer configuration

A dedicated *Transfer Daemon* is the only process with simultaneous access to both profile namespaces. All cross-profile data flow goes through it; nothing bypasses it. Whenever data flows out of a profile, a non-dismissable toast appears for three seconds identifying the source profile, destination, and file name, with an Undo option. The transfer proceeds but can be cancelled within the window. All transfers are recorded in the Audit Log with source, destination, transfer type, and data description.

Locked profiles have transfer hardcoded to off in all directions. This cannot be overridden by the user or by an organization's configuration package; it is a system-level invariant with no configuration surface.

Signed profile packages. Profiles can be deployed via a *Signed Profile Package*, a .lenv file created and signed by an organization. The user installs it themselves and retains full control over their device. The organization gets a tamper-evident, versioned configuration. This is the BYOD model: the organization controls their context, the user controls their device.

```
[environment]
id = "com.acme.work-profile"
name = "Acme␣Work"
version = "2.1.0"
publisher = "Acme␣GmbH"
type = "persistent" # or "locked" for exam/controlled sessions

[transfer]
clipboard = "in-only"
drag_drop = "in-only"
shared_folder = "off"

[network]
```

```
require_vpn = "acme-vpn"
allow_outbound = ["*.acme.com", "*.office365.com"]
block_outbound = ["*"]

[apps]
required = ["com.acme.timetracker", "org.mozilla.firefox"]
blocked = ["com.valve.steam"]

[graph]
ai_access = "off"
export = "off"

# Project templates bundled with this profile.
# Each entry is a path to a .project file relative to the package root.
# These are written to the profile's home directory on install and
# registered as active projects in the Knowledge Graph.
[projects]
templates = [
    "projects/acme-onboarding.project",
    "projects/acme-infrastructure.project",
]
```

Listing 46 Signed profile package format

Project templates bundled in the [projects] block are installed into the profile's home directory at ~/.projects/<template-name>/ on first login and registered as active Project nodes in the profile's Knowledge Graph. Each template is a standard .project TOML file (Section 4.7) and follows the same format; the organization can pre-configure name, description, tags, linked accounts, AI scope, and Focus Mode rules. Project templates do not contain user data and are not synchronized back to the organization; they are a starting configuration only. The user can modify or delete them after installation without affecting the profile package itself.

The package is signed by the organization's private key. Key distribution uses TOFU by default: on first install the user sees the publisher name and key fingerprint and confirms trust. For managed onboarding, IT can send an enrollment link so that the organization's key is imported before any package arrives, eliminating the TOFU prompt entirely.

Before installation, a full summary of every policy and restriction is shown. Nothing is hidden. There are certain things an organization cannot do through a signed package, enforced technically rather than by policy: the organization cannot read the user's Personal profile or its Knowledge Graph, cannot access any data outside the profile's Linux user boundary, cannot monitor activity in other profiles, cannot remotely wipe or modify the device, and cannot prevent the user from uninstalling the package. If notify_on_uninstall was declared and accepted at install time, the organization receives a notification when the package is removed, but the removal proceeds regardless.

8.3 Gaming and Windows Compatibility

Gaming and Windows application compatibility are first-class features, not afterthoughts. The required components are pre-installed and pre-configured; no manual setup is required.

Wine and Proton. *Wine* provides compatibility for Windows applications on Linux. *Proton* [117] is Valve's fork of Wine, optimized for gaming and maintained with extensive patches from Valve's Linux gaming team; for gaming specifically it is substantially better than vanilla Wine. Both are pre-installed alongside *DXVK* [32], which translates Direct3D 9, 10, and 11 API calls to Vulkan. Direct3D is the Windows graphics API; most games targeting Windows use it rather than Vulkan directly. By translating at the API boundary, DXVK allows these games to run on Linux via the Vulkan driver. *vkd3d-proton* [118] is Valve's fork of the Wine project's vkd3d library; it translates Direct3D 12 (the modern Windows GPU API) to Vulkan, with extensive optimizations for gaming workloads that upstream vkd3d does not include. Winetricks is included for installing Windows runtime components into Wine prefixes.

Steam is available in the Store and works out of the box with Proton enabled. Steam manages its own Proton version per game; the system does not interfere with Steam's internal Proton management.

Non-Steam Windows games and applications use Wine directly, managed through the first-party *Wine Manager* application. The underlying components are shared with Steam's installation rather than duplicated. Wine Manager is a thin UI over the existing tooling providing prefix creation and deletion, per-prefix Winetricks access, launch shortcuts, and per-prefix Proton and Wine version selection.

DXVK and vkd3d-proton require Vulkan. Mesa provides Vulkan support for AMD and Intel GPUs by default. Nvidia requires the proprietary driver for Vulkan support; the Store provides explicit guidance for Nvidia users during setup.

Wine theming. By default Wine applications render with a Windows Classic or Windows 7-era appearance that is visually inconsistent with a modern desktop. A first-party Wine theme is maintained as part of this project, generated from the same design token system used for GTK and Qt theming. It updates automatically when the system theme changes. Wine applications do not look native, but they look consistent with the rest of the system. The Wine theme is applied automatically to all Wine prefixes managed by the system; users who prefer the default Windows appearance can opt out per prefix.

Performance tooling. *Gamemode* [39] by Feral Interactive is pre-installed and enabled by default. It applies system-level performance optimizations when a game is running: the CPU governor switches to performance mode, the I/O scheduler is tuned, and kernel scheduler hints are applied. Proton handles the Gamemode activation call automatically; no manual configuration is required.

MangoHud [76] is pre-installed as an in-game overlay showing frames per second, GPU and CPU usage, temperatures, and frame timing. It is disabled by default and toggled via a keyboard shortcut.

Knowledge Graph integration. Gaming activity is tracked in the Knowledge Graph like any other activity: which games were played, when, and for how long; which Wine and Proton versions were used per game; and performance data from MangoHud when enabled, linked to sessions. This makes queries such as ”how many hours did I play last week” or ”which Proton version worked best for this game” available locally without relying on any external service.

8.4 Security and Privacy

Security is not a feature added on top of this system. It is a foundational design constraint that shapes every architectural decision.

Core principles. Five non-negotiables drive everything in this section. First, the Knowledge Graph, Audit Log, and AI context are local by default and never leave the machine without explicit user action; there is no opt-out dark pattern, the default is that nothing goes anywhere. Second, the system collects no usage data, crash reports, or any other information without explicit user consent. Third, the system works fully offline and without registration; cloud features are opt-in. Fourth, every component gets exactly the permissions it needs and nothing more, applying to filesystem access, network access, graph access, and syscalls alike. Fifth, security must not destroy usability: defaults are chosen to protect without requiring constant interaction, because a system so locked down that users disable its protections is less secure than a well-calibrated one.

A sixth principle follows from the others: data minimization is itself a security property. Data that does not exist cannot be stolen. The retention policy defined in Section 4.6 is not only a privacy decision but an active security measure. Every decision to extend retention windows is a deliberate expansion of attack surface and must be understood as such.

8.4.1 Threat Model

Security measures without a named adversary are hard to evaluate. This section states explicitly what this system is and is not designed to resist, so that the measures described in the sections that follow can be judged against a concrete baseline.

In scope. The primary adversary is a *local attacker with user-level privileges*: a malicious or compromised application running as the logged-in user that attempts to exceed its declared permissions, exfiltrate graph data, escalate privileges, or persist across sessions. The secondary adversary is an *opportunistic remote attacker* who exploits a network-reachable vulnerability in a running service or a browser-class application. The tertiary adversary is a *physical attacker* with brief unsupervised access to a powered-off or locked device. Managed-environment scenarios add a fourth adversary:

a *malicious or over-reaching administrator* who attempts to access user data beyond the boundaries described in Section 8.5.

Out of scope. This system does not attempt to resist a *root-level attacker*: an adversary who already holds superuser privileges on the running system. Root can modify kernel modules, bypass all userspace controls, and read any unencrypted memory. Full Disk Encryption and TPM-sealed keys protect data at rest against root on a different boot, not against root in the current session. *State-level adversaries* with hardware implant capabilities, supply-chain compromise of the base distribution, or CPU microcode vulnerabilities are similarly out of scope; no single OS can address these without hardware-level attestation infrastructure outside the scope of this project.

Knowledge Graph as a high-value target. The Knowledge Graph introduces a risk that does not exist on a conventional Linux system: a structured, queryable, cross-application timeline of user activity. On a conventional system, an attacker who gains access to a user's files reads unindexed data spread across many directories. An attacker who gains access to the running Graph Daemon gains a pre-indexed, graph-traversable record of everything the user has done. This is structurally more valuable than a raw filesystem breach, and the security model of this system must account for it explicitly.

The measures described in Section 8.4.5 reduce the probability of such a breach. Section 8.4.5 addresses what happens when prevention fails.

8.4.2 Full Disk Encryption

Full Disk Encryption means that everything stored on disk is encrypted at rest. On Linux this is implemented via *LUKS* on top of the *dm-crypt* kernel module. The entire partition is encrypted and the OS decrypts it transparently at boot after credentials are provided.

FDE is enabled by default. A stolen or lost device is a realistic threat for any user; without FDE, physical access means full access to all data. The performance concern that historically argued against FDE no longer applies: all modern x86 and ARM processors have hardware AES acceleration [61] that makes encryption effectively free at current workloads.

The primary unlock mechanism is a passphrase entered at boot. On systems with a *TPM* (Trusted Platform Module) [19], a dedicated security chip on the motherboard (or firmware-emulated since TPM 2.0) that provides hardware-backed cryptographic operations, secure key storage, and tamper-evident measurements of the boot process. The FDE key can be sealed to the TPM and released automatically if the system boots in a known-good state. This allows passwordless boot on trusted hardware while still protecting the drive if it is physically removed. If the boot configuration changes due to tampering, the TPM refuses to release the key and falls back to the passphrase. TPM-based unlock is opt-in; passphrase is always the fallback.

Hibernate requires additional care because the entire RAM contents are written to the swap partition. That swap partition must also be LUKS-encrypted, and key management must allow the system to resume without manual intervention. This is

a known-solved problem on Linux but requires deliberate configuration during the installation process.

8.4.3 Secure Boot, Unified Kernel Images, and Measured Boot

Secure Boot [20] is a UEFI [115] firmware feature that verifies the cryptographic signature of every piece of code executed during the boot process. If any component is not signed by a trusted key, the firmware refuses to execute it. The practical effect is that an attacker who modifies the bootloader or kernel cannot boot the modified system without also possessing the signing key.

Both candidate base distributions support Secure Boot via a Microsoft-signed shim bootloader. Custom kernel modules built for this project must be signed with a Machine Owner Key enrolled in the firmware.

A traditional Linux boot chain consists of multiple separate files: the bootloader, the kernel image, and the initramfs. A *Unified Kernel Image* [21] packages all three into a single signed EFI binary, reducing the attack surface to a single component requiring verification. `systemd-boot` supports UKI natively and the broader Linux ecosystem is moving in this direction. This project targets UKI as the boot format.

Measured Boot goes further than Secure Boot: every boot component is cryptographically hashed and the hash is stored in the TPM's Platform Configuration Registers. This creates a tamper-evident record of exactly what ran during boot. If any component changes, even legitimately due to an update, the PCR values change. Locally, the TPM can be configured to release the FDE key only if the PCR values match an expected set, so that a modified bootloader causes the TPM to refuse key release. In managed environments, a remote server can request a signed attestation report from the TPM, and a device that fails attestation can be denied network access.

Secure Boot combined with *Measured Boot* makes the Evil Maid Attack significantly harder. An Evil Maid Attack is the scenario where an attacker has brief physical access to a powered-off device and modifies the bootloader to capture the user's FDE passphrase on the next boot. Without the signing key, Secure Boot blocks the modified bootloader from executing entirely.

8.4.4 App Sandboxing

Every application runs in a restricted environment that limits what it can do even if fully compromised. Three mechanisms operate simultaneously and independently; bypassing one does not bypass the others.

Table 19 Sandboxing mechanism layers

Mechanism	Function
Linux Namespaces [13]	Changes what the application can see
AppArmor [6]	Blocks actions on resources the application can see
seccomp [15]	Blocks specific syscalls regardless of resource visibility

Linux Namespaces give each application its own restricted view of the system before any permission check occurs. A Mount Namespace means the application sees only its

own data directory, explicitly granted paths, and a minimal set of system paths; other applications' data directories do not exist from its perspective. A Network Namespace means applications without declared network access see no network interface at all; applications with declared network access see only a filtered virtual interface. A PID Namespace means the application can only see its own process tree.

AppArmor profiles are defined externally in the package and reviewed during Store submission; an application cannot define or modify its own profile. A text editor does not need `ptrace`, `mount`, or `kexec`; a seccomp profile that excludes these syscalls causes the kernel to kill the process immediately if a compromised application attempts them.

IPC is restricted to the Event Bus, the Graph Daemon, and explicitly declared inter-application communication channels. An application cannot open a socket to another application without that channel being declared and approved.

Clipboard hygiene. The clipboard is an underappreciated data-leak vector. On a conventional Wayland desktop, any application with a `wl_data_device` interface can read the current clipboard contents without restriction; a password manager and a text editor have equal access to whatever the user last copied. This system changes that.

Every clipboard entry carries a sensitivity label. Labels are assigned in two ways. Manual labeling is available from the clipboard history UI: the user can mark an entry as sensitive, causing it to be hidden from clipboard history and restricted from low-trust applications. Automatic labeling applies heuristics at copy time: content that matches the pattern of an API key, an IBAN, a private key, a password-manager field, or a credit card number is labeled sensitive automatically without user interaction.

Applications declare a clipboard permission level in their manifest, analogous to filesystem or network permissions. An application without the `clipboard.read.sensitive` permission receives an empty string when it requests clipboard contents that carry a sensitive label. The application is not informed that content exists but was withheld; from its perspective the clipboard is empty. This prevents a compromised application from silently harvesting credentials that were copied from a password manager.

Clipboard entries labeled sensitive are excluded from clipboard history storage entirely and are not written to the Knowledge Graph. They exist only in the compositor's in-memory clipboard buffer and are cleared when the entry is superseded or when the screen locks.

8.4.5 Knowledge Graph Security

The Knowledge Graph is the most sensitive component in the system. It accumulates a detailed record of everything the user does and its access control must be correspondingly rigorous.

No application has direct access to the Ladybug database files. All graph access goes through the *Graph Daemon*, a privileged system process that is the sole reader and writer of the graph. Every other component communicates only with the daemon over a defined IPC interface, and the daemon checks permissions before executing any query.

Three axes of permission. Graph permissions are defined along three independent axes. The first axis is node types: which categories of data the caller can read, such as File, App, NetworkConnection, or Session. A calendar application may read Session nodes but not NetworkConnection nodes. The second axis is relations: which connections between nodes the caller can traverse. An application permitted to read File nodes may still be blocked from traversing the OPENED_BY relation that connects files to the applications that opened them. The third axis is instances: whether the caller can see only the nodes it generated itself or nodes generated by other applications as well.

```
[app."com.example.notes"]
allow_read = ["File.path", "File.last_modified", "Session.started_at"]
deny_read = ["File.content_snippet", "NetworkConnection.*"]
allow_relations = ["PART_OF_SESSION"]
instance_scope = "self" # only nodes this app generated
```

Listing 47 Graph permission entry

Two permission layers. Permissions are assigned in two layers. At install time, the package declares the maximum graph permissions it can ever receive; the user reviews this declaration as part of installation. At runtime, the application can request permissions within its declared maximum, with a user prompt explaining what is requested and why. A runtime request that exceeds the install-time declaration is rejected outright. Some fields within permitted node types are additionally restricted regardless of node-level permissions: file content snippets, for example, require an explicit extra permission separate from general File node access.

Ladybug file protection. The Ladybug database files are protected by three independent layers. At the filesystem layer, the Graph Daemon runs as a dedicated system user graph-daemon and the database files are owned exclusively by this user with 600 permissions. A successful sandbox escape by an application process running as the logged-in user does not grant access to these files. At the AppArmor layer, a separate deny rule in every application's AppArmor profile prohibits access to the graph directory even if filesystem permissions were bypassed. At the encryption layer, the graph directory lives inside a LUKS-encrypted loopback image that the daemon mounts at startup using a key retrieved from the Kernel Keyring. When the daemon stops, cleanly or via crash, the image is unmounted and the key is discarded from memory. During any period when the Graph Daemon is not running, the Ladybug files are LUKS-encrypted ciphertext unreadable even to root.

Cypher injection prevention. The Graph Daemon never accepts query strings with embedded values. All queries must use named parameters, and the daemon runs a validator on every incoming query that detects embedded values before execution.

```
// Rejected - string interpolation
let query = format!("MATCH (f:File {{name: '{}'}}) RETURN f", user_input);

// Required - parameterized
daemon.query(
```

```
    "MATCH (f:File {name: $name}) RETURN f",
    params!{ "name" => user_input }
)
```

Listing 48 Parameterized graph queries

Applications without arbitrary_query permission in their manifest cannot send free-form query strings at all; they select from pre-approved query templates and supply only the parameters. Only explicitly trusted components such as the AI daemon and system daemons receive arbitrary_query permission, and even they go through the parameterization validator.

Capability tokens. Instead of looking up permissions in an ACL table on every query, applications receive a cryptographically signed capability token at startup. The token encodes the application's allowed scopes directly, and the daemon verifies the *HMAC* (Hash-based Message Authentication Code) signature on each query without a database lookup: the verifier recomputes the HMAC and checks it matches without needing to contact any authority. This is faster at high query rates and more robust against a corrupted ACL table. The AI daemon can derive restricted sub-tokens for specific query scopes without involving the daemon. Token revocation when permissions change mid-session is handled via a small in-memory revocation list that is cleared on daemon restart.

Token issuance and the Policy Store. Token issuance is separate from token verification and requires careful design: if the component that signs tokens can be influenced to grant excessive scopes, the entire permission model is undermined.

Tokens are signed by the Account Daemon at application startup. The permitted scopes for each application are not decided by the Account Daemon itself; instead, the daemon reads them from a *Policy Store*, a static TOML file owned by root with 644 permissions (world-readable, root-writeable only). The Policy Store maps application IDs to their maximum permitted graph scopes:

```
[app."com.example.notes"]
graph_read = ["File.path", "File.last_modified", "Session.started_at"]
graph_write = []
instance_scope = "self"
arbitrary_query = false
```

Listing 49 Policy Store entry (read-only by Account Daemon)

The Account Daemon can only issue a token whose scopes are a subset of what the Policy Store declares for that application ID. It cannot grant scopes that are absent from the Policy Store entry. A compromised Account Daemon can sign fraudulent tokens, but only within the ceilings the Policy Store enforces — because the Graph Daemon independently reads the Policy Store and rejects any token whose scopes exceed the registered maximum for that application. Scope expansion requires a root-level file write, which is an observable, auditable action. New applications receive a restrictive default scope at install time; privilege elevation is explicit and logged.

Graph Daemon hardening. The daemon runs with the minimum privileges needed for its function, enforced by a systemd service unit with PrivateNetwork=true, MemoryDenyWriteExecute=true, PrivateTmp=true, and a tightly scoped SystemCallFilter. PrivateNetwork=true prevents a compromised daemon from making outbound connections to exfiltrate data. A systemd watchdog monitors the daemon; an unresponsive or crashed daemon triggers a clean shutdown sequence that unmounts the LUKS image and discards the key before restarting.

Clients verify that they are speaking to the legitimate Graph Daemon before sending queries. At startup, the daemon signs a challenge with a private key that only it holds; clients verify the signature against the public key distributed with the system package. A connection that fails verification is dropped, preventing a trojanized replacement daemon from intercepting graph traffic.

Read/write process split. A single daemon process that both writes graph data and answers queries is a larger attack surface than necessary. The Graph Daemon is therefore split into two cooperating processes: graph-daemon-write accepts events from the Event Bus and writes to Ladybug, and graph-daemon-read answers queries from applications and daemons.

graph-daemon-read opens the Ladybug database in read-only mode. A compromised read process cannot modify graph data. graph-daemon-write does not expose a query interface to external callers; a compromised write process cannot be queried for data it has written. The two processes communicate only via a restricted internal channel to coordinate schema changes and to propagate index updates. Both remain subject to the full systemd sandboxing described above.

This split does not eliminate the risk of a compromised read daemon leaking data it has legitimate access to. It does prevent an attacker who gains code execution in one path from trivially pivoting to the other.

Compromise scenario and damage control. The measures above reduce the probability of a Graph Daemon compromise. They do not make it impossible. An attacker who successfully executes arbitrary code inside graph-daemon-read can issue arbitrary Cypher queries against the open database handle and receive responses over the Unix socket, regardless of the capability token the calling client presented. This is the honest consequence of any architecture where a process must be running and reachable to serve data: the running process is the trust boundary.

Several controls bound what an attacker can do from this position. PrivateNetwork=true means the data cannot be sent over the network from the daemon process itself; exfiltration requires additional steps through another compromised component. The read-only Ladybug handle prevents writing or deleting data. The Anomaly Detector monitors the Audit Log for query volume spikes and unusual node-type access patterns; a compromised daemon issuing a full-graph dump will generate anomaly alerts unless the attacker also compromises the Audit Daemon, which is a separate process. The LUKS-encrypted graph image means the data is protected at rest the moment the daemon stops.

What this system cannot offer is runtime confidentiality against a compromised read daemon with an open database handle. No local architecture that provides queryable history can offer this guarantee: it is a structural property of the threat model, not a gap specific to this design. The correct response is layered mitigation and honest disclosure to the user that the Knowledge Graph, while protected against external breach, relies on the integrity of the Graph Daemon process itself.

Rate limiting and query complexity. Two attack vectors are addressed: denial of service via query flooding, and timing side-channels. Rate limiting is enforced per client identity using a token bucket: normal applications are limited to 100 queries per second with a burst limit of 200; the AI daemon has a higher sustained limit appropriate to its workload; system daemons are unlimited. Exceeding the rate limit generates an event for the Anomaly Detector, and repeated violations trigger an alert.

Query complexity limits prevent expensive traversals from being used as denial of service. Traversal depth is capped at 5 hops by default and never more than 10. Every query must include a `LIMIT` clause; queries without one are rejected before execution. Queries are aborted after 500 milliseconds; the client receives `QueryTimeout`. Response times have a small random noise value added to make statistical timing attacks on the graph impractical.

8.4.6 AI Security

The AI daemon has broader graph access than most applications by necessity, making it a higher-risk component with its own security model.

AI access levels. The user configures a global AI access level as the primary control over how much the AI can see. Table 20 lists the five levels. Action permissions, meaning what the AI can do rather than what it can read, are configured separately. Reading data and taking actions are fundamentally different risk categories and are therefore independently controlled. Suggest only mode means the AI proposes actions and the user manually executes them. Supervised mode means the AI executes actions but shows a preview with a cancellation window before proceeding. Autonomous mode means the AI can act within specific applications without per-action confirmation, and each application must be individually enabled for this; autonomous mode is never a global setting.

Prompt injection. Prompt injection is an attack where malicious instructions are embedded in content the AI processes with the intent of manipulating its behavior: a PDF might contain hidden text instructing the model to send files to an external server. Current language models cannot reliably distinguish between content they should process and instructions they should follow. This is a structural limitation, not a fixable bug. The defense is therefore architectural.

Every piece of content the AI receives is tagged with its origin before entering the prompt, so the model sees explicit markers distinguishing user input, external content such as PDFs, and graph data. Before external content reaches the main model, it passes through a small local model whose sole function is to detect AI-directed

Table 20 AI read access levels

Level	Name	What the AI can access
0	Minimal	Nothing from the graph; only what the user explicitly pastes
1	Session-scoped	Only data from the current active session
2	Project-scoped	All data associated with a user-selected project. Applied automatically by Focus Mode when a project is active (Section 6.8).
3	Time-scoped	Historical data up to a configurable lookback window
4	Full read	Everything the user has read access to

instructions; this is a probabilistic first pass that catches obvious injection attempts cheaply and locally.

As a hardcoded rule with no configuration surface, any action the AI wants to take that was triggered by external content requires explicit user confirmation regardless of the configured action permission level. An AI that read a malicious PDF cannot use that as a basis to execute commands without the user seeing exactly what it wants to do and why.

Documents are parsed in an isolated subprocess before their text content is passed to the AI. The subprocess has no network access and no access to the graph; only the extracted text leaves the sandbox, with no embedded scripts, no active content, and no metadata that could carry instructions.

These layers substantially reduce the attack surface and raise the cost of a successful injection, but they do not solve the underlying problem. Prompt injection is an open, unsolved problem across the AI industry [53]. No current system can guarantee that a sufficiently sophisticated injection embedded in external content will be detected or neutralized. This system's position is to minimize exposure, make AI-triggered actions visible and reversible, and be transparent about the remaining risk rather than to claim a solution that does not exist.

AI network proxy. The AI daemon has no direct network access. All cloud API calls go through a dedicated system proxy daemon that is the only component allowed to make outbound connections to AI provider endpoints. The AI daemon constructs a request, hands it to the proxy, and receives only the response. This prevents a compromised AI daemon from independently exfiltrating data over arbitrary network connections, ensures all AI-related network traffic is logged and attributable, and allows the proxy to enforce that only approved provider endpoints are contacted.

8.4.7 Audit Log

Every access to the Knowledge Graph, every AI action, and every permission grant or denial is recorded in the Audit Log. This serves two purposes: giving the user visibility into what is happening on their system, and providing the data needed for behavioral anomaly detection.

The Audit Log is an append-only ledger. Entries cannot be modified or deleted; only new entries can be added. Each entry is cryptographically linked to the previous one via a hash chain, so tampering with any historical entry is detectable by any reader that verifies the chain.

Two log tiers coexist. The Structural Log is always active and records interaction metadata without content: timestamp, application identity, which node types were queried, which relations traversed, result count, duration, and status. The query string itself, result contents, and concrete node IDs are never recorded in this tier. This is sufficient to detect suspicious behavior without being sufficient to reconstruct what data was accessed.

The Forensic Log is opt-in and time-limited. When activated in Settings, it additionally records the full query string, query parameters (not result contents), and the calling process stack trace. Forensic Mode is always limited to a maximum of 24 hours and then automatically deactivates. It is shown prominently as active in Settings while enabled and cannot be activated remotely by any application or administrator; only the user, locally.

A dedicated Audit Daemon is the sole writer to the log files. Other components send audit events over a restricted IPC channel and cannot write directly to the log. The Forensic Log is accessible exclusively to the user; no administrator, no daemon, and no application can read it. The Structural Log is readable by the user and by the Anomaly Detector for pattern analysis; in managed environments, administrators receive aggregated anomaly alerts only, never raw log contents.

Project-scoped export. The Audit Log supports filtering by project context for export. Settings exposes a date-range export that can be scoped to a specific project: the export includes all graph access records, AI actions, and permission events that occurred during sessions where that project was active in Focus Mode. The output is a structured JSON document suitable for time tracking, client billing, or project handover. The Structural Log tier is used for this export; the Forensic Log is never included in project exports. The export is produced locally and handed to the user as a file; it is not transmitted anywhere automatically.

Hash chain integrity and TPM anchoring. The hash chain makes retroactive tampering with individual log entries detectable by any reader that verifies the chain. It does not, however, prevent an attacker with early system access from deleting the log and constructing a new chain from scratch over fabricated entries, because without an external anchor the chain-start value is self-referential.

To close this gap, the HMAC key used to sign Audit Log entries is derived from a secret sealed to the TPM against the measured boot state (TPM 2.0 PCR sealing). The TPM releases the key only when the system booted in the exact state recorded at seal time: the same bootloader, the same kernel, the same initramfs. An attacker who wants to forge a clean audit chain must reproduce the exact measured boot state that authorized the signing key, which requires the Secure Boot signing key. The TPM sealing described here uses the same PCR infrastructure already required for FDE key release in Section 8.4.2; no additional hardware is needed.

On systems without a TPM the HMAC key falls back to a software-derived secret stored in the Kernel Keyring. This preserves the chain-integrity guarantee against post-hoc entry modification but not against early-access forgery. The Security section of Settings indicates which protection level is active; this difference is disclosed to the user.

8.4.8 Anomaly Detection

The Anomaly Detector is a system daemon that reads the Audit Log and identifies unusual patterns. It is not a traditional antivirus and does not match file signatures or scan for known malware. It is a behavioral analysis tool that establishes a baseline of normal behavior on this specific machine and flags deviations.

Signature-based detection requires a continuously updated database of known threats and cannot detect novel attacks. Because this system already maintains a rich picture of normal behavior via the Knowledge Graph and Audit Log, behavioral detection is the natural fit: the data needed to define normal behavior already exists.

The Anomaly Detector watches for patterns including an application suddenly querying graph node types it has never queried before; query volume from an application spiking above its historical baseline; an AI action executed without a preceding user interaction in the same session; a network connection from a process that has never previously made network connections; a new eBPF program being loaded outside system update windows; and permission escalation requests at unusual times.

Alerts are surfaced via the notification system with enough context for the user to make a decision: which application, what it did, why it is unusual, and options to investigate or dismiss. The detector does not automatically block anything; it informs. Automatic blocking based on anomaly scores is an opt-in feature for managed environments where an administrator configures the policy. The Anomaly Detector runs locally and alert data does not leave the machine unless the user or administrator explicitly configures remote alert forwarding.

8.4.9 Network Security

On a conventional Linux system, any process can open a network connection to anywhere by default. This is the wrong default. On this system, outbound network access is denied unless explicitly permitted. Every application that needs network access declares it in its package manifest by protocol, destination, and direction; connection attempts outside the declaration are blocked at the kernel level via eBPF-based network filtering and logged as security events.

VPN enforcement can be configured in managed environments such that certain network destinations are reachable only when a VPN is active. *WireGuard* [33] is the preferred VPN protocol. WireGuard is a modern VPN implemented directly in the Linux kernel (merged in 5.6), using Curve25519 for key exchange and ChaCha20-Poly1305 for encryption. Its design is deliberately minimal: the entire implementation is under 5,000 lines of kernel code, compared to tens of thousands for OpenVPN, making it easier to audit and maintain. Key management is simpler than IPsec: each peer has a static key pair, and peers are added by exchanging public keys. *OpenVPN* [90] support is provided for compatibility with existing enterprise infrastructure.

Tor and i2P. Two anonymity networks are supported as opt-in features configurable in the Advanced section of Network Settings. Both are pre-installed and require no manual setup beyond enabling them; the complexity that typically prevents adoption is handled by the system.

Tor [112] (The Onion Router) is an anonymity network that routes traffic through a series of volunteer-operated relays, encrypting it in layers so that no single relay knows both the origin and the destination of a connection. Each relay knows only its predecessor and successor in the chain; the full path is visible to no single party. Tor is widely used by journalists, activists, and privacy-conscious users to circumvent surveillance and censorship, and is operated by the non-profit Tor Project.

When Tor routing is enabled in Advanced Settings, the system configures a transparent proxy using nftables: all outbound TCP traffic is redirected through the local tor daemon, and DNS resolution is also routed through Tor to prevent DNS leaks. UDP traffic cannot traverse the Tor network and is blocked entirely while Tor routing is active rather than allowed to bypass it, which would leak information about the user's network activity. This is the correct behavior and is disclosed clearly in the Settings UI. Tor routing does not provide anonymity against an adversary who controls the entry and exit relays simultaneously, against JavaScript fingerprinting in the browser, or against application-level identifiers such as cookies and logged-in accounts. The Settings UI states these limitations plainly; Tor routing is presented as a tool, not a guarantee.

i2P [109] (Invisible Internet Project) is a separate anonymity network optimized for internal services rather than general internet access. Unlike Tor, which is primarily an anonymizing proxy for the regular internet, i2P is a self-contained network where participants host and consume services within the network itself. Websites within i2P (called *eepsites*) are reachable only from within i2P; external internet access through i2P is possible but secondary to its design intent. i2P uses a distributed hash table for routing and encrypts traffic in layers analogously to Tor, but with a different threat model and different traffic analysis resistance properties.

The system uses *i2pd* [58], a C++ implementation of the i2P router. The original i2P router is written in Java and carries a significant runtime overhead; i2pd provides the same network participation and tunnel routing in a fraction of the memory footprint, with no Java dependency. i2pd runs as a system service managed by systemd and is controlled through a first-party Tauri management UI that exposes tunnel configuration, bandwidth limits, and network participation status without requiring the user to edit configuration files manually.

Both Tor and i2P are entirely disabled by default and leave no listening services or background processes active until explicitly enabled. Enabling either requires navigating to Advanced Network Settings, not a quick-access toggle, reflecting that these tools require the user to understand their purpose and limitations.

8.4.10 Physical Security

Full Disk Encryption handles the most common physical threat, but physical access threats go further.

A Cold Boot Attack exploits the fact that DRAM does not lose its contents instantly when power is removed. At low temperatures, an attacker who obtains a running or recently suspended device could potentially read the FDE key from memory. Two mitigations apply: the kernel is configured to overwrite RAM contents before suspend-to-disk, and AMD SME and Intel TME are hardware features available on modern CPUs that encrypt the entire RAM contents with a key that lives inside the CPU die and never leaves it. Even if an attacker physically removes the RAM modules, they read only ciphertext. Both are enabled by default where hardware supports them.

USBGuard [116] enforces a policy on USB device connections. When the screen is locked, no new USB devices are permitted to become active; only devices connected before the screen locked remain accessible. This blocks BadUSB attacks, where a specially crafted USB device presents itself as a keyboard and injects keystrokes. Enabled by default; specific devices such as hardware security keys can be whitelisted in Settings.

IOMMU [14] (Input-Output Memory Management Unit) restricts which memory regions each hardware device can access via Direct Memory Access (DMA). With IOMMU enabled, a malicious Thunderbolt or PCIe device can only access the memory regions the OS has explicitly mapped for it, not arbitrary system RAM. Enabled by default via kernel parameters. For Thunderbolt specifically, the default security level requires newly connected devices to be explicitly authorized by the user before they are activated; previously authorized devices are remembered.

TPM Remote Attestation allows a device to cryptographically prove to a remote server that it booted in a known-good state by sending a signed report of its measured boot PCR values. In a managed deployment this can gate network access: a device that fails attestation is denied VPN access until an administrator investigates. This is inspired by the *BeyondCorp* [121] model (Google's zero-trust network architecture, where no device or user is trusted by default based on network location alone) and represents a meaningful differentiator for enterprise deployments.

8.4.11 Memory Safety

Memory safety bugs are the most common source of exploitable vulnerabilities in system software and are structural problems with C and C++ that persist even with experienced developers. All system components developed for this project are written in Rust. The Rust compiler enforces memory safety at compile time through its ownership and borrow checker; an entire class of bugs that would be possible in C simply cannot compile in Rust. This is a compile-time guarantee, not a runtime check with overhead.

Third-party applications cannot be rewritten. Sandboxing limits what a compromised application can do; hardware-level mitigations provide a further layer. *Intel CET* [60] (Control-flow Enforcement Technology) adds a hardware-enforced shadow stack: a second, protected copy of return addresses that the CPU maintains separately from the regular call stack. On every function return, the CPU compares the return address from the regular stack against the shadow stack; a mismatch causes an immediate fault. This defeats the most common class of control-flow hijacking, return-oriented programming (ROP), where an attacker overwrites saved return addresses

to chain together existing code fragments. CET is supported by the Linux kernel since version 6.6 and is enabled by default for new processes on supported hardware with no application changes required. For any C or C++ components compiled as part of this project, Clang's *CFI* [75] (Control Flow Integrity) instrumentation is applied. CFI adds compile-time instrumentation that validates at runtime that every indirect function call and virtual method dispatch targets only a function with the expected type signature; an attacker who corrupts a function pointer to point to an arbitrary location will cause an immediate abort rather than code execution.

8.4.12 Multi-User Isolation

When multiple users share a system, their isolation goes deeper than traditional Unix file permissions because the Knowledge Graph, AI configuration, and Audit Log introduce sensitive state beyond the home directory. Each user has a separate Ladybug instance, a separate Graph Daemon process running under their own system user, and a separate Audit Log. These are not shared resources with access control layered on top; they are separate instances that other users' processes cannot reach.

Home directories are managed via `systemd-homed`: each user's home directory is stored as a LUKS-encrypted portable image. When a user logs out, their home image is unmounted and the encryption key is discarded from memory. No other user, including root, can access the home image without the user's credentials. The Knowledge Graph lives inside the home directory and is therefore protected by this encryption as a structural consequence rather than an additional policy.

In a multi-user or managed environment, an administrator can install and remove system-wide software, configure network policies, set password and screen lock policies, view aggregated anomaly alerts, manage USBGuard whitelists, and manage Secure Boot keys. The administrator explicitly cannot read any user's Knowledge Graph, access any user's home directory contents, view any user's Audit Log, or intercept any user's AI sessions. These limitations are enforced by `systemd-homed` and the per-user Graph Daemon architecture, not by policy alone. The administrator sees that something unusual happened via anomaly alerts; they do not see what.

8.5 Managed Environments

For enterprise deployments where this system runs on many machines administered centrally, additional infrastructure is required beyond what individual Signed Profile Packages provide. This section describes the managed environment architecture for organizations that need fleet-level control.

Policy Engine. A system daemon receives signed policy packages from a central management server and applies them locally. The daemon verifies the cryptographic signature of every package before applying any policy; unsigned or incorrectly signed packages are rejected outright. Policies can control permitted and blocked applications, network access rules beyond per-app declarations, AI provider restrictions including complete disabling of cloud AI, the USBGuard device whitelist, screen lock timeout and password complexity requirements, update scheduling so that updates are applied

only during configured maintenance windows, and a minimum audit log retention period for compliance purposes.

LDAP and Active Directory. Most enterprises already have an identity management system, typically Microsoft Active Directory or an LDAP-compatible directory. Integration with these systems is a baseline requirement for enterprise adoption. *LDAP* (Lightweight Directory Access Protocol [98]) is the standard protocol for querying and modifying directory services such as Microsoft Active Directory and OpenLDAP. It stores organizational information: users, groups, computers, and their attributes. *Kerberos* [83] is the authentication protocol layered on top: it issues time-limited tickets that prove identity without sending passwords over the network; Active Directory uses Kerberos as its primary authentication mechanism.

User authentication, group membership, and basic policy delivery are handled via these standard protocols, which existing enterprise infrastructure already supports without proprietary agents on the management server side.

MDM-style management. Beyond LDAP, a modern MDM-style management layer provides richer capabilities: device enrollment, remote policy push, health status reporting, and remote wipe. This is a longer-term goal; LDAP compatibility is the foundation and MDM-style management is built on top of it once the base is stable.

Remote attestation as network gate. TPM Remote Attestation, described in Section 8.4.10, can be used as a network access gate in managed deployments. The workflow is:

1. The device boots and generates a TPM attestation report containing signed PCR values.
2. The device connects to the management server's attestation endpoint.
3. The server verifies the report against the expected baseline for that device's configuration.
4. If verification passes, the device receives its VPN credentials and network access.
5. If verification fails, the device is quarantined pending administrator review.

This ensures that a device with a modified bootloader, disabled Secure Boot, or an outdated security policy cannot silently access company resources. It is inspired by the BeyondCorp model [121] and is a meaningful differentiator for enterprise deployments.

GDPR and compliance tension. In managed environments there is an inherent conflict between user privacy rights and organizational compliance requirements. GDPR grants users rights over their personal data; many compliance frameworks in financial services, healthcare, and legal contexts require organizations to retain logs and audit trails. This system does not resolve the conflict, because it is a legal and organizational question rather than a technical one.

What the system does is make both sides technically possible. Users retain full visibility into and control over their own Knowledge Graph and personal Audit Log. Organizations can configure a minimum retention period for compliance-relevant

audit events, which do not include Knowledge Graph contents. The separation between personal graph data, which is private to the user, and system audit events, which may be subject to organizational retention requirements, is maintained architecturally rather than by policy. Organizations deploying this system should seek legal counsel to define appropriate policies. The system provides the technical primitives; the legal framework is out of scope.

9 Companion App and Health

9.1 Companion App

The system ships with a first-party companion app for Android and iOS that bridges phone and desktop into a single coherent experience. It is built with Tauri 2.0 using the same Rust backend, the same design token system, and the same UI component library as the desktop. The companion app looks and feels like the OS because it is built from the same parts.

The protocol is custom. *KDE Connect* [65] is a widely used open-source protocol and application that links Linux desktops and Android phones: clipboard sync, notification mirroring, file transfer, and remote input. It was evaluated and rejected as the foundation for this project. Its protocol encodes assumptions appropriate for a bolt-on integration layer – such as per-feature capability negotiation at the session level rather than at pairing time – that would obstruct the deeper integration features planned for later tiers. Building on top of it would mean working around its assumptions rather than designing for the actual requirements.

Feature tiers. Features are organized into three trust tiers established at pairing time. Table 21 shows the breakdown. The distinction between paired and trusted exists because Tier 2 features involve access to graph context and health data, which require a higher trust level than file transfer and notification mirroring. Tier 2 is restricted to the official app; a third-party client that implements the pairing protocol only gets Tier 1.

Table 21 Companion app feature tiers

Tier	Trust level	Features
1	paired	File Drop; clipboard sync (manual trigger only, not automatic); notification mirroring; media playback control
2	trusted	Handoff (continue reading, browsing, or editing across devices; project-aware when Focus Mode is active); Unified Timeline (cross-device activity, local only); graph presence (phone sends active app and content context); health data forwarding
3	Planned	Wearable data pipeline; extended context sharing

Protocol and transport. Discovery uses *mDNS* [27] (Multicast DNS) service advertisement on a shared WLAN. mDNS is the same zero-configuration discovery protocol used by Apple's Bonjour and by Avahi on Linux; it allows devices to announce themselves by name on a local network segment without a central DNS server. Without a shared network, the desktop and phone locate each other via *BLE* (Bluetooth Low Energy [7]) presence broadcast carrying the device ID. BLE is a mode of Bluetooth designed for low-power periodic broadcasting; a device can transmit small advertisement packets continuously at minimal battery cost without maintaining a connection. Transport is selected by availability at connection time, shown in Table 22.

Table 22 Companion app transport selection

Transport	Use case
HTTPS over LAN	Primary transport; full feature set
WiFi Direct [126]	Large transfers without a shared router. WiFi Direct allows two devices to form a peer-to-peer 802.11 connection without a router or access point, at full WiFi throughput.
Bluetooth	Small data only: clipboard, notifications, control commands

Pairing uses a Trust on First Use model. Both devices generate an *Ed25519* [63] keypair on first launch. Ed25519 is an elliptic-curve digital signature scheme based on Curve25519; it produces 64-byte signatures, is fast to compute and verify, and has no known weaknesses from weak random number generation (a historical problem with earlier elliptic-curve schemes like ECDSA [25]). The first connection uses *TLS* [94] (Transport Layer Security, the protocol underlying HTTPS) with self-signed certificates rather than certificates issued by a certificate authority. Self-signed certificates cannot be validated by the standard CA chain, but here that is intentional: the validation happens out-of-band via a six-digit confirmation code displayed on the desktop and confirmed on the phone. This code is a hash of both devices' public keys; confirming it proves that the TLS session is between the two expected devices and not an attacker in the middle. After pairing, reconnection is automatic with no further confirmation. Unpair deletes the key on both sides with no trace.

Auto-connect. Paired devices broadcast BLE presence in the background. When a known device is detected, the system connects automatically using the best available transport. No user action is required after initial pairing. Android supports full background BLE advertising reliably. iOS imposes *CoreBluetooth Background Mode* [3] restrictions: background BLE scanning is not permitted continuously, only in periodic system-managed windows. Auto-connect works but is less reliable and carries a marginally higher battery cost on the iPhone side.

Project-aware Handoff. When Focus Mode is active on the desktop (Section 6.8), the Handoff feature carries project context to the phone. The companion app receives the active project's name, description, linked resources, and recent files from the graph, and surfaces them as a continuation card on the phone's lock screen and in the

app. Tapping the card opens the project's issue tracker link, most recently accessed document, or notes entry depending on which was last active. The project context is transmitted as structured metadata over the existing Tier 2 channel; no additional pairing or permission is required beyond the trusted trust level.

When Focus Mode is not active, Handoff operates in its baseline mode: the phone receives the currently active application and document without project context.

Theme synchronization. Because the companion app uses the same design token system as the desktop, the active theme propagates to the phone automatically: accent color, dark and light mode, typography. When a project accent color is active via Focus Mode, that override propagates to the companion app as well, giving the phone the same ambient context signal as the desktop shell. This is a structural side effect of the shared token infrastructure, not a feature that was separately implemented.

iOS limitations. iOS imposes constraints that cannot be worked around. There is no persistent background BLE scanning, which makes auto-connect less reliable and requires more manual reconnects than on Android. WiFi Direct is not available to third-party iOS apps. USB-C wired transport is inaccessible to third-party iOS apps without *MFi* (Made for iPhone/iPad/iPod) certification [5], Apple's hardware licensing program. Apps without MFi certification cannot use the Lightning or USB-C accessory protocol for direct data transfer. Health data forwarding via *HealthKit* [4] (Apple's on-device health data framework, accessible to third-party apps with explicit user consent) is feasible with explicit user permission.

Tier 1 works well on iOS. Tier 2 is functional but degraded. iOS is explicitly a secondary platform and this limitation is communicated honestly in the app and in documentation rather than obscured.

9.2 Health and Wellbeing

The system includes a first-party health data component: a local daemon that aggregates, stores, and optionally analyzes health and wellbeing data from connected sources. This is not a fitness application. It is health data infrastructure: an open, local equivalent of Apple Health or Google Fit, built as a system component with the same privacy guarantees as the rest of the platform.

Data architecture. Health data lives in a dedicated local store managed by a *Health Daemon*, separate from the Graph Daemon. Data never leaves the device without an explicit user action. Clinical data uses *FHIR R4* [57] (Fast Healthcare Interoperability Resources, the international standard for structured health data interchange maintained by HL7). FHIR represents health records as typed REST resources – Observation, Condition, MedicationRequest, and others – with a defined JSON and XML serialization. Hospital and clinic systems across the EU are increasingly FHIR-capable, particularly following regulatory pressure in Germany (MIO/DiGA) [45] and the broader eHealth interoperability mandates [36]; using FHIR natively enables direct import and export without format conversion. Non-clinical data such as steps,

sleep stages, heart rate variability, and nutrition uses an open JSON schema with documented field definitions. There is no proprietary format; full export is always available.

Data sources. The initial release supports manual input and phone sensors forwarded via the Companion App: steps, GPS, and screen time. The plugin-based architecture is designed from the start to accommodate further sources without structural changes: wearables via the Companion App, hospital and clinic FHIR import, and third-party applications via a documented Health API gated by the standard permission system.

Wearable pipeline. Wearables connect to the system through the Companion App rather than directly, using the wearable manufacturer's existing phone SDK on the phone side. The data path is:

```
Wearable -> Companion App (phone, manufacturer SDK)
         -> OS Health Daemon (via Companion protocol)
```

Listing 50 Wearable data path

No direct wearable protocol implementation is required in the OS. The Health Daemon's plugin interface is designed to accept forwarded wearable data from the start, so that adding wearable sources later does not require structural changes. This is planned architecture, not initial release scope.

Graph integration. Health data integration with the Knowledge Graph is strictly opt-in and split into three explicit levels, all of which default to off. Table 23 lists the levels. Each level requires the previous one; the Settings UI enforces this and does not allow incompatible states. The AI layer's health-adjacent features are gated entirely on these toggles. With all three off, the health module is a pure local viewer and aggregator with no AI involvement whatsoever.

Table 23 Health graph integration levels

Level	What it enables
1 - Health Nodes	Health events appear as standalone nodes in the Knowledge Graph
2 - Correlations	Graph links health data to activity data by time window (e.g. sleep quality correlated with focus session length; active focus time per project derived from Focus Mode session records)
3 - Pattern Recognition	Full cross-domain analysis across health, productivity, and communication data

Project focus time. Focus Mode session records (Section 6.8) are written to the Knowledge Graph with project context attached. The Health Daemon reads these records at Level 2 integration and derives a focus time metric per project per day: the

total duration of sessions where a given project was active and the user was interacting with the system rather than idle. This metric feeds two Health Dashboard views. The *work distribution* view shows time split across active projects over a configurable rolling window, useful for identifying whether time allocation matches priorities. The *concentration signal* view flags extended periods of single-project focus, which correlates with both deep work and with burnout risk depending on duration and frequency. Neither view interprets the data for the user; both present the signal and let the user draw conclusions.

The focus time metric requires only Level 2 graph integration; no health sensor data is involved. It is derived entirely from session and project activity already present in the graph.

Settings and permissions. Health has its own dedicated section in Settings rather than being buried under general privacy settings. Controls include per-source enable and disable for each data source independently; per-data-type enable and disable so the user can, for example, share step data with an app but not sleep data; the graph integration toggles from Table 23 presented as a grouped sub-section with clear explanations of what each level enables; project focus time tracking as a sub-toggle within Level 2 integration; a configurable data retention period per data type; full FHIR and JSON export; and full deletion with a confirmation step.

Third-party applications request health data access per data type through the standard permission system, but health-specific confirmation dialogs are more explicit about data sensitivity than generic permission prompts. A fitness app requesting sleep data gets a dialog that names the data type and explains clearly that this is health information, not a generic file access request.

10 Developer Experience, Sustainability, and Roadmap

10.1 Developer Experience and Infrastructure

The development infrastructure is designed around one constraint: a project of this scope spans many components with cross-cutting concerns, and the tooling must make that manageable without forcing artificial coupling between parts that should evolve independently.

Repository structure. The project uses a multi-repo layout under a single GitHub organization. Each component has its own repository, its own release cycle, and its own CI pipeline. Components can be updated independently as long as they do not introduce breaking changes to the interfaces they expose. A breaking interface change requires a coordinated update across all affected components, enforced by the versioning mechanism described below.

```
# Documentation and architecture
foundation <- architecture document (this document)
wiki <- user-facing wiki, API reference landing page,
```

```
                                rustdoc/typedoc aggregation script

# OS image and distribution
distro <- mkosi configuration, ISO generation,
                                meta-packages, VM scripts, CI image pipeline
store-backend <- Store API service, PostgreSQL schema,
                                object storage integration

# System layer
compositor <- cosmic-comp fork (standalone repo, not a
                                GitHub fork; upstream tracked via git remote)
kernel-layer <- eBPF programs, Event Bus daemon
knowledge <- SQLite write store, Ladybug query store,
                                Graph Daemon, promotion pipeline
ai-layer <- provider abstraction, MCP client, AI daemon
notification-daemon <- Notification daemon (Rust)
install-daemon <- Install Daemon, Integration Package matching

# Desktop and shell
desktop-shell <- taskbar, Waypointer, Notification Center,
                                Quick Settings (Tauri + TypeScript)
app-lock-screen <- Lock Screen (privileged context, own repo)
app-greeter <- Greeter (runs before session, own repo)

# First-party applications
app-settings <- Settings
app-files <- File Manager
app-terminal <- Terminal
app-store <- Store
app-text-editor <- Text Editor
app-system-monitor <- System Monitor
app-wine-manager <- Wine Manager
app-viewers <- PDF Viewer, Image Viewer, Archive Manager

# Shared libraries
ui-kit <- component library, design token system
os-sdk <- Rust crate + TypeScript shell API
                                (first-party apps)
module-sdk <- shell API subset, module scaffolding
                                (third-party modules)
themes <- GTK4 theme, Wine .msstyles, token source

# Community
community-integrations <- community Integration Packages (PR-based,
                                CI validates schema)
community-adapters <- community Settings Adapters (PR-based,
                                CI validates schema)
```

Listing 51 Repository structure

On GitHub. The project is hosted on GitHub. This requires an honest acknowledgment: GitHub is owned by Microsoft, hosted in the United States, and therefore structurally

at odds with the project's European digital autonomy position. The decision to use it is deliberate and time-limited, not an oversight.

Three factors make GitHub the right choice for the build phase. First, network effects: the overwhelming majority of open-source contributors are already on GitHub. Reducing the contribution barrier at the point when the project most needs contributors is worth the sovereignty trade-off temporarily. Second, tooling maturity: GitHub Actions, Dependabot, and the PR review infrastructure are more reliable than the alternatives at this stage. Building community around infrastructure problems is a distraction the project cannot afford early on. Third, trust signal: a public GitHub repository with visible activity, issues, and contributions is a credibility marker for a new project asking users to run it as their operating system.

The migration path is planned and not open-ended. *Forgejo* [40] – a fully open-source, self-hostable Git forge – is the target for self-hosted source control. *Codeberg* [28], a Forgejo-based hosting service operated by a German non-profit, is a viable intermediate step. Git repositories are trivially portable; the harder migration is Issues, PRs, and CI pipelines. That migration will be executed when the project has the operational capacity to absorb it without disrupting contributor workflows.

The distro repository. distro is responsible for everything that turns individual component builds into a running operating system. It contains the mkosi configuration declaring which packages and overlays go into the image, build scripts, VM tooling for development and testing, meta-package RPM spec files, and the CI pipeline that builds images on release candidates.

Two image build targets exist. The *production image* contains no SSH server, no debug symbols, and optimized binaries. This is what users receive. The *developer image* adds an SSH server, debuginfo packages, Rust binaries compiled with debug symbols, and a known SSH key injected at build time. It is only buildable via an explicit flag and is never distributed as a release artifact.

```
# Production image (default)
distro/build-image.sh

# Developer image with SSH and debug symbols
distro/build-image.sh --target developer

# Start a normal development VM
distro/run-vm.sh

# Mount a local directory into the VM via virtiofs
distro/run-vm.sh --share ~/projects/kernel-layer

# Start a persistent VM for update scenario testing
distro/vm-update-test.sh --init # create persistent qcow2 disk
distro/vm-update-test.sh --start
distro/vm-update-test.sh --apply-update
distro/vm-update-test.sh --list-snapshots
distro/vm-update-test.sh --rollback
```

Listing 52 distro build targets

The update test target uses a qcow2 disk image that persists across VM restarts, enabling reproducible testing of the full Snapper snapshot, health-check, and rollback cycle. The developer image is used as the base for update tests so that SSH access and debug tooling are available during the scenario.

A strict rule applies to eBPF development: eBPF programs are developed and tested exclusively inside the VM, never directly on the development machine. A bad eBPF program can destabilize the kernel; the VM isolation is not optional.

Build system. *mkosi* [104], maintained by the systemd team, is the target tool for image generation. It builds bootable OS images from a declarative configuration file that specifies which packages to include, which overlay files to add, and how to configure the system. Unlike Kickstart or Kiwi, mkosi is designed around the systemd ecosystem: it integrates directly with systemd-repart for partition layout and produces well-formed UEFI images that support Secure Boot and Unified Kernel Images without custom glue scripts. It supports multiple output formats: ISO, raw disk image, and container. It integrates cleanly with the Open Build Service. Confirming that mkosi handles the full build pipeline in practice is an open item pending actual use.

The *Open Build Service* [86] (OBS) handles packaging for all custom components. OBS is an openSUSE-hosted build infrastructure that takes RPM spec files and produces binary packages for multiple distributions and architectures in parallel, without requiring the developer to provision build machines for each target. Packages are signed with the project GPG key – *GPG* (GNU Privacy Guard [51]) is the standard open-source implementation of the OpenPGP signing and encryption standard; package signatures let end-user systems verify that a package was built and signed by this project and has not been tampered with. OBS hosts the resulting package repository that end-user systems pull from. Both Fedora and OpenSUSE targets can be built from the same spec files, which is one practical reason both remain viable candidates.

Update system. Updates arrive from two independent streams managed separately. The base system, meaning the kernel and base distribution packages, is handled by the native package manager (dnf or zypper depending on the chosen base), with Snapper automatically taking a Btrfs snapshot before every update. Project components, meaning the compositor, shell, event bus, graph daemon, AI layer, and first-party applications, are distributed via the project's own OBS repository as RPM packages managed as a coordinated meta-package.

The update behavior is designed to be non-intrusive. Security updates for both streams are downloaded automatically in the background and applied at the next shutdown or reboot, never during an active session. No forced reboots, no countdown timers. All other updates are downloaded silently in the background; a single calm notification appears when they are ready. The user applies them when convenient. In managed environments, administrators can configure mandatory update windows; even then, no mid-session interruption occurs.

Rollback is available for every update: Btrfs and Snapper record a snapshot before each change, and the user can boot into a previous snapshot from the boot menu and roll back the entire system without any additional tooling.

Project components communicate over defined IPC interfaces. A breaking interface change increments the major version number and must be shipped as a coordinated meta-package update, so that incompatible component versions cannot coexist. Components verify interface compatibility at startup:

```
fn check_compatibility() -> Result<()> {
    let daemon_version = get_graph_daemon_version()?;
    if daemon_version.major != EXPECTED_MAJOR {
        bail!("Graph Daemon interface version mismatch. Run system update.");
    }
    Ok(())
}
```

Listing 53 Interface compatibility check at startup

A component that detects a version mismatch refuses to start and surfaces a clear message. The only resolution is applying the pending system update.

Hosting infrastructure. The project operates two distinct distribution channels with different infrastructure requirements.

System package repository. All core system components – the compositor, shell, daemons, and first-party applications – are distributed as RPMs via the Open Build Service. OBS builds, signs, and hosts the repository. End-user systems add the repository once:

```
dnf config-manager --add-repo \
    https://packages.[project].dev/fedora/$releasever/
```

After that, updates arrive through the normal `dnf` update mechanism. No custom infrastructure is required for this channel; OBS provides everything.

Store backend. The Store requires purpose-built infrastructure because its requirements are specific to this project: Integration Package matching, adapter schema validation, the rating-based Install Daemon policy, signed package verification, and payment processing. The target stack is a Rust backend service with PostgreSQL for metadata and S3-compatible object storage for package binaries. *Hetzner Object Storage* is the preferred object storage provider: it is operated in European data centres, priced predictably for outbound traffic, and S3-compatible without vendor lock-in. A CDN sits in front of the object storage for binary downloads; *BunnyNet* is the preferred provider for the same reasons.

Payment processing uses *Stripe*, which covers European payment standards, handles VAT for digital goods across EU jurisdictions, and provides a well-documented API. Stripe's commission is accounted for separately from the Store's own commission tiers described in Section 7.6.

The Store backend is self-hosted, not dependent on any US-based platform service for its critical path. This is a direct consequence of the project's European digital autonomy stance: the infrastructure that distributes software to users should be under the project's control and subject to European law.

CI/CD pipeline. Tests run at three levels of integration. Table 24 shows when each level runs.

Table 24 CI trigger summary

Trigger	What runs
Every push or pull request	Unit tests for the affected repository
Pull request merge to main	Unit tests, lint, and build check
Daily at 02:00	Cross-component integration tests against all main branches
Release candidate	Full VM tests and smoke tests
kernel-layer push	Unit tests plus VM-based eBPF tests

Unit tests run per repository against mocks from os-sdk. Rust components use cargo test; TypeScript components use *Vitest* [119], a Vite-native unit test framework with Jest-compatible API and fast execution via native ES modules. No PR merges without green unit tests. The kernel-layer is a special case: eBPF logic that can be tested without a real kernel runs on the host as normal Rust unit tests, while eBPF programs themselves run in a minimal QEMU instance that CI spins up automatically.

Cross-component integration tests verify that two or more components communicate correctly over their real IPC interfaces. They run on the host without a full system boot, starting only the directly involved daemons as real processes against real IPC sockets. A failing integration test with all unit tests passing means interface drift between repositories, which is the primary failure mode this level exists to catch. Examples: the Event Bus emits a FileAccessEvent and the Graph Daemon writes correctly to SQLite; a theme token change in Settings propagates to all running Tauri processes; the shell sends a keybinding action and the Input Manager calls the correct handler.

Full VM tests boot a complete system image in QEMU and run smoke tests: system boots without crash; Waypointer opens, finds an application, and launches it; eBPF events arrive and land in the Knowledge Graph; theme switch propagates correctly; module install and uninstall behave as expected. These tests are slow and run on release candidates only. The VM image is built by the same script used for real releases; no separate test image exists.

Local development workflow. Development happens at three levels of integration, each suited to a different class of work. The common case is developing a single component in isolation. Each repository has its own development setup with cargo run, tauri dev, or hot-reload as appropriate. Components that use the mocks from os-sdk need no external dependencies running. Fast feedback loop, runs entirely on the developer's machine.

For shell and compositor work, the cosmic-comp fork supports nested mode: the compositor runs as a regular Wayland window inside the developer's existing desktop. The full shell appears inside that window. No VM or extra hardware is needed. This covers the majority of shell and compositor development; eBPF and kernel-level components are not active in nested mode.

Full system work requires *QEMU* [24], a machine emulator and virtualizer. QEMU can emulate a complete x86-64 machine – CPU, RAM, storage, network, and display –

entirely in software, or accelerate execution via KVM (the Linux kernel's hardware virtualization support) to near-native speed when running on matching hardware. This covers eBPF development, the boot sequence, the update system, and anything that requires a real kernel or a complete system environment. The distro repository contains the full script set for this workflow, described in the distro paragraph above.

Mock strategy. Every IPC interface in the system has an official mock implementation that lives in os-sdk directly alongside the interface definition. Mock and real implementation share the same type definitions. An outdated mock is a build error, not a runtime surprise: if an interface changes, the mock must be updated in the same pull request.

```
#[cfg(test)]
use os_sdk::graph_daemon::mock::GraphDaemonMock as GraphDaemon;

#[cfg(not(test))]
use os_sdk::graph_daemon::client::GraphDaemonClient as GraphDaemon;
```

Listing 54 Conditional mock usage in tests

There are no per-repository custom stubs and no fixture files that can drift from the real schema. The mock is the contract.

Project planning tooling. A project of this scope has cross-cutting concerns that span multiple repositories simultaneously: an Epic about the Waypointer Module API touches desktop-shell, os-sdk, module-sdk, and potentially compositor. Standard per-repository issue tracking is not sufficient for this. The requirements are hierarchical issues or Epics that span multiple repositories, dependency tracking between issues, a roadmap view across all components, and cross-repository milestone coordination.

Three options are under evaluation. The first is GitHub Projects v2 with a meta-repo convention: cross-cutting Epics live as issues in the foundation repository and component repositories link back to the relevant Epic; GitHub Projects v2 aggregates issues from all repositories into one board. The advantage is zero additional tooling and everything staying in GitHub. The disadvantage is that dependency tracking is honor-system only with no enforcement, there is no proper roadmap view, and Epic-to-child linking is manual.

The second option is *Linear* [74], a proprietary hosted tool with strong multi-repo GitHub integration, real Epics with child issues, a dependency graph, and a roadmap view. It has the best user experience for exactly this use case but is proprietary, hosted externally, and costs money at team scale.

The third is *Plane* [92], an open-source Linear alternative that can be self-hosted. Feature parity with Linear is growing but not yet complete. Its advantage is that self-hosting fits the project's digital sovereignty philosophy with no external dependency; the disadvantage is additional operational overhead and some features that are still maturing.

The decision is not yet made. The requirement is clear: cross-repository Epic tracking with dependency enforcement and a roadmap view. GitHub Projects alone is likely insufficient for a project of this scope.

Documentation. Two documentation sites are maintained, both self-hosted. The user-facing wiki at wiki.[project].dev is written by hand in the spirit of the Arch Wiki: prose that explains intent and reasoning, not only what a setting does. It covers concepts, how-tos, troubleshooting, and architecture explanations. Community contributions are accepted via pull requests reviewed by the core team. The wiki source lives in the wiki repository, separate from foundation which contains only the architecture document.

The developer API reference at api.[project].dev is auto-generated from code comments: rustdoc for Rust components, typedoc for TypeScript components, aggregated and hosted under a single domain so there is one place to search across all components. The aggregation script and the API reference landing page live in wiki alongside the user-facing documentation; both sites share the same visual language and are maintained together. Keeping the API reference current is a CI gate: a pull request that removes or modifies a public API without updating its documentation comment does not merge.

10.2 Sustainability and Business Model

This is an open source project. The code is and remains open; that is a foundational commitment, not a marketing position. At the same time, building and maintaining a full operating system is a serious engineering effort that requires sustained funding. The goal is not to extract value from users but to find revenue streams that are structurally aligned with the project's values: user sovereignty, transparency, and European digital autonomy.

Principles. Four constraints rule out entire categories of business models upfront. Core OS functionality is not feature-gated behind a paywall; the desktop, the permission system, the Knowledge Graph, and the AI layer are open and available to everyone. There is no advertising, no telemetry sold to third parties, and no data brokering. Venture capital with growth-at-all-costs expectations is incompatible with a project built on user trust and is therefore excluded. Enterprise features are an acceptable source of revenue; making the core worse for non-paying users is not.

Store revenue. The Store's tiered commission model, described in Section 7.6, generates revenue proportional to ecosystem health. The project earns meaningfully only if developers earn well first. At early stage this is negligible. At meaningful adoption it becomes a significant funding stream. The Store handles application sales, theme sales, module distribution, and profile package distribution under a single infrastructure.

Enterprise services. The Signed Profile Package system described in Section 8.2 is already designed with organizations in mind. The .lenv format is open and documented; anyone can author packages. Organizations deploying managed environments at scale will often want professional help: authoring and maintaining custom packages, deployment tooling, update management, and support SLAs. This is a natural B2B

offering that does not touch the open source core. The analogy is Red Hat selling support for software that anyone can download for free.

OEM and hardware partnerships. At sufficient maturity and hardware compatibility, small European laptop manufacturers may want to ship the system pre-installed. The structural alignment is good: the manufacturer gets a differentiated product, users get a well-tested hardware and software combination, and the project gets sustainable revenue from certification and validation work and an ongoing support relationship. This requires a stable release and a polished out-of-box experience first and is a longer-term play. Relevant candidates in the European market include *Tuxedo Computers* [114], the *PINE64* [91] ecosystem, and potential *Framework* [41] partnerships.

Support tiers. A support subscription aimed at power users, small businesses, and developers who want guaranteed response times and direct access offers prioritized bug reports, LTS release guarantees with defined support windows, and a direct communication channel with the core team. No features are locked behind this; it is a service layer on top of the open product, priced modestly enough to be accessible to individual developers.

Public funding and grants. The European open source and digital sovereignty funding landscape is underutilized by most projects. The *Sovereign Tech Fund* [101] in Germany funds open source infrastructure projects and explicitly values work that reduces dependency on US technology monopolies. The *NLnet Foundation* [84] funds internet and open source projects with a privacy and user rights focus and has supported numerous small to medium projects. The *NGI* (Next Generation Internet [37]) programme runs multiple EU funding calls per year for open source, privacy, and decentralization work.

The European digital autonomy angle is not only a philosophical position; it is a genuine funding argument. A European open source operating system built on sovereignty principles is precisely what these programmes exist to support. Grants do not provide recurring revenue but they can fund specific development phases without creating investor obligations or governance compromises.

Path to incorporation. A company makes sense when there is external money to manage, not before. Premature incorporation adds administrative overhead with no corresponding benefit. A realistic sequence: the project gains traction and a community forms; grant applications are filed; Store revenue starts, providing the first external money flow; the first enterprise inquiry for deployment support arrives as the first B2B signal. At that point incorporation becomes appropriate, probably as a GmbH in Austria or Germany, or as a European cooperative structure if community governance is a priority.

The company's role is to employ core contributors, manage the Store infrastructure, and sell enterprise and support services. The open source project remains governed independently of the company: contribution policy, release decisions, and core ar-

chitecture are not dictated by commercial interests. This separation must be written down explicitly before the company is incorporated, not negotiated afterward under time pressure.

10.3 Roadmap

The roadmap is intentionally sparse at this stage. Defining detailed phase scope before the foundational components are proven would produce a plan that reflects assumptions rather than reality. Phases 1 through the public release will be scoped as Phase 0 concludes and the first components are built and measured.

Phase 0 - Foundation (current). Architecture and design decisions are documented in this blueprint. The proof of concept consists of the Event Bus and Knowledge Graph running as a daemon and collecting basic system events. The base distribution decision between Fedora and OpenSUSE Slowroll [89] is pending and will be made before Phase 1 begins. A sustained write benchmark of the SQLite write store under simulated eBPF load targeting 5,000 events per second, combined with validation of the promotion pipeline into Ladybug for graph queries, is a Phase 0 exit criterion.

Phase 1 and beyond. Phase scope will be defined once Phase 0 is complete. The ordering of subsequent phases follows the architectural dependency graph: kernel layer and event infrastructure before the graph, the graph before the AI layer, the compositor before the shell, the shell before Store and module infrastructure. Public release comes after a stable, tested combination of all layers exists.

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